

Masterarbeit

Benthic invertebrate primary colonization of two experimental channels in Lunz am See, Austria

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Abstract

The master thesis deals with the primary colonization of two man-made experimental channels in Lunz am See, Lower Austria by benthic invertebrates. These channels were created to investigate the effects of pulse-release on aquatic organisms. Water is supplied directly from the lake Lunzer See via two pipelines and flow through the channels started on the 9th of September 2011. Macrozoobenthos-sampling was conducted monthly until the 10th of April 2012 using a box-sampler. During the sampling period the channels were subject to repeated pulse-releases, which occurred more often in the orographically right channel. Moreover, the channels were stocked with young fish several times, which were removed by electro-fishing after pulse-release experiments.

During the first three months only few macrozoobenthic specimens were found in the samples. The period from December until March is characterized by growth of total individual biomass (average biomass per specimen) and abundance of a few taxa. Abundance continued growing from March until the end of the study period, but biomass declined. All samples from the channels showed explicitly lower abundance and biomass than the reference samples from the Unterer Lunzer Seebach which parallels the channels. However, development of colonization was very similar in both channels. Chironomids dominated the fauna in the experimental channels. These and the stonefly genera *Nemoura/Nemurella* were the only taxa reaching abundances similar to those in the Seebach by the end of the study period. Identification of Chironomid larvae of the last sampling date showed distinct dominance of several species of the genus *Diamesa*.

This study showed that colonization through oviposition most probably was the quantitatively most important path for colonization and that pioneer taxa show characteristic traits which support this way of colonization. Drift, upstream migration and colonization from the sediment are of minor importance in this particular case. Dominance of *Diamesa* spp. could either be a result of limitation in sources of colonists or disturbances in the channels.

Zusammenfassung

Die vorliegende Arbeit befasst sich mit der Erstbesiedlung zweier künstlich angelegter Fließwasserrinnen in Lunz am See, Niederösterreich durch Makrozoobenthos. Bei diesen Rinnen handelt es sich um Experimentierrinnen zur Untersuchung der Auswirkungen von hydrologischen Schwankungen auf aquatische Organismen. Die mit Flusssediment gefüllten Rinnen werden über zwei Rohrleitungen direkt aus dem Lunzer Untersee gespeist und wurden am 9. September 2011 erstmals durchflutet. Die Makrozoobenthos-Probenahme erfolgte mit einem Kasten-Sampler in Abständen von rund zwei bis vier Wochen bis zum 10. April 2012. Während des Untersuchungszeitraums wurden in beiden Rinnen Schwallereignisse initiiert, in der orographisch linken jedoch deutlich seltener als in der rechten. Des Weiteren wurden mehrmals Jungfische ausgesetzt und nach den Schwallversuchen durch Elektrofischung wieder entfernt.

Während der ersten drei Monate befanden sich nur wenige Individuen in den Proben. Von Dezember bis März war ein Anstieg der durchschnittlichen individuellen Biomasse und der Abundanz einiger Taxa zu verzeichnen. Zwischen 20. März und 10. April 2012 wuchs die Abundanz weiter, die Biomasse sank jedoch wieder. In allen Proben waren sowohl Abundanz als auch Biomasse deutlich geringer als in den Proben aus einem Vergleichsgewässer (Unterer Lunzer Seebach), die beiden Rinnen unterschieden sich aber kaum in ihrer Besiedlung. Die Fauna beider Rinnen wurde von Chironomiden dominiert. Diese und die Steinfliegen-Gattungen *Nemoura/Nemurella* waren die einzigen Taxa, welche zum Ende des Untersuchungszeitraums in den Rinnen eine ähnliche Abundanz aufwiesen wie im Seebach. Eine Bestimmung der Chironomidenlarven der letzten Probenahme ergab eine ausgeprägte Dominanz verschiedener Arten der Gattung *Diamesa*.

Es konnte gezeigt werden, dass Besiedlung durch Eiablage sehr wahrscheinlich die quantitativ wichtigste Form der Besiedlung war und dass Pionier-Taxa charakteristische Eigenschaften haben, welche diesen Besiedlungsweg begünstigen. Besiedlung durch Drift, Flussauf-Wanderung und Besiedlung aus dem Flusssediment von Larven kann in diesem konkreten Fall eine untergeordnete Bedeutung beigemessen werden. Die festgestellten Dominanzverhältnisse könnten ein Resultat der eingeschränkten Besiedlungsmöglichkeiten oder der regelmäßigen Störungen in den Rinnen sein.

1 Introduction

This work is based on benthic samples of two newly human-made experimental channels that are part of the scientific facility HyTEC constructed in the summer of 2011 in Lunz am See, Lower Austria to simulate pulse-release and its effects on aquatic organisms. The channels with a length of 40 m are supplied directly from a lake via two pipelines and flow into a nearby stream (ZEIRINGER, unpublished). Discharge and water temperature are regulated independently for each channel. Sampling was carried out monthly during the first eight months after the start-up of the facility on the 9th of September 2011. Consequently, this study analyzes primary colonization of newly created habitats under two different hydrologic regimes and subject to various restrictions and disturbances.

Disturbance is an integral part of natural stream ecosystem functioning and hence, disturbance and recolonization usually are the main factors forming stream invertebrate communities (TOWNSEND, 1989). However, anthropogenic disturbance regimes can differ from natural ones in terms of frequency and impact. Especially pulse-releases induced by intermittent hydropower generation may occur more frequently and abruptly than natural discharge fluctuations and influence a larger part of a river compared to other disturbances caused by human activities (SCHMUTZ et al., 2011). Since life cycles of benthic invertebrates are adjusted to natural discharge fluctuations such alteration of the hydrologic regime consequently has severe adverse effects on the benthic fauna. The loss of benthic invertebrate biomass through pulse-releases lied between 75 and 95 % in the permanently wetted channel area within the first few affected kilometers according to case studies in four Austrian rivers (Drau, Bregenzerach, Salzach and Enns). In the following 20 to 40 kilometers loss of biomass lied between 40 and 60 % (MOOG, 1993).

Colonization of newly created habitats or recolonization following any type of disturbance is crucial for the success of river restoration. The most elaborate revitalization measures bring only little increase of diversity if colonization is hindered (e.g. FUCHS & STATZNER, 1990; BREDERVELD et al., 2011; HUGHES, 2007). Possible constraints to stream invertebrate colonization could be barriers like dams and weirs that block colonization paths (BILTON et al, 2001; WILLIAMS & HYNES, 1976), inadequate habitat quality (WILLIAMS & SMITH, 1996) or a lack of source populations (e.g. JÄHNIG et al., 2010).

2 Objectives

This study shall verify or falsify the following hypotheses concerning different aspects of stream invertebrate colonization:

- Ecological profile of colonists
Pioneers show specific traits, which distinguish them from other species that colonize later.
- Sources of colonists
Aerial colonization is the main source for colonization of the experimental channels, since other sources are restricted.
- Time scale of colonization
It will take a longer time to achieve an equilibrium benthic community composition in the channels than in other habitats, where colonization is less restricted.
- Seasonal aspects of colonization
Primary colonization of the experimental channels is restricted by the start-up in late summer.
- Colonization and disturbance
Benthic colonization is influenced by discharge fluctuations and thus, colonization patterns differ between the two channels, since they were subject to pulse-releases of different frequency and intensity.

3 Macrozoobenthic colonization

The following chapter is based on literature research work. It gives important definitions and describes why and how benthic invertebrates colonize habitats. Furthermore, it characterizes important influences on macrozoobenthic colonization.

3.1 Definitions

Colonization, in the sense of MAYR (1964), is “the invasion of species into disturbed and newly created habitats.”

The term **recolonization**, as a particular case of colonization, refers to colonization of a habitat by “original species [...] which had been wiped out by the perturbations” (WILLIAMS & HYNES, 1976).

A **disturbance** is every factor, which is not part of the normal environment of organisms, populations or ecosystems, and induces reversible or irreversible changes of the properties of a system. It can be a unique event or a permanent condition, natural or anthropogenic (SCHAEFER, 2003).

3.2 Ecological relevance of colonization

Populations colonize new habitats because it is ecologically beneficial for them to disperse. Advantages include the avoidance of inbreeding, the chance of locating a new site with low density of individuals and therefore few resource competitors, and the possibility of escaping from unfavorable conditions (BILTON et al., 2001). This ensures genetic exchange between subpopulations within a metapopulation and enables dispersal of a species into new habitats or recolonization if a subpopulation is wiped out by disturbance (SCHAEFER, 2003).

3.3 Colonization mechanisms of stream benthos

WILLIAMS & HYNES (1976) recognized four main sources of colonizing stream invertebrates:

- Aerial colonization
- Drift
- Upstream migration within the water
- Vertical upward movement from deeper zones of the substrate

3.3.1 Aerial colonization

Aerial colonization can be divided into reproductive colonization, which means oviposition, and nonreproductive colonization. The latter includes invasion by flying or springing insects and passive dispersal through wind or hitchhiking on other animals (TRONSTAD et al., 2007).

Reproductive colonization may occur in form of MÜLLER's (1954) colonization cycle. This concept describes adult insects flying upstream before oviposition to compensate downstream drift. Eggs and larval instars are then distributed over the area of occurrence by drift. The so-called compensatory flight is known from species of Ephemeroptera, Plecoptera, Trichoptera as well as Chironomidae and Simuliidae (RUSSEV, 1972). In contrast, lateral flight of insects between streams seems to be limited (BILTON et al., 2001). However, ELLIOTT (1967) found that flight direction of mayflies was influenced by the direction of wind rather than by water flow direction.

Wind may also be responsible for the dispersal of eggs or other desiccation-resistant stages. However, the relevance of this mechanism, especially over long distance, is questionable due to the high risk of deposition in terrestrial environment. Nevertheless, small weakly flying adult insects like stoneflies, mayflies, caddisflies and some dipteres can be dispersed over longer distances by wind (BILTON et al., 2001).

Animal vectors, water birds for instance, can either carry benthic organisms or their eggs outside on their body, or inside their guts, followed by defecation. For example, there is evidence for transport of bivalves, gastropods, leeches, water mites, crustaceans and bryozoans by animal vectors (BILTON et al., 2001).

3.3.2 Drift

Drift is estimated to displace a maximum of 2 % of benthic organisms in streams (HYNES, 1970). According to WATERS (1972) three types of macrozoobenthic drift can be distinguished: catastrophic drift, behavioral drift and constant drift. Catastrophic drift, also called passive drift, is caused by an involuntary loss of contact to the substrate induced by disturbances such as spate, pollution, change in water temperature or substrate disturbance (e.g. through construction works). In contrast, behavioral drift or active drift occurs voluntarily in order to escape competition or predators or to locate food or mates. Constant drift is also involuntary but occurs constantly since it is not induced by extraordinary events. Both behavioral and constant drift are important for genetic exchange within a metapopulation.

It was observed that many stream insects show a maximum drift rate at night, which is assumed to be the result of their response to light. During daylight they tend to seek shelter from predators and when night falls they leave their shelter and are therefore easier dislodged by the current. This diurnal drift rhythm was found to persist even during spates with higher total drift rates (HYNES, 1970).

3.3.3 Upstream migration

Upstream movements of benthic invertebrates on the streambed also seem to be of relevance for the colonization of stream habitats, at least at a small spatial scale (WILLIAMS & HYNES, 1976). For several taxa, upstream is the only possible direction of voluntary movement in lotic aquatic habitats due to their body shape, a phenomenon known as positive rheotaxis (SCHAEFER, 2003). Nevertheless, upstream crawling can probably not compensate for downstream drift (WATERS, 1972).

3.3.4 Recolonization from the substrate

During spates, benthic invertebrates are known to move into deeper zones of the interstitial to avoid going into drift and recolonize the streambed afterwards (WILLIAMS & HYNES, 1976). However, human-induced disturbances like pulse-release (MOOG, 1993) or mechanical dislocation of the sediment (MATTHAEI et al., 1997) occur more suddenly which may impede escape into the interstitial.

In several cases stream invertebrates are also able to survive drought within the substrate, either in an active or dormant state. For instance, there are Chironomid larvae that form protective cocoons and Oligochaetes which produce desiccation-resistant eggs

(TRONSTAD et al., 2005). Moreover, the eggs of several insects like Ephemeroptera (BAUERNFEIND & HUMPESCH, 2001), Plecoptera (NILSSON (ed.), 1996) and Trichoptera (WARINGER & GRAF, 1997) are embedded in a matrix, which prevents them from drying out for some time. Thus, dry substrates can be recolonized without immigration if they are flooded again (TRONSTAD et al., 2005). However, benthic recolonization after desiccation depends on biological as well as on climatic factors. Firstly, it is a good precondition for quick recolonization if a drought event coincides with the emergence of the animals. Moreover, recolonization depends on the periodicity of desiccation and another important factor is the stability of the stream bed (SAGAR, 1983). Finally, the abundance of drought refuges such as perennial reaches or ponds and moist sediment influences the recolonizing benthic community (CHESTER & ROBSON, 2011).

3.4 Factors that influence colonization

3.4.1 Dispersal ability

Dispersal is crucial for recolonization of disturbed patches or colonization of new habitats, since it provides potential colonists on a regular basis. Thus, species that are able to disperse also over longer distances have better chances to colonize new habitats.

For active dispersers, part of dispersal ability is supposed to be determined by the length of their wings (BILTON et al., 2001). Indeed, rare species of stoneflies and mayflies were discovered to have shorter wings than widespread ones, although wing length cannot be considered the only important parameter for dispersal ability (MALMQVIST, 2000). Since the development of a strong flight musculature and flight itself are energetically expensive, as is the production of eggs, good flight performance is not always the most important qualification for a species. Thus, there is a trade-off between reproduction and dispersal ability. In some taxa this results in the so-called oogenesis-flight syndrome, which describes active dispersal in early adult life and subsequent autolysis of flight musculature to regain energy for reproduction (BILTON et al., 2001).

Differences in active dispersal ability have also been detected within species. Even within populations there is dimorphism in wing length (BILTON et al., 2001). An example is sexual dimorphism in many species of Trichoptera. In most species, females seem to be the dispersing sex considering their morphology. However, there are also species where males are morphologically better adapted to dispersal as well as species without sexual dimorphism (MÜLLER-PEDDINGHAUS, 2011).

Another observed adaptation for dispersal is the production of large numbers of dispersible small stages (r-strategy), either eggs or larvae to be dispersed by drift or adults to disperse actively or be transported by wind. Furthermore, passive dispersal stages increase their survival rate by resistance to shorter or longer exposure to drought, fluctuations in temperature and other terrestrial or aerial conditions (BILTON et al., 2001).

3.4.2 Habitat requirements

Stream invertebrates have different preferences of habitat features like substrate particle size, water temperature, current velocity, radiation and nutrient concentration (WILLIAMS & SMITH, 1996). Different functional feeding groups (e.g. shredders, grazers, detritivores, active or passive filter-feeders, predators) prefer different habitats. Grazers, for instance, need epilithic algae, shredders depend on allochthonous food resources like leaf litter or wood debris and passive filter-feeders cannot feed without current (JUNGWIRTH et al., 2003). Generalists, which tolerate a wide range of features, have better chances to colonize new habitats and therefore tend to be typical pioneers (TOWNSEND & HILDREW, 1994).

A recent study on caddisflies (MÜLLER-PEDDINGHAUS, 2011) showed that dispersal ability is probably linked to habitat preferences. Species known as ecological generalists display a dispersal promoting morphology whereas specialists were found to be less adapted to dispersal. The reason for this relation may be a higher chance for generalists to locate an adequate habitat by dispersal. Therefore the first colonizers of a new habitat tend to be generalists, specialists arrive later (MÜLLER-PEDDINGHAUS, 2011). BREDERVELD et al. (2011) argue a similar relation. Hence, they suggest dispersal ability and habitat specialization to be limiting factors for colonization.

The influence of a particular habitat feature on colonization is different for each species and therefore not easy to investigate. For instance, RAE (2004) found a positive correlation between mean sediment particle size and species richness of colonizing Chironomids. In contrast, WILLIAMS & SMITH (1996) discovered that the amount of detritus, which many stream invertebrates feed on, accumulating in the substrate, had a more significant increasing effect on species diversity than particle size. However, the accumulation of sand had a significant negative effect on diversity. Thus, RAE (2004) concluded that particle size and consequently porosity of the sediment influenced

retention of detritus and sand which in turn were important criteria for invertebrates to colonize.

3.4.3 Life cycles

Benthic colonization depends on the flight periods of aquatic insects, which are a result of their life cycles (TRONSTAD et al., 2007). Several benthic invertebrates are polyvoltine, which means that they have more than one generation per year. Some taxa of Diptera and Ephemeroptera (*Baetis quilleri*, *Callibaetis montanus*, *Leptohyphes packeri* and *Tricorythodes dimorphus*) were observed to develop in less than two weeks from egg to adult stages (GRAY, 1981). Other animals just bear one or two generations per year or even need a few years to develop one generation, which restricts their emergence and consequently their ability for aerial colonization to certain periods. Differences in the number of generations per year also exist within species, depending on thermal regimes and hence on the latitude and altitude of occurrence. A mayfly species common to both Canada and Florida was found to emerge throughout the year in Florida, but only in late spring and summer in Canada. Short life cycles and polyvoltinism support resilience of stream communities. Thus, invertebrate biomass can be expected to recover faster in warm climate zones, even after severe disturbance (WALLACE, 1990).

Compared to temperatures on land, water temperatures in streams are very stable throughout the year. In the hyporheic zone, temperatures over 1°C can be measured in the winter, even if there are tens of centimeters of surface ice (SCHWOERBEL, 1967). Several species use this lack of frost combined with continuous food supply through autumn leaf-fall to be active and grow fast during the winter. Moreover, in the winter they are less endangered to be predated by fish, which are less active in cold times. Thus, the biggest biomass of stream benthos is often present in spring. On the other hand, there are many stream animals which outlast the winter in a non-growing state. They start their period of growth in the early summer (HYNES, 1970). The resulting peaks of adult insect emergence in spring and autumn are the seasons when aerial colonization rates are high. In other times of the year, colonization by drift may become more important (WALLACE, 1990).

3.4.4 Spatial aspects

As observation of dispersal distances is an elaborate matter, there are only few studies providing such data (BILTON et al., 2001). HERSHEY et al. (1993) found adult *Baetis* sp.

mayflies flying about 1 km upstream. McNEALE et al. (2005) observed adult stoneflies flying up to 700 m upstream and also dispersing laterally over a distance of up to 500 m. Due to very limited lateral dispersal, especially of weakly flying insects, distance is an important factor of isolation between streams rather than within streams. However, there is the possibility for a species to invade a new stream by drifting downstream of the confluence of two streams and an upstream flight into the other stream (BILTON et al., 2001).

BUNN & HUGHES (1997) investigated dispersal in stream insects by studying the genetic structure of populations in Australian streams, assuming that low genetic differentiation indicated high dispersal activity. They came to the surprising result that genetic difference generally was higher within reaches and streams than between streams of the same subcatchment and between different subcatchments. However, they supposed these relations to be different in temperate climate zones, since insects there are not emerging year-round but synchronously only for a certain period of the year. This synchronized emergence promotes intense dispersal into the surrounding habitats, decreasing with distance (BUNN & HUGHES, 1997; BILTON et al., 2001).

These findings are to be regarded with caution because dispersal limitation by distance differs between taxa. In the study of BUNN & HUGHES (1997), mayflies of the genus *Baetis*, for instance, showed low genetic differentiation among adjacent sites and high differentiation at wider spatial scales. *Baetis* spp. are known as good swimmers (MACKAY, 1992), thus it was concluded that they are more likely to disperse by swimming between close sites or drift within limited distance of a stream than by flying between different reaches or streams. For the caddisfly genus *Tasiagma*, which is believed to disperse by adult flight and rather not to change sites by crawling or swimming within the stream, it was the other way around (BUNN & HUGHES, 1997).

HUGHES (2007) found that stream animals with aquatic stages only manifested higher genetic differentiation among different catchments than animals with a flying adult stage. Hence, she concluded that dispersal distance was strongly related to the ability of flight. Different catchments seemed to be the strongest limitation to dispersal, irrespective of the altitude of the catchment. Within catchments, however, altitude had a significant effect on genetic differentiation, being higher in the upland. This is reinforced by JÄHNIG et al. (2010) and BREDERVELD et al. (2011), who found dispersal over far distances more common in lowland rivers than in the upland.

3.4.5 Different approaches in recolonization studies

There are different approaches for studying benthic recolonization. One can take samples before and after a natural disturbance. The disturbance can also be self-made by stirring up the sediment, which provides better knowledge about its extent and intensity. On the other hand, artificial disturbance tends to be small-scale compared to a natural one. Recovery can occur much faster though in small, almost completely depopulated patches than after a large-scale spate or drought (see MATTHAEI et al., 1996, 1997 and ELSER, 1999).

Another typical method to study colonization is the placement of artificial or semi-natural substrates such as Hester-Dendy samplers (MEIER et al., 1979), bricks or single rocks (CASEY & KENDALL, 1997) or baskets filled with gravel (ELSER, 1999; OFENBÖCK & MOOG, 2000) in the stream bed. MACHADO DE CARVALHO & SANCHES UIEDA (2004) compared invertebrate colonization of an artificial and a near-natural substrate and found no significant difference between colonization patterns. However, the usefulness of artificial substrates for investigating recolonization was doubted by WILLIAMS & SMITH (1996) and ELSER (1999), who stated that the effect on a stream was not comparable to a spate and hence, it was not considered to be an adequate method to study recolonization.

3.4.6 Presence of intact source populations

Recolonization of restored stream or river reaches by benthic invertebrates does not only depend on the quality of the restored habitats, but also on the proximity of possible sources of colonizers (FUCHS & STATZNER, 1990; BREDERVELD et al., 2011; JÄHNIG et al., 2010). A study that compared recovery time of two different restored stream reaches in Germany suggests that the degree of isolation from natural or near-natural stream ecosystems has a great influence on recolonization success. Upstream and downstream of the first experimental reach the benthic invertebrate fauna was intact and populations in this reach recovered in a bit more than one year. The other experimental reach, however, was extremely isolated. Intact benthic communities could only be found 20-25 km away in different streams. Hence, this reach was colonized mainly by ubiquitous species not typical for that type of stream. In addition, two oil spills about five years after restoration reduced or even wiped out some of the species (FUCHS & STATZNER, 1990).

These findings are supported by HUGHES (2007), stating that a species that is extinct in a whole river may not recolonize just by habitat restoration. Therefore it seems reasonable to regard a greater spatial scale in the course of restoration projects to ensure abundance of possible recolonizers and sufficient connectivity between sources and restored reaches (BREDERVELD et al., 2011). JÄHNIG et al. (2010) came to the same conclusion comparing hydromorphology and benthic invertebrate communities in 26 central and southern European rivers. While diversity of morphology was significantly higher in the restored sections than in the non-restored control sections, there was statistically no difference in diversity of benthic communities. Thus, they stated that restoration of small reaches was successful in improving habitat diversity of rivers, but increasing biodiversity needed intact source populations and solutions to large-scale problems such as bad water quality or fine sediment input. A similar study (SUNDERMANN et al., 2011) that compared 25 restored river sites in Germany also found no significant difference in density and diversity of benthic communities between restored and unrestored sites. Minor differences in the numbers of individuals and taxa were not related to changes in habitat quality. Only 3 of the 25 sites were stated to have good ecological quality according to the criteria of the European Water Framework Directive. Thus, SUNDERMANN et al. (2011) also support the conception that recolonization is hindered by a lack of near-natural species pools and by poor habitat quality through land use practice.

Sources of recolonizing invertebrates do not necessarily have to be completely intact reaches upstream of the restored channel. KORSU (2004) showed the role of aquatic mosses as refugia for invertebrates and therefore recommended to keep a good part of them undisturbed during reconstruction works. The importance of aquatic and riparian vegetation for the recolonization process is also emphasized by BREDERVELD et al. (2011). This supports the suggestion of JÄHNIG et al. (2010), that cost-intensive constructing of optimal habitats in small reaches, with side effects such as degradation of the riparian zone, may be less effective than large-scale passive restoration.

The catastrophic character of in-stream restoration works is pointed out by MUOTKA et al. (2002). Compared to natural disturbance events such as spates, channel reconstructions alter the stream beds very profoundly, and since channel reconstruction is an evolutionary young event, there was no possibility for biocoenosis to adapt. Thus, recovery times for benthic communities can be expected to be longer after restoration works than after spates. MUOTKA et al. (2002) found that benthic communities in

restored streams stabilized after eight years, although densities seemed to recover within two years. Similar to KORSU (2004), they recommended protecting mosses as far as possible and adding large deadwood elements to increase retention potential in the course of restoration operations.

Concerning timing of restoration works, KORSU (2004) suggests that operations in the stream bed have more harmful effects on benthic communities in winter, because then invertebrate densities are high and drift rates as well as aerial colonization are low.

FELD et al. (2011) investigated the success of restoration measures in different European countries and therefore divided the literature into three groups of measures: restoration of buffer strips, in-stream habitat improvement and removal of weirs and dams. Buffer strips retain fine sediments and pollutants, keep water temperature more stable and supply food and large wood debris as a habitat. However, riparian vegetation like black alder or willow takes 30 to 40 years to mature, which means that buffer strips need about that time to become effective. Moreover, the habitat improvement by small-scale buffer strip restoration may remain ineffective due to large-scale impacts in the watershed or the lack of intact source populations. As mentioned above, the same applies for in-stream habitat enhancement. FELD et al (2011) point out the importance of large wood debris for the short-term recovery of macroinvertebrate communities.

4 Characterization of the study area

The study was conducted in Lunz am See, Lower Austria, in the experimental facility HyTEC (Hydromorphological and Temperature Experimental Channel). The facility was constructed in the summer of 2011 for research on reactions of aquatic organisms to pulse-releases, which is a common measure in the Alps to generate additional electricity in times of high consumption (SCHMUTZ et al., 2011).

The village Lunz am See (Figure 1) is located at an altitude of 599 to 608 m above the Adriatic Sea level. The area belongs to the bioregion of the Northern Limestone Alps (BUNDESMINISTERIUM FÜR LAND- UND FORSTWIRTSCHAFT, UMWELT UND WASSERWIRTSCHAFT, s.a.). The stream closest to the research facility is the Unterer Lunzer Seebach (below: Seebach), outflow of the lake Lunzer See. It passes by in a distance of about 12 m. Its Strahler number is 3 and it drains a catchment area of 25.5 km² (WIMMER & MOOG, 1994). Its natural basic saprobic condition is 1.5 (OFENBÖCK et al., 2010) and the associated fish region is the Metarhithral (BUNDESMINISTERIUM FÜR LAND- UND FORSTWIRTSCHAFT, UMWELT UND WASSERWIRTSCHAFT, s.a.). The Seebach flows into the river Ybbs, which is also fed by another stream called Bodingbach from the opposite side of the village. Moreover, there are some small brooks draining into the river Ybbs and into the lake Lunzer See (BUNDESAMT FÜR EICH- UND VERMESSUNGSWESEN, s.a.).

4.1 The experimental channels

The main part of the facility is formed by two man-made channels (Figure 2, Figure 3, Figure 4 and Figure 5), each 40 m long, 6 m wide and 1.4 m deep. The channels were filled with dry gravel extracted from a bank above the water level of the river Ybbs between the 6th and the 9th of September 2011. In order to avoid loss of water into the ground, the bottom of the channels is sealed with a bentonite blinding. A wooden piling wall bordering one side of each channel (Figure 6) guarantees for wide bank geometry at the other side in spite of scarce space and also during low discharge, especially in the winter (ZEIRINGER, unpublished).

Moreover, between the 31st of October and the 3rd of November 2011 base slope was changed from 1 % to 0.5 %. A 10-20 cm layer of new, smaller gravel was added to alter the shape of the beds. Gravel was withdrawn from the river Ybbs and stored for two

days before positioning into the channels. Furthermore, nets were mounted over the channels to protect fish from being predated by birds. All these activities were attended by people walking on the gravel bed and disturbing the sediment.

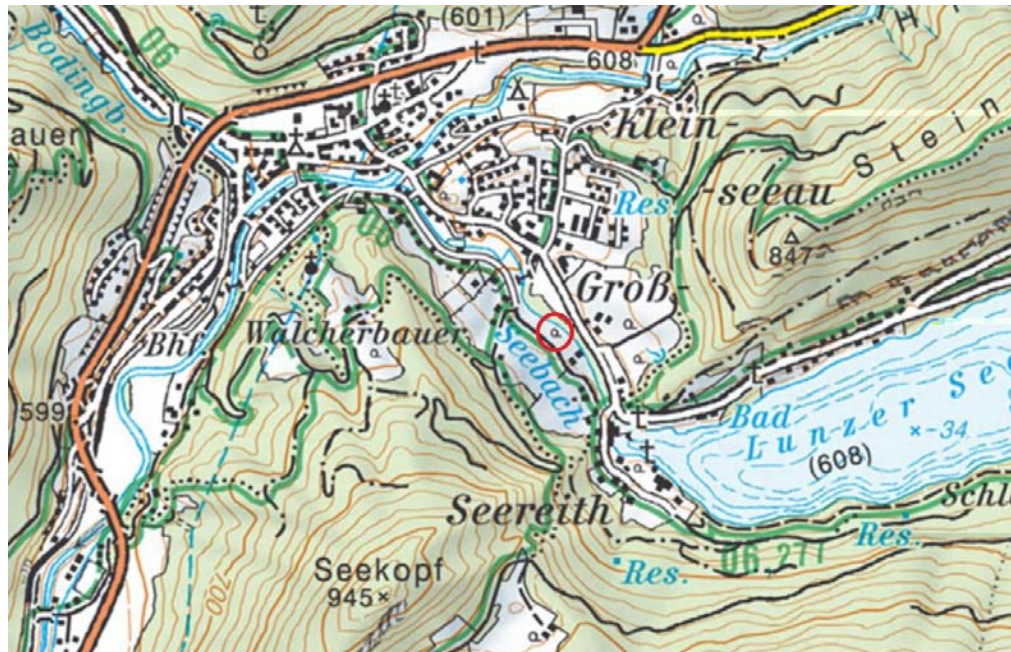


Figure 1: The study area (marked with a red circle) and the surrounding 1 000 m (BUNDESAMT FÜR EICH- UND VERMESSUNGSWESEN, s.a.)



Figure 2: Overview of HyTEC, 29th February 2012.

Left: the Unterer Lunzer Seebach and right: the experimental channels. Direction of the photo: upstream.

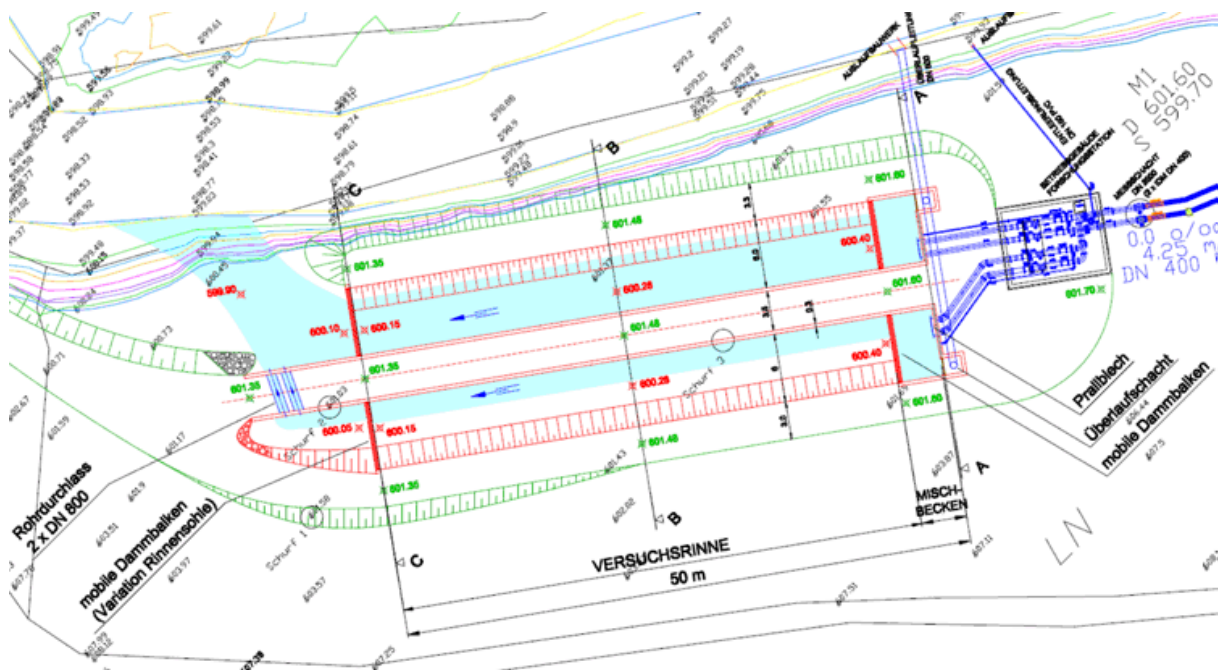


Figure 3: Horizontal plan HyTEC
(SCHMUTZ et al., 2011)

Water supply for the experimental channels is obtained directly from the lake Lunzer See, which is located about 400 m upstream and the difference in altitude between the water level of the lake and the water level at the inlet of the channels is at least 6 m. Two independent pipelines supply the channels with water from the lake, each with a maximum capacity of 300 l/s. One of them withdraws water from the surface (“warm water line”). The intake of the other one is in a depth of 10 m, close to the ground (“cold water line”). Water intakes are equipped with screen baskets of 10 mm width between the bars. Both pipelines divide into two ways so that two mixing ponds (Figure 7) are supplied with cold and warm water independently. Each mixing pond is the start point of one experimental channel. Thus, the two experimental channels can be operated with different discharge- and temperature- regimes respectively at the same time. Wooden dam and ramp constructions at the lower end of the channels (Figure 8) enable controlled discharge of water into the Seebach via a 10 m long outflow passage. These constructions can be equipped with nets for catching fish that are flushed down (ZEIRINGER, unpublished).



Figure 4: The orographically right channel – upstream perspective.



Figure 5: The orographically left channel – upstream perspective.



Figure 6: Gravel streambed with wide bank and piling wall.



Figure 7: Mixing pond and start of the channel.



Figure 8: Lower end of a channel with dam / ramp and driftnet for fish.

4.2 Operation of the experimental channels

Flow through the channels was started on the 9th of September 2011 with a base flow of 35 l/s. About 10 times, discharge in each channel was raised to a maximum of 500 l/s for a few seconds until the datalog-system was launched on the 13th of October. Since the datalog-system was out of order between the 1st and the 26th of January 2012, data from the concerning period were derived from the experimental protocols. Discharge graphs of the channels from the 13th of October 2011 until the 10th of April 2012 are shown in Figure 9 and Figure 10 together with the sampling dates. These figures show

that one of the channels, the orographically left one, was hydrologically less disturbed than the right one.

Apart from discharge fluctuations the channels were subject to various disturbances throughout the study period. Juvenile graylings (*Thymallus thymallus*) were introduced for experiments 21 times in the right channel and 20 times in the left channel (300 specimens each) and removed again by electro-fishing (18 times in the right and 17 times in the left channel). Twice, in December and in March, electro-fishing was conducted only two hours before sampling. The timeline in Figure 11 shows gravel addition and fish experiments together with the sampling dates.

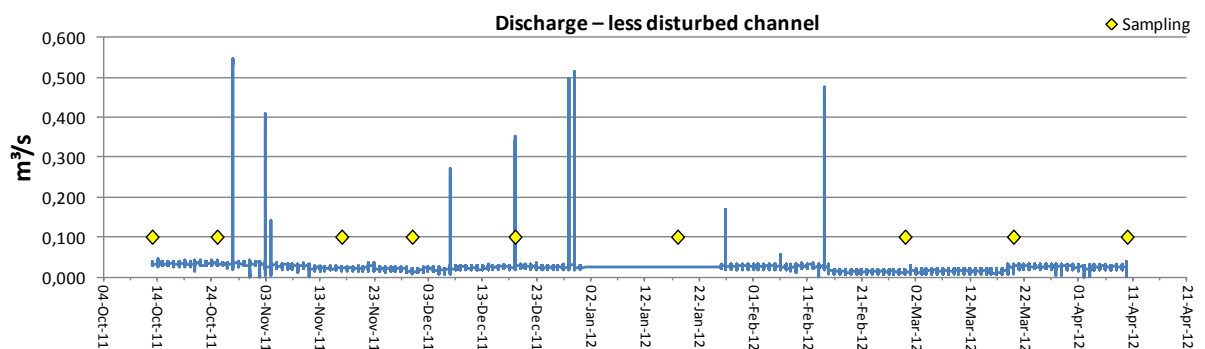


Figure 9: Discharge of the orographically left channel
– below: “less disturbed channel” – from the 13th of October until the 10th of April (ZEIRINGER 2012, unpublished data).

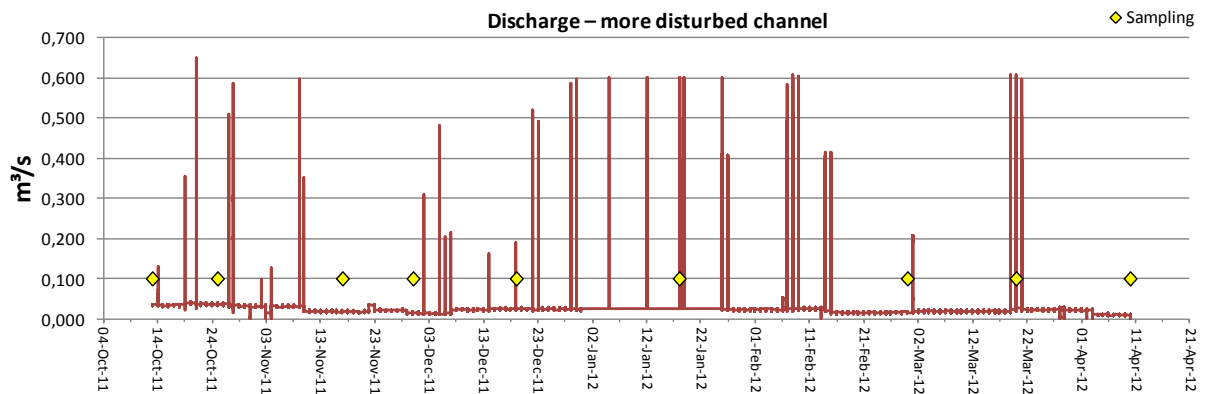


Figure 10: Discharge of the orographically right channel
– below „more disturbed channel“ – from the 13th of October until the 10th of April (ZEIRINGER 2012, unpublished data).

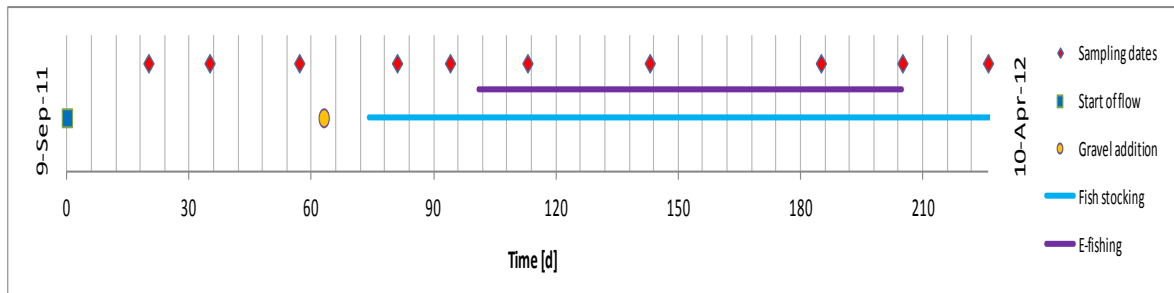


Figure 11: Timeline of the study period with disturbances and sampling dates.

Water temperature in the channels between the 13th of October 2011 and the 10th of April 2012 is depicted in Figure 12. In the winter, water temperature was more stable than in autumn and spring. Since the online-measurement was out of order for a part of January, there is a gap in the graphs. Measured water temperature on the 18th January was at 2.5°C in both channels.

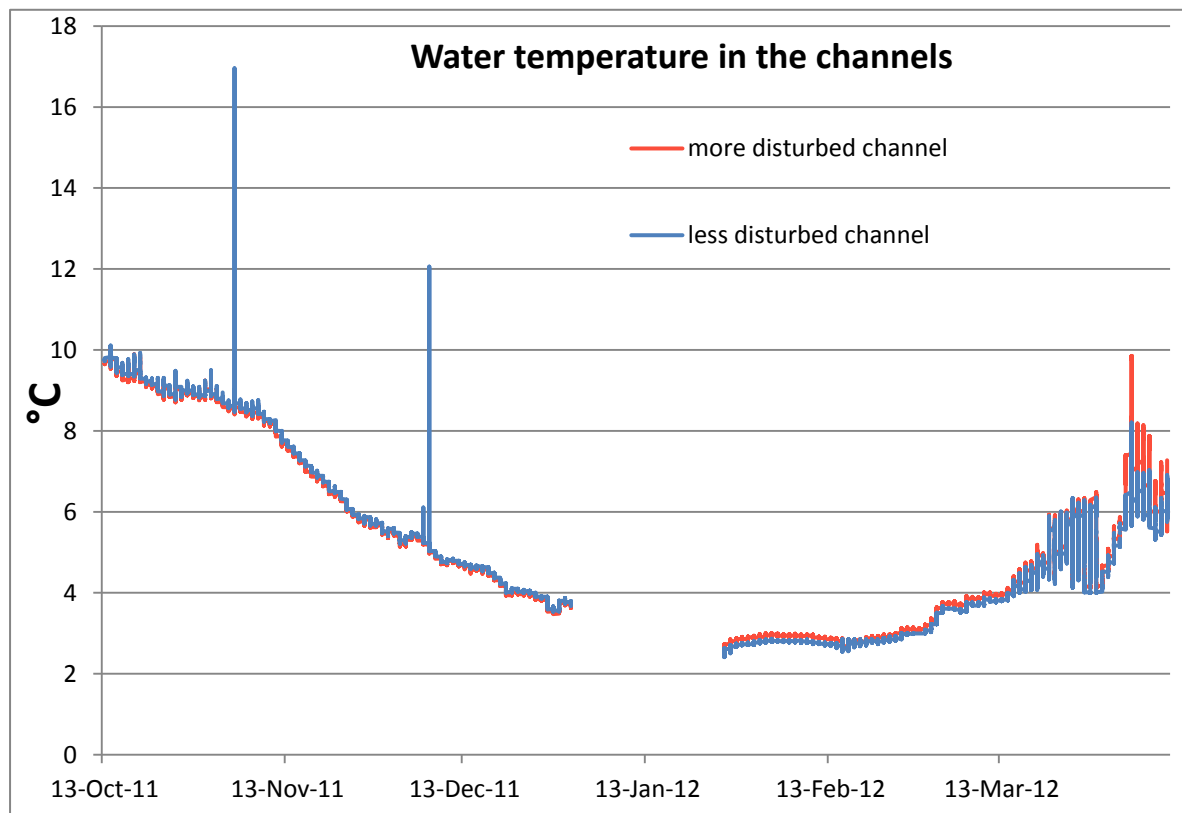


Figure 12: Water temperature in the experimental channels.
(from the 13th October until the 10th April.)

Considering this general set-up, the possible paths of colonizing animals are:

- Aerial colonization from nearby sources (e.g. the Seebach)
- Drift from the lake through the pipelines
- Upstream migration from the Seebach by overcoming the wooden dam and ramp construction
- Survival of eggs or animals within the gravel brought in during construction and again during reshaping works between the 31st of October and 3rd of November 2011
- Passive transport attached to birds or other animals

5 Material and methods

5.1 Sampling

Samples were taken between the 28th of September 2011 and the 10th of April 2012, as shown in Figure 13 and Table 1. On each sampling occasion, six box-samples were taken in each of the two channels. The first sampling spot was located about 4 m upstream of the lower end of the channels, the second about 8 m upstream. Samples three and four were taken 16 and 20 m upstream of the lower end and samples five and six about 28 and 32 m upstream. All box-samples were taken within an imaginary ribbon of 1 m width in the middle of the wetted bottom of the channel.

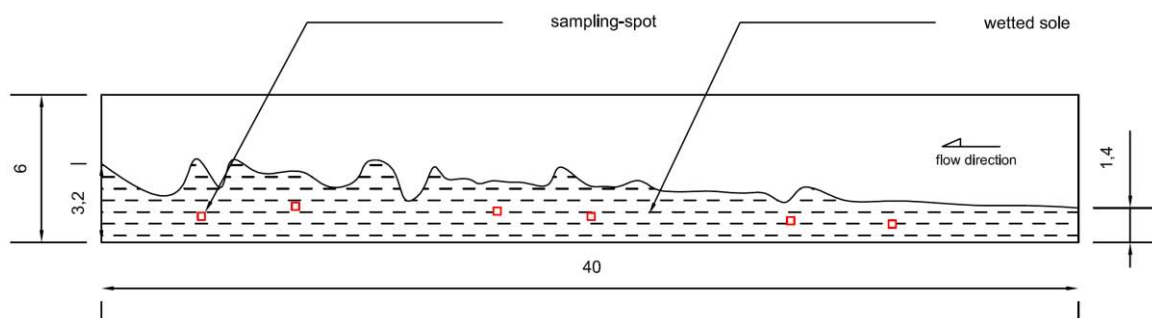


Figure 13: Sampling design in the channels
Horizontal plan – all dimensions in m.

Additionally, on the first, the seventh and the ninth sampling date, six box-samples and one Multi-Habitat-Sample (MHS) containing 20 single units were taken in the reach of the Unterer Lunzer Seebach paralleling the study channels. However, only three of the box-samples of each date and one MHS respectively were examined in the laboratory. This should provide rough information about potential recolonizers and be a reference for biomass and abundance in a “natural” stream. In order to avoid introducing animals from the Seebach into the samples from the experimental channels by accident, channel samples were always taken prior to the Seebach samples.

Table 1: Sampling design.

If not all samples were sorted in the laboratory the numbers in brackets refer to the total amount of samples, the numbers without brackets to the samples actually examined.

Sampling date	Less disturbed channel		More disturbed channel		Seebach	
	Box	MHS	Box	MHS	Box	MHS
28-Sep-11	6		6		3 (6)	1
13-Oct-11	6		6			
25-Oct-11	6		6			
17-Nov-11	6		6			
30-Nov-11	6		6			
19-Dec-11	6		6			
18-Jan-12	6		6		3 (6)	(1)
29-Feb-12	6		6			
20-Mar-12	6		6		3 (6)	(1)
10-Apr-12	6		6			

Before disturbing the sediment, each sampling point was documented photographically and the following habitat features of each sampling location were recorded in a sampling protocol.

- The dominant substrate particle sizes were estimated on the field. The documented size classes were Akal (0.2 – 2 cm), Microlithal (2 – 6.3 cm), Mesolithal (6.3 – 20 cm) and Makrolithal (20 – 40 cm) (see MOOG et al., 1999).
- Percentages of coverage by mosses, filamentous algae, micro algae and bare stones were estimated on the field. *Hydrurus* coverage was estimated on the basis of the photos taken from the sampling spots (see Appendix).
- Flow velocity was estimated roughly by timing a leaf or small piece of wood passing the length of the box-sampler.
- Water depth was measured in the center of the box with a plastic folding rule.
- Electric conductivity and water temperature of each channel were measured with a WTW MultiLine® P4 Universal Meter.

Box-sampling was conducted using a box made of mesh at three sides, open at the bottom and top and a 1 m long net with 300 µm mesh width attached to the fourth side (Figure 14). The base area of the box was 0.1 m². After locating a sampling spot, the box was set onto the streambed with the net in downstream direction, making sure every edge was as close as possible to the ground. Then the sediment was sampled by different means, depending on the type of habitat. Large and medium-sized stones were lifted and brushed off at all sides. Mosses were scratched off the stones and put into the

sample as a whole. Smaller stones and fine sediment were disturbed by stirring with a large screwdriver up to a depth of about 15 cm. If the current was not strong enough to move all fine particles and animals into the net, it was compensated by hand paddling movements. In order to avoid resampling disturbed sediment, sampling locations were documented and the next sample was taken from a different spot. The content of the net was carefully washed into a flat white plastic dish (Figure 15) and then filled into labeled wide-neck plastic bottles and preserved in 4 % Formalin. Before examination in the laboratory, the samples were stored for a minimum of two weeks.



Figure 14: Box-sampler.



Figure 15: Content of a box-sample, 20th March 2012.

Multi-Habitat-Sampling was carried out according to OFENBÖCK et al. (2010). The proportions of area covered by different habitat types were estimated and 20 sampling spots were located so that they reflected habitat diversity in the reach, using a documentation scheme by MOOG (2004) (see Appendix). Sampling started at the downstream end of the reach. A standardized 1,5 m long hand net with 25 cm edge length of the inflow opening and 500 μ m mesh width was set to the ground vertically below each sampling spot. An area of approximately 25 x 25 cm directly in front of the net was then sampled equal to the Box-sampling procedure, except for stirring with the screwdriver, which was replaced by foot kicking. Since there were hardly any large wood debris and leaf accumulations in the reach and none of them in the experimental channels, these habitat types were not sampled.

5.2 Examination of the samples

Each sample of the experimental channels was screened through two steel sieves. The first one with 1 mm mesh width was set upon the second one with 500 µm mesh width to obtain two different fractions. For the samples from the Unterer Lunzer Seebach five sieves were used, ranging from 500 µm to 10 mm mesh width. The content of each sieve was then washed into a flat white plastic dish to be searched for animals. The smallest fraction was poured portion by portion into a petri dish and sorted under a binocular. Plant fragments and coarse wood debris were washed in water and searched for invertebrates before removing them from the samples.

The Multi-Habitat-Sample was sub-sampled as suggested by OFENBÖCK et al. (2010) to avoid excessive time consumption for sorting. For this purpose, the sample was put into a rectangular 500 µm steel sieve subdivided into 30 squares of 6x6 cm. Large particles were removed prior to this, washed and disposed of, with the washed-off material mixed into the sample again. The sieve was then set into a suitable rectangular basin and filled with water to enable homogenizing the sample. After this, the sieve was lifted out to empty the basin and replaced again. Five of the 30 squares were drawn by lot and the material on those five squares, representing one sixth of the whole sample, was sorted as described above. Finally, the number of specimens of each taxonomic unit was multiplied by six to obtain a number for the whole sample.

All invertebrates were identified to the Austrian-Screening-Level (OFENBÖCK et al., 2010) with the binocular, using MOOG et al. (2010), BAUERNFEIND & HUMPESCH (2001), GLÖER & MEIER-BROOK (2003) and WARINGER & GRAF (1997). Additionally, Chironomids of the September- and April-box-samples were identified by Dr. Berthold Janecek.

Specimens of each taxonomic unit were dried off roughly before being weighed with a high-precision scale and finally preserved in 70 % Ethanol. Minimum measurable mass of the scale is 0.1 mg. If the mass was lower than that, the value of 0.05 mg was recorded in the protocol. Abundance and biomass of each taxonomic unit in each sample were noted in an Excel-table. Invertebrates that obviously had dried out before sampling were not taken into account for abundance and biomass since it was assumed that they had been introduced with the gravel and not colonized the channels. Specimens of the Multi-Habitat-Sample were not scaled, since there is no exact reference surface in the MHS method.

5.3 Data analysis

For all calculations and the composition of diagrams Microsoft® Office Excel 2007 was used. Box-plots and scatter-plots as well as correlation tables were created with IBM SPSS Statistics 15.0 and 17.0.

Community composition was compared using the SØRENSEN (1948) - Index and the RENKONEN (1938) - Index. The SØRENSEN-Index takes into account presence or absence of taxonomic units, independent of the number of specimens. It is calculated by the following formula.

$$S_A = \frac{2 * b}{c + d} * 100$$

Formula 1: SØRENSEN-Index.

b = Number of taxa occurring in both coenoses

c = Number of taxa occurring in the first coenosis

d = Number of taxa occurring in the second coenosis

In contrast, the RENKONEN-Index takes into account the relative abundance of the taxonomic units to assess similarity of dominance. It is obtained by summing up the minima of relative abundances of each taxon.

$$R_e(\%) = \sum_{i=1}^G \min D_{A,B}$$

Formula 2: RENKONEN-Index.

i = Taxon i

G = Number of taxa found in both communities

D_{A,B} = Relative abundance of taxon i in community A or B

6 Results

In this chapter the results from the box-samples are described and visualized graphically. Abundance and biomass is always given per m^2 (100 specimens per m^2 stand for 10 specimens per sample). Complete unprocessed data of box-samples and MHS are provided in the Appendix.

6.1 Total abundance

Total abundance of invertebrates in the experimental channels was increasing throughout the study period, as shown in Figure 16. After a phase of single findings from September until November, total abundance was higher in the less disturbed channel until February. However, in March and April it was the other way around. Figure 17 shows the wide statistical dispersion of abundance in the channels. Despite rapid growth of abundance in the channels between February and March density was about six times less than in the Seebach in the samples of March, as displayed by Figure 18.

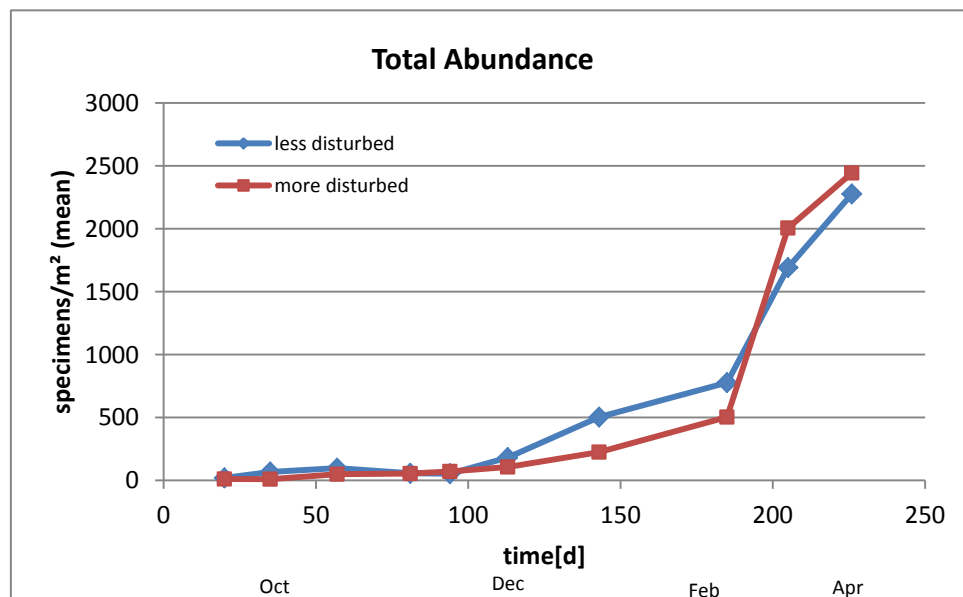


Figure 16: Total abundance in the channels.

The horizontal axis refers to the time after first flow in the channels on the 9th of September 2011.

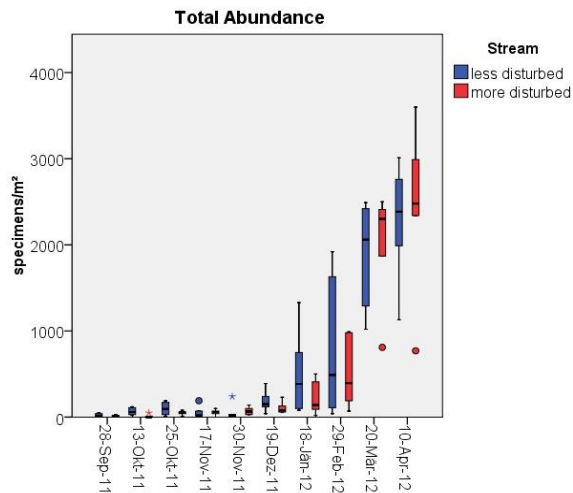


Figure 17: Total abundance without Seebach.

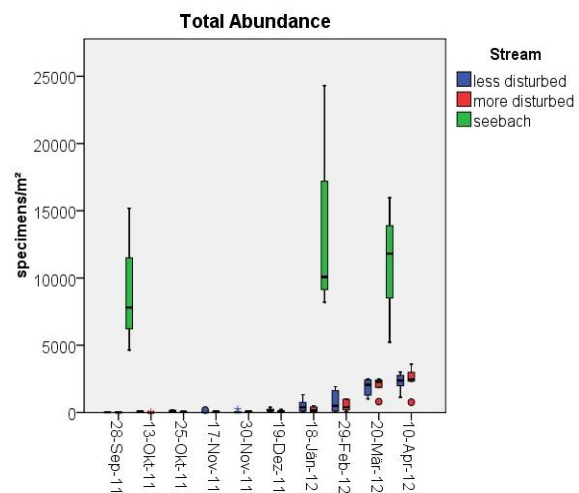


Figure 18: Total abundance with Seebach.

6.2 Total biomass

Total biomass developed similar to total abundance until February. Between the 29th of February and the 20th of March it increased by the factor five but then declined until the 10th of April 2012 (see Figure 19). Statistical dispersion of total biomass, plotted in Figure 20, was very high. Total biomass was higher in all the reference samples from the Seebach than in the channels. In March, total biomass in the Seebach exceeded that in the channels by the factor three, as depicted by Figure 21.

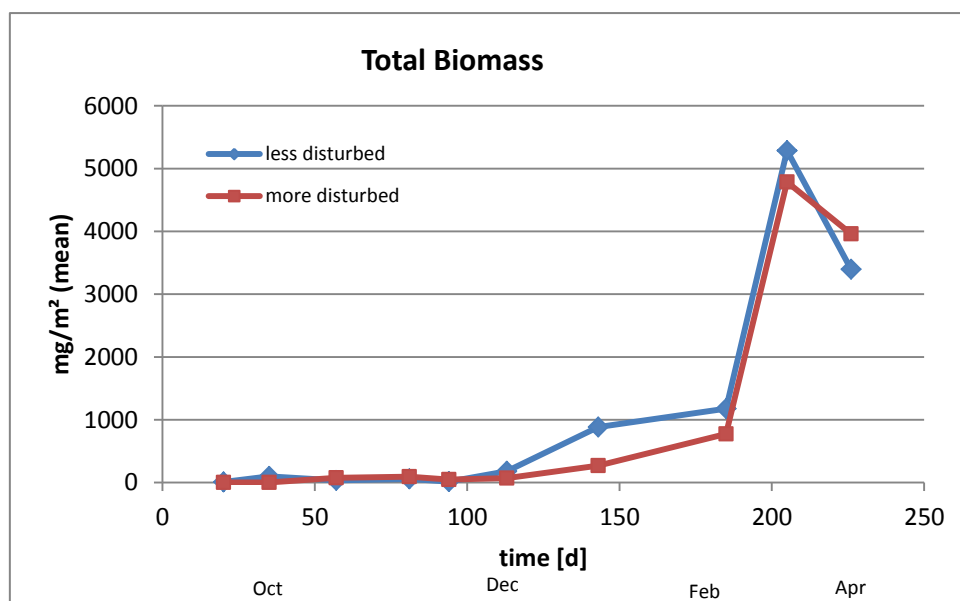


Figure 19: Total biomass in the channels.

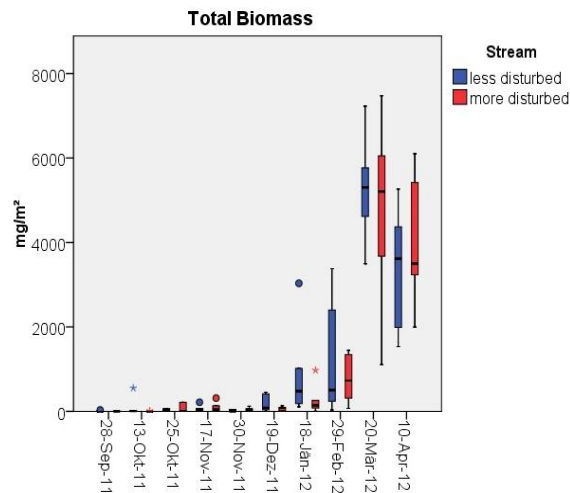


Figure 20: Total biomass without Seebach.

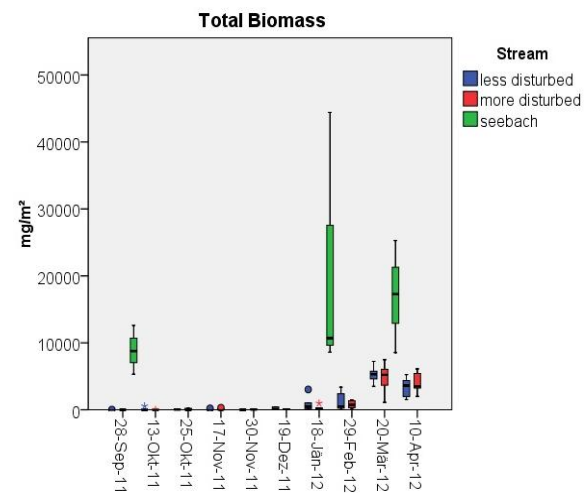


Figure 21: Total biomass with Seebach.

6.3 Individual biomass

Individual biomass is obtained through dividing biomass by abundance. Between September and November total individual biomass was fluctuating as a result of low abundance in the samples and a big difference in weight of invertebrates. For instance, the Oligochaete *Eiseniella tetraedra* or different species of Gastropoda, which were found only sporadically, are heavier than all the insects that had been appearing in the channels. From December, when abundance started to increase, until March there was growth in individual biomass before it then declined between March and April, as represented by Figure 22.

Individual biomass excluding Gastropoda and *Eiseniella tetraedra* is depicted in Figure 23. This reveals an outlier of individual biomass on the 17th of November, the first sampling date after the introduction of new gravel. From December onwards, the two graphs do not differ significantly, since total abundance was higher and Gastropods and *E. tetraedra* were hardly occurring in the samples.

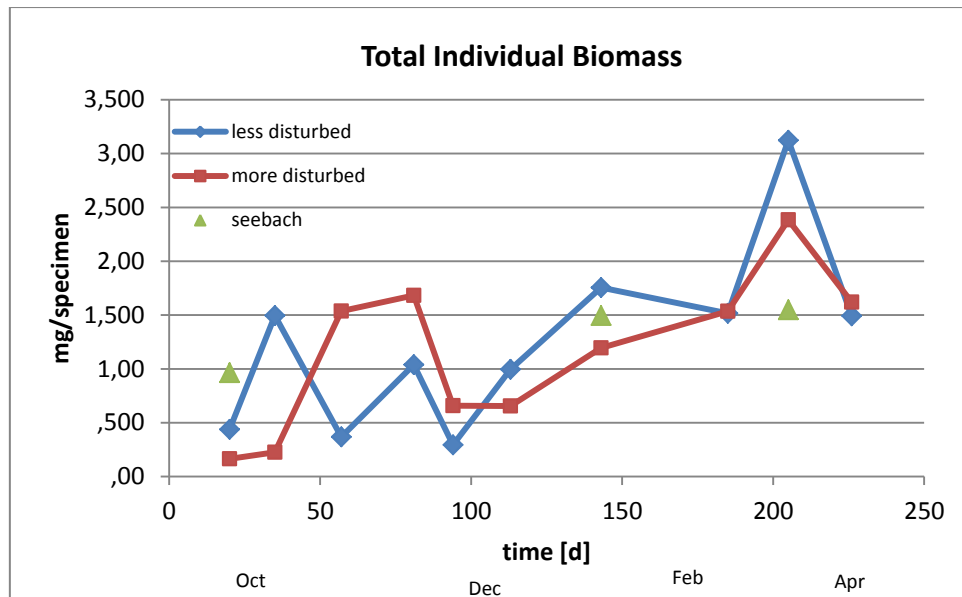


Figure 22: Individual biomass in the channels and the Seebach.

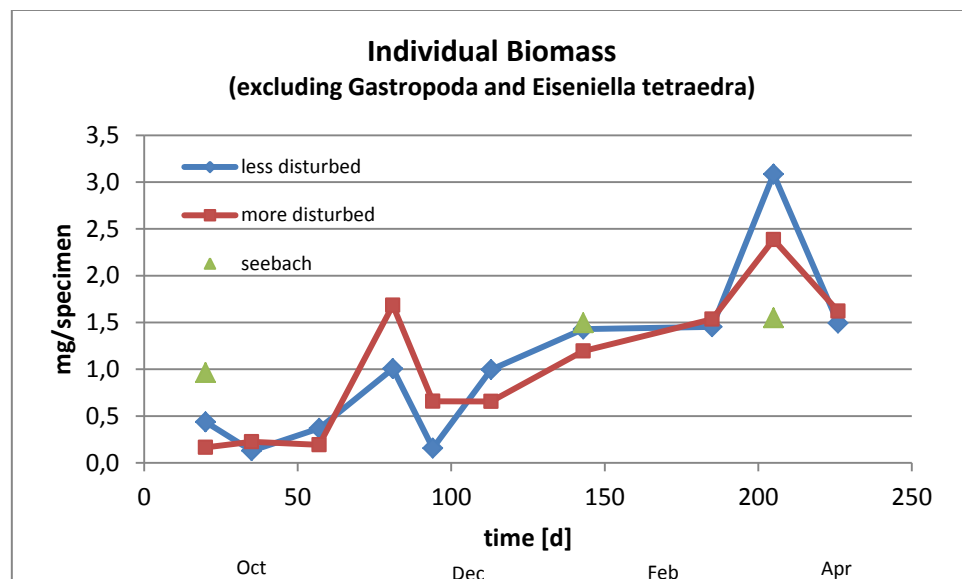


Figure 23: Corrected individual biomass in the channels and the Seebach. Gastropoda and *Eiseniella tetraedra* were excluded from the calculation to eliminate fluctuations resulting from their size.

6.4 Number of taxa

The number of taxa identified at the Austrian-Screening-Level increased from six in the end of September to twelve in the less disturbed and 15 in the more disturbed channel in April. In contrast, the number of taxa identified at the same level in the samples from the Seebach was around 40 at all three reference sampling dates. As shown by Figure 24, the number of taxa did not grow continuously, but phases of growth and phases of decline seem to be similar in the two channels. Between October and November the

number of taxa increased to remain quite stable until January. During February it declined a little, but then started to increase again.

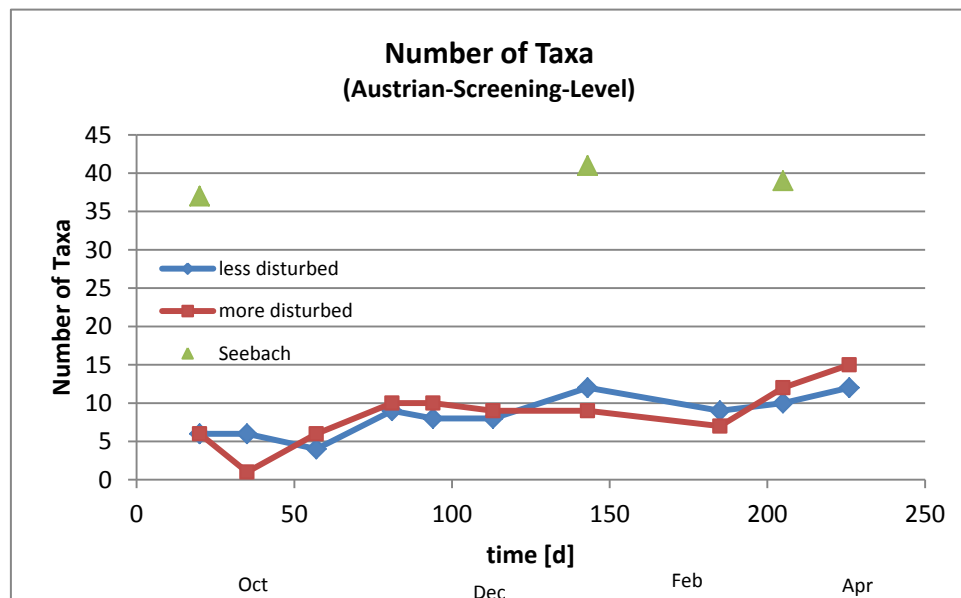


Figure 24: Number of taxa in the channels and the Seebach. (Austrian-Screening-Level)

6.5 Community composition

6.5.1 Abundance

Figure 25 displays abundances of all benthic invertebrates in the box-samples summed up to groups (mainly orders). Invertebrates in the experimental channels were clearly dominated by the Diptera family Chironomidae throughout the study period, especially from January onwards. Relative abundance of Chironomids highly fluctuated throughout October, increased from around 30 % in November to over 90 % in February to remain at this level until the end of the study period in April.

Since abundances of the further groups are difficult to recognize due to their low numbers, Figure 26 shows abundances excluding Chironomids. It is visible at first sight that worms (Oligochaeta) were more abundant in the less disturbed channel and stoneflies (Plecoptera) in the more disturbed one. Furthermore, abundances of these other groups show a strong fluctuation in contrast to the constant growth of Chironomid abundance.

In the reference samples from the Seebach community was more heterogeneous and not dominated by one family or order of invertebrates. Figure 27 indicates quantitative

importance of mayflies (Ephemeroptera), caddisflies (Trichoptera) and beetles (Coleoptera), the latter two of which were rarely found in the experimental channels. Chironomids were also important, but this group accounted for 16 % of total abundance in March in the Seebach, as opposed to 95 % and 93 % respectively in the channels.

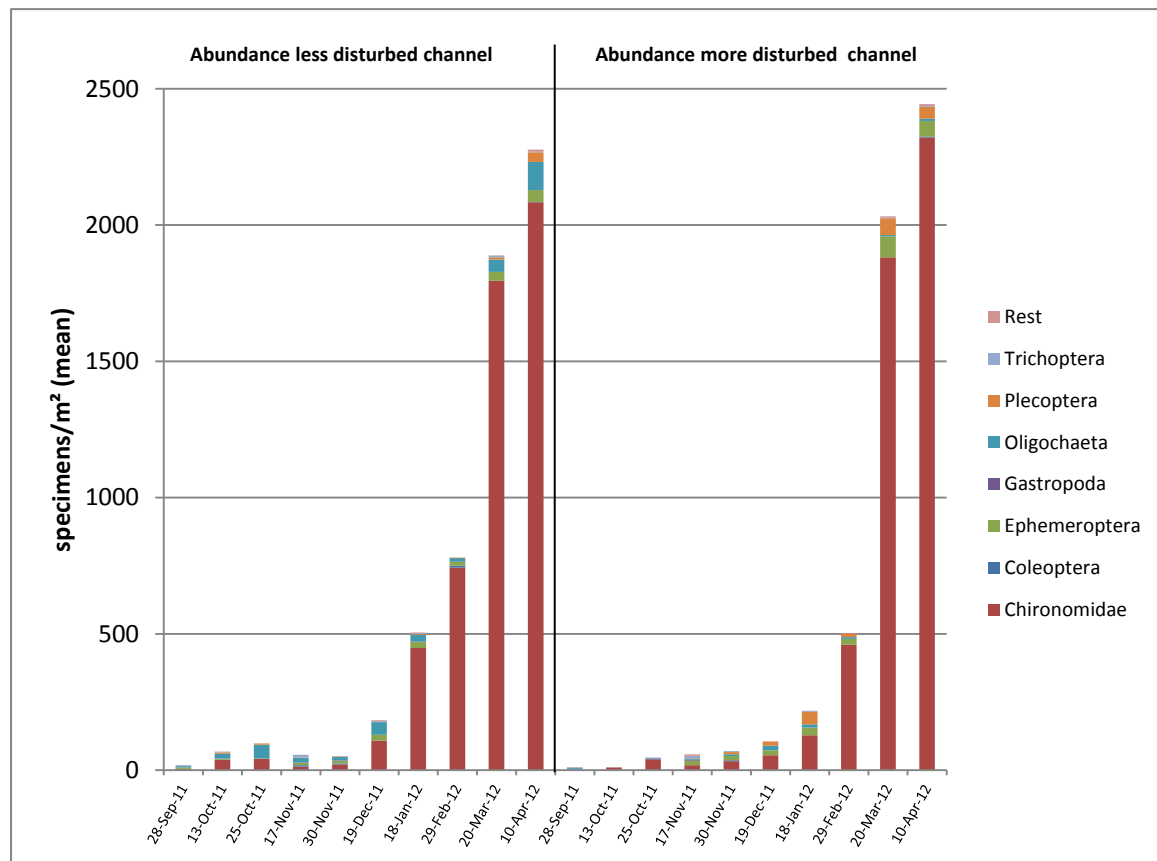


Figure 25: Community composition in the channels.

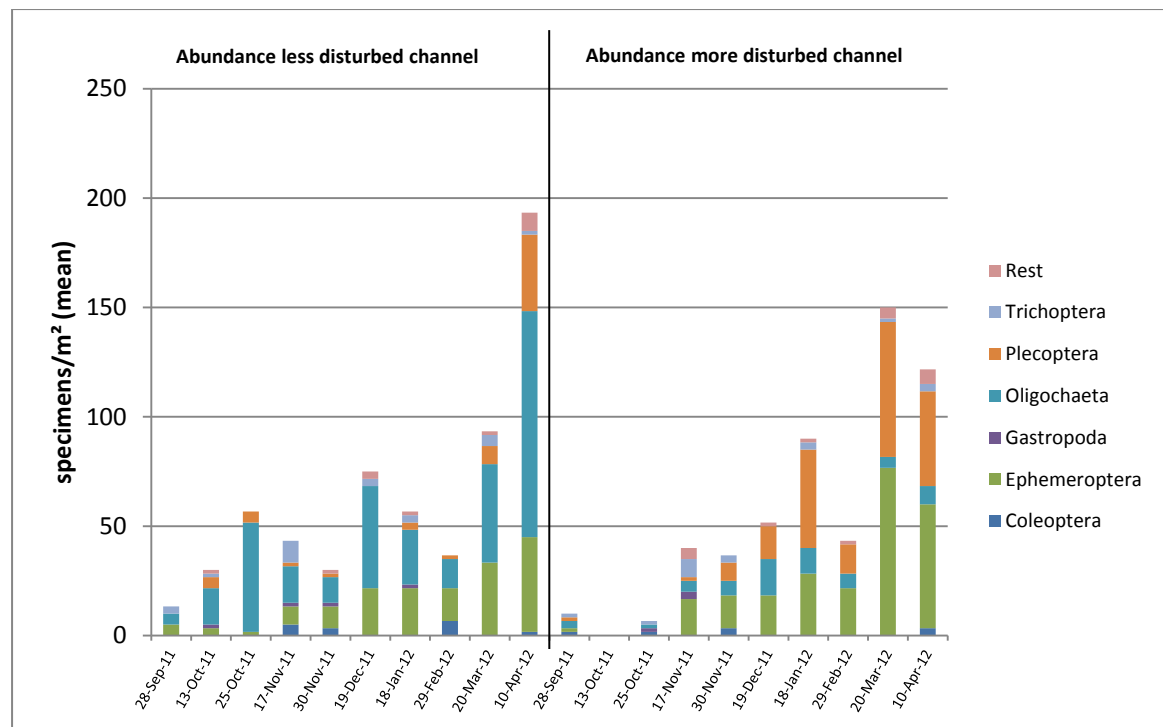


Figure 26: Community composition in the channels without Chironomidae.

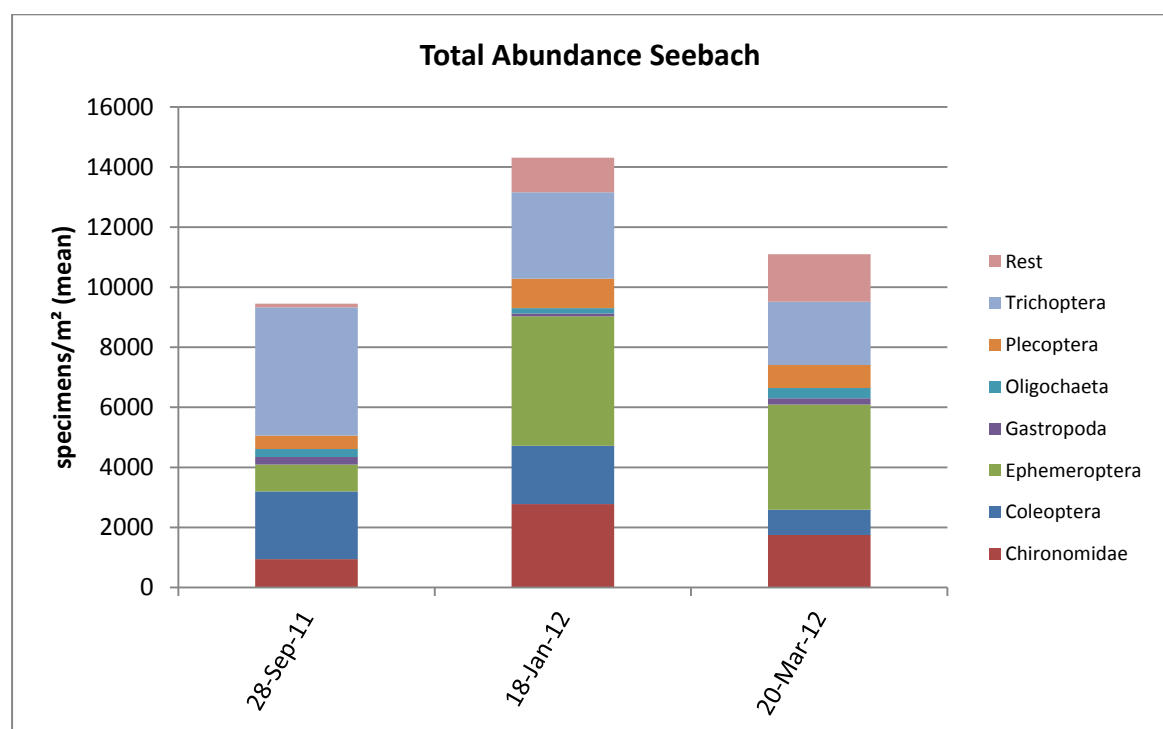


Figure 27: Community composition of the Seebach.

Similarity of dominance structures between communities in the channels, expressed by the RENKONEN-Index, increased from less than 50 % in September to more than 90 % in February, March and April, when Chironomids accounted for the greatest part of

invertebrate fauna. Hence, dominance was very similar in both channels since February. The SØRENSEN-Index for communities of the channels showed a strong fluctuation throughout the study period, which indicates a fluctuation in presence and absence of taxa, but on the long term it also increased (see Figure 28).

Similarity between a channel and the Seebach, given in Table 2, was never higher than 34 % in case of the RENKONEN-Index and 41 % in case of the SØRENSEN-Index. This emphasizes the fundamental difference between the communities of the channels and the community of the Seebach.

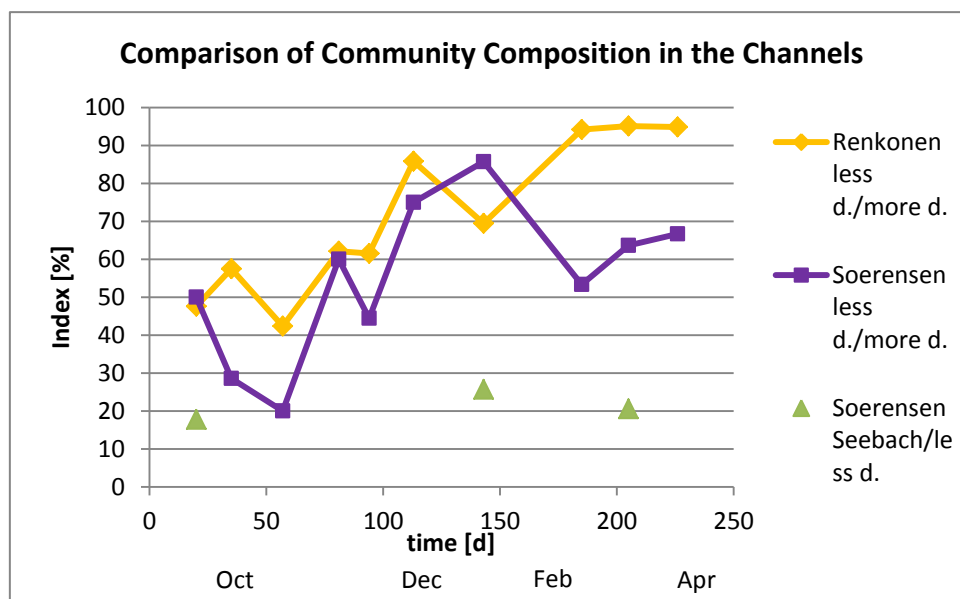


Figure 28: Similarity between the channels.

Table 2: Similarity between the channels and with the Seebach.

Index	Compared streams	28-Sep-11	18-Jan-12	20-Mar-12
RENKONEN	less disturbed more disturbed	48	69	95
	less disturbed seebach	18	26	21
	more disturbed seebach	34	33	20
SØRENSEN	less disturbed more disturbed	50	86	64
	less disturbed seebach	18	41	37
	more disturbed seebach	27	31	31

6.5.2 Biomass

Looking at the composition of total biomass, dominance of Chironomids was at a similar extent as in abundance structure, as displayed by Figure 29. Until January the share of Chironomid biomass in the channels was highly fluctuating between 10 % and 100 %. From February until April Chironomids accounted for more than 90 % of total biomass in the channels whereas in the Seebach in March this portion was little more than 1 % (Figure 31).

Figure 30 shows how relatively large representatives of Gastropoda, Oligochaeta, Plecoptera and Trichoptera induced fluctuations in biomass. Especially Gastropoda and Trichoptera accounted for large proportions of total biomass, compared to small proportions of abundance. This figure also demonstrates that Ephemeroptera accounted for a larger share of total biomass on the last two sampling dates in the more disturbed channel. As shown in Figure 26, this is partly due to higher abundance, but the difference in biomass is obviously higher than the difference in abundance.

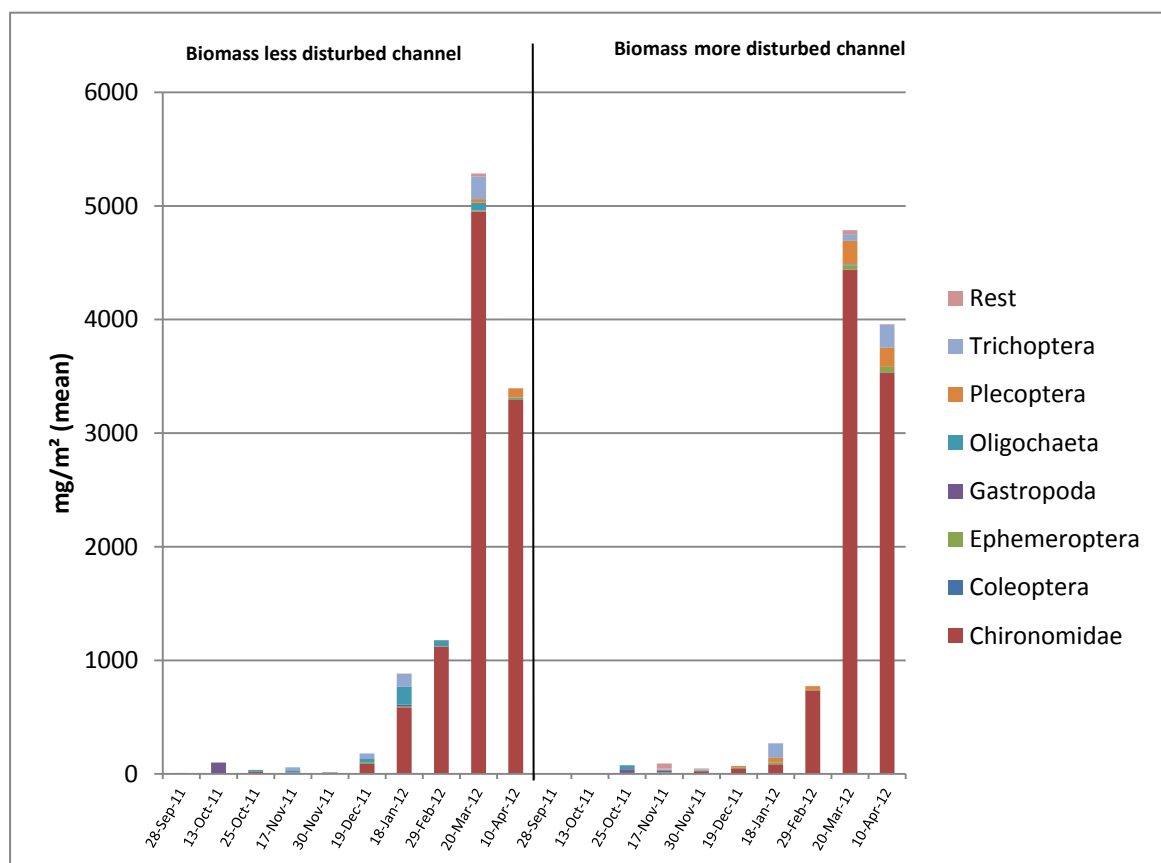


Figure 29: Composition of total biomass in the channels.

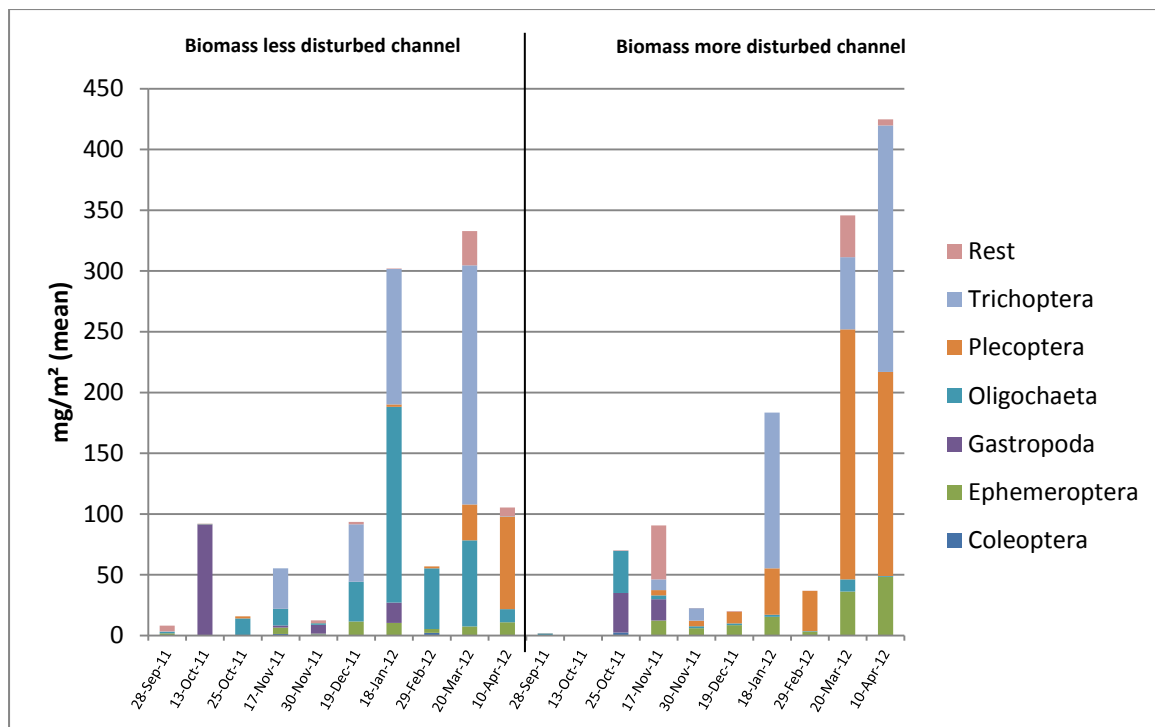


Figure 30: Composition of total biomass in the channels without Chironomidae.

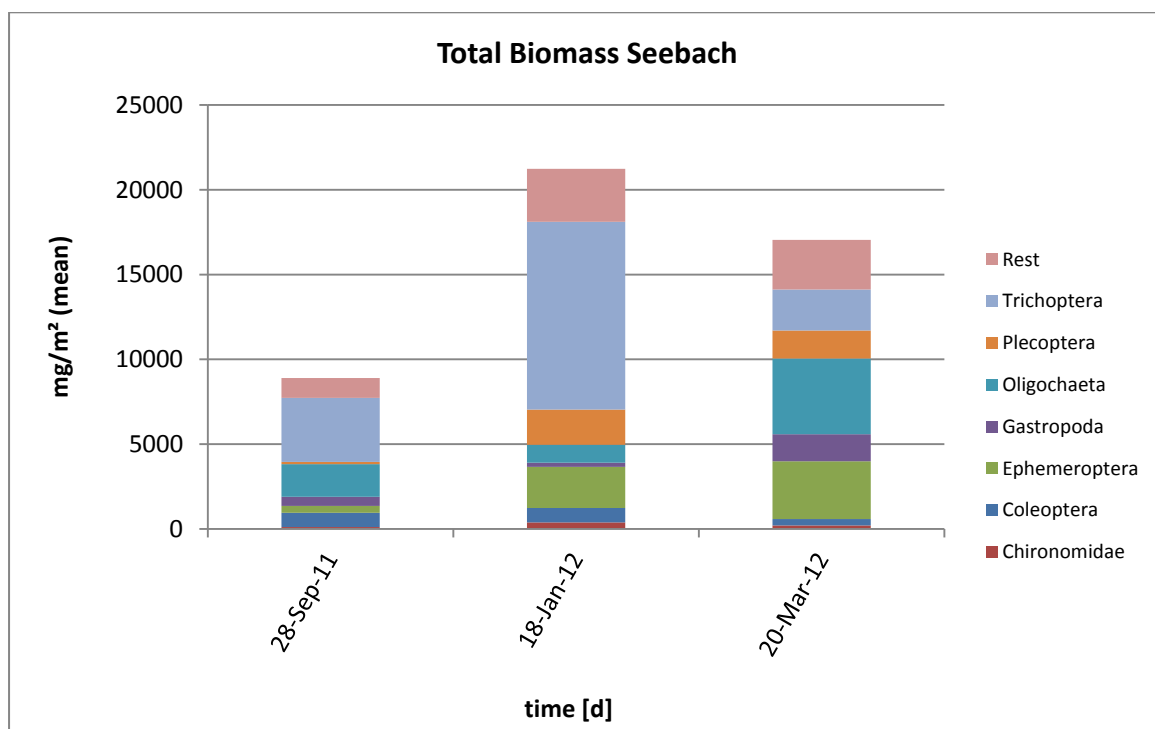


Figure 31: Composition of total biomass in the Seebach.

6.6 Analysis of selected taxa

6.6.1 Chironomidae

Growth of abundance and biomass could only be observed of few taxa in the channels. The most important taxonomic unit, as mentioned above, was the Dipteran family Chironomidae. As demonstrated by Figure 32, Chironomid abundance in the channels increased to the same level as in the Seebach until March but Chironomid biomass was much higher in the channels at that time (Figure 33). From December onwards, Chironomid pupae were constantly found in the samples, at first more abundant in the less disturbed channel but in March and April at high numbers in both channels. Although, numbers of Chironomid pupae declined sharply between March and April while total Chironomid abundance (larvae and pupae) still increased, as pointed out by the scatter-plots in Figure 34 and Figure 35. On the contrary, Chironomid pupae were only found sporadically in the samples from the Seebach.

Abundance varied extremely between the six samples of each sampling date. The scatter-plots show data of all six box-samples in each channel on each sampling date. One sample could contain four times as many Chironomids as another one on the same date in the same channel, which indicates patchy distribution of specimens in the channels.

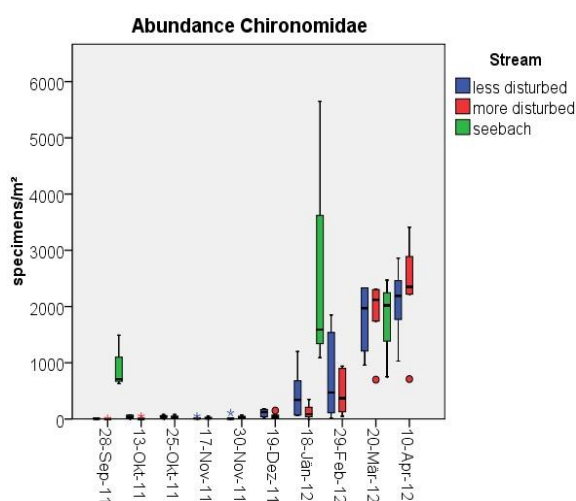


Figure 32: Abundance of Chironomid larvae and pupae.

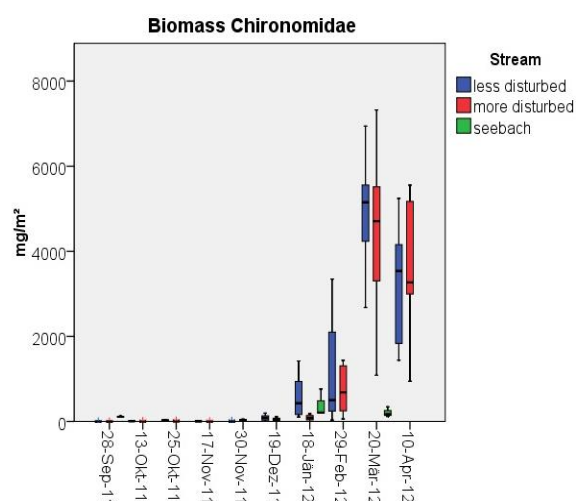


Figure 33: Biomass of Chironomid larvae and pupae.

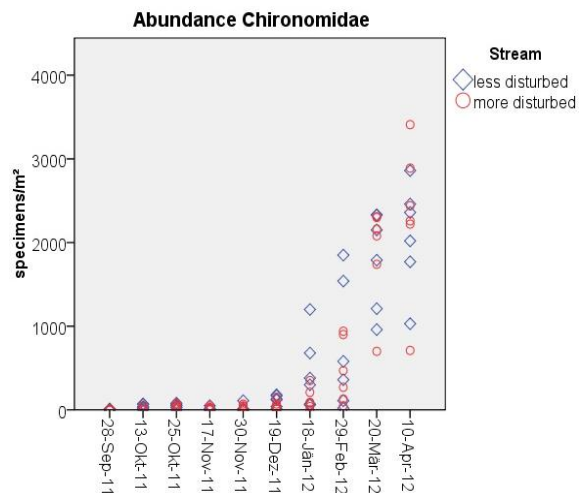


Figure 34: Scatter-plot - abundance of Chironomid larvae and pupae.

Please note: Each point represents one box-sample; 1000 specimens per m² refer to 100 specimens per sample.

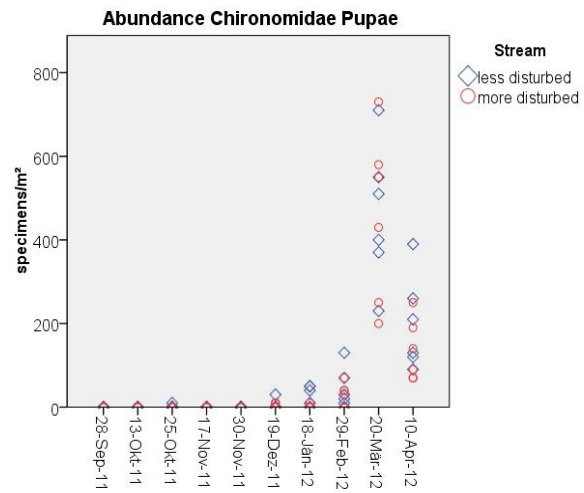


Figure 35: Scatter-plot - abundance of Chironomid pupae.

Chironomids did not only grow quantitatively, they also gained weight during the study period. Average individual biomass, displayed in Figure 36, increased from 0.1 mg in September to 2.4 mg in the more disturbed and 2.8 mg in the less disturbed channel in March. In April, however, it had declined again to the level of February. Growth curves were similar in both channels.

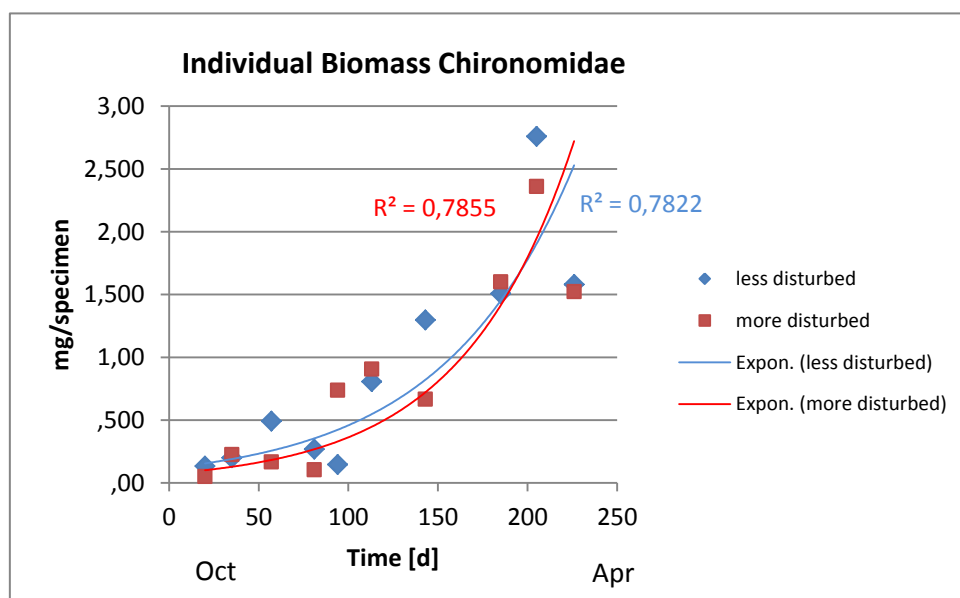


Figure 36: Average individual biomass of Chironomids in the channels. (larvae and pupae)

On the 28th of September 2011 only four specimens of Chironomidae were found in the samples, one of which could not be identified (Figure 37). The other three belonged to the subfamilies of Orthocladiinae and Chironominae. 53 different taxa of Chironomids could be identified in the samples of the 10th of April 2012. As demonstrated by Figure 38, about 90 % belonged to the genus *Diamesa* (subfamily Diamesinae). The rest was represented by the subfamilies Orthocladiinae and Chironominae, the latter dominated by the tribe Tanytarsini. *Micropsectra atrofasciata* accounted for the greatest part of Tanytarsini. Only one specimen of the tribe Pentaneurini (subfamily Tanypodinae) was identified. Most abundant within the genus *Diamesa* were the *Diamesa cinerella* group, *D. hamaticornis* and *D. insignipes* (see Figure 39). Chironomid pupae were also predominantly represented by the genus *Diamesa*.

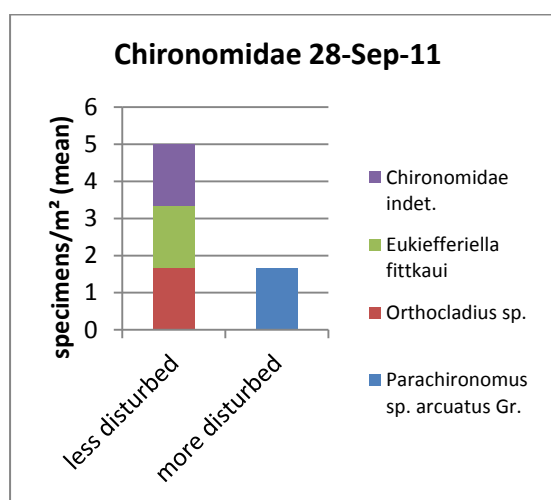


Figure 37: Chironomid fauna in the channels on the 28th of September 2011.
(Larvae and pupae)

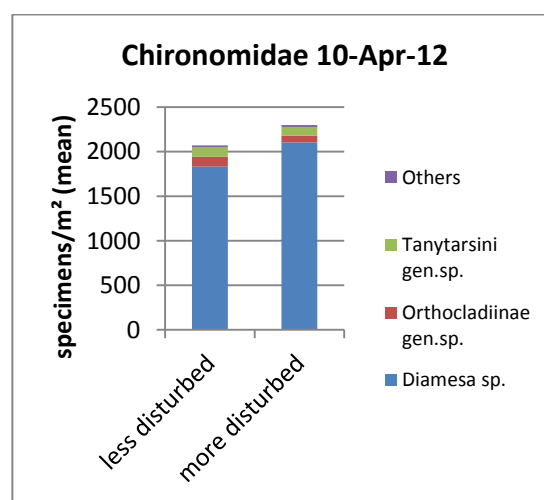


Figure 38: Chironomid fauna in the channels on the 10th of April 2012.
(Larvae and pupae)

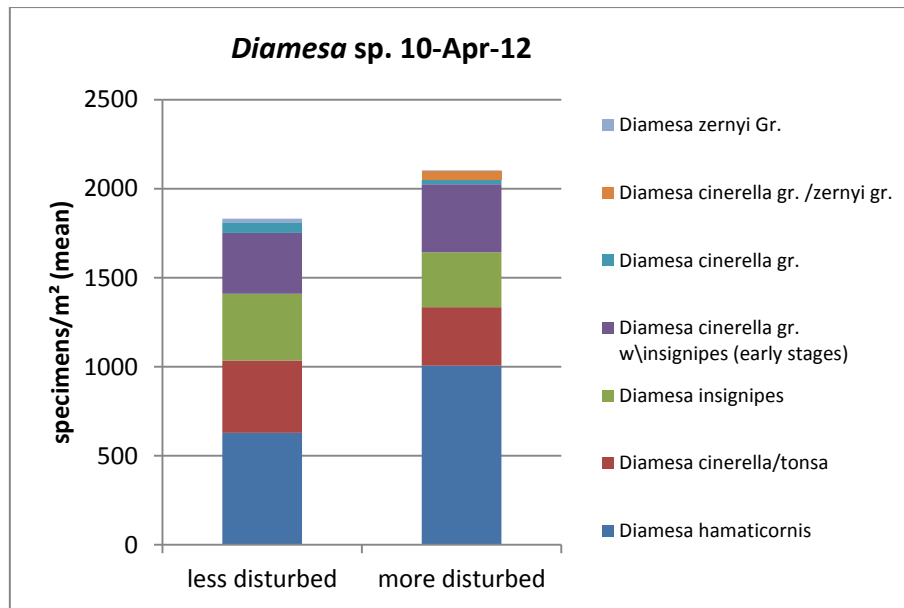


Figure 39: Abundance of the genus *Diamesa* in the channels on the 10th of April 2012. (Larvae and pupae)

Calculation of similarity of communities in the channels under consideration of Chironomid community composition demonstrated high similarity in April and low similarity in September. The SØRENSEN-Index was at 29 in September and at 73 in April, the RENKONEN-Index at 33 in September and at 79 in April.

6.6.2 *Nemoura* / *Nemurella* sp.

Another group of invertebrates with growing abundance and biomass were the stonefly genera *Nemoura* and *Nemurella*, grouped to one taxonomic unit in the Austrian-Screening-Level. However, Figure 40 indicates stronger growth of abundance in the more disturbed channel and complete absence in the samples from the less disturbed channel on some dates. It also has to be pointed out that on each sampling date there were samples completely lacking *Nemoura/Nemurella* sp. in both channels, which indicates patchy distribution of this taxon. Figure 41 shows that abundances of *Nemoura/Nemurella* in the channels compared well to abundances in the Seebach.

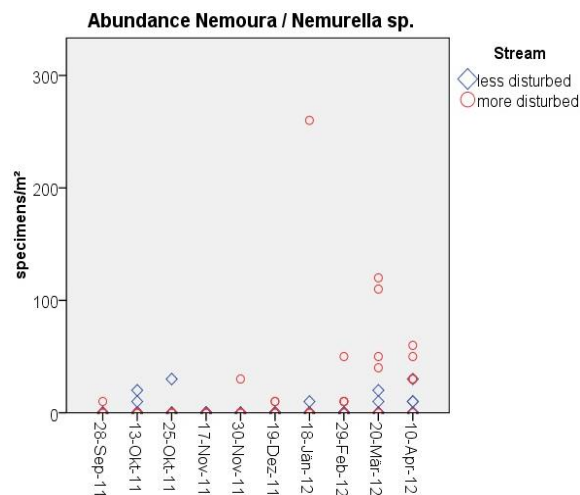


Figure 40: Scatter-plot - abundance of *Nemoura/Nemurella* without Seebach.

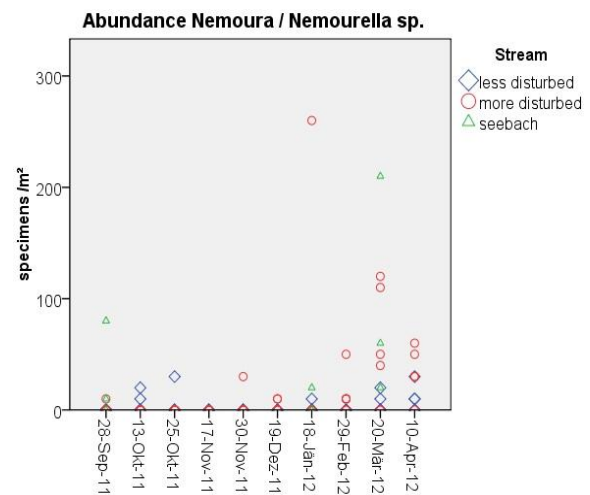


Figure 41: Scatter-plot - abundance of *Nemoura/Nemurella* with Seebach.

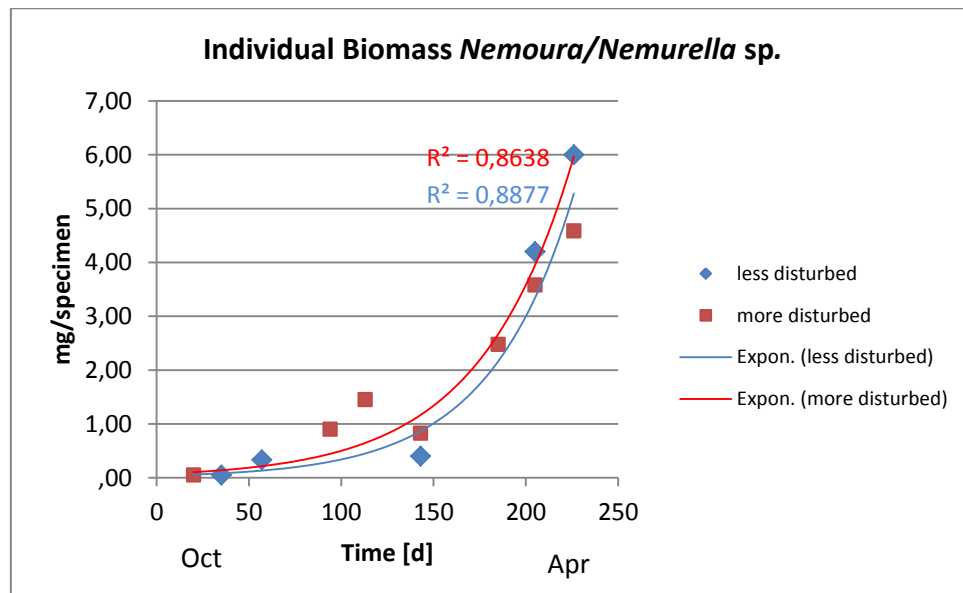


Figure 42: Average individual biomass of *Nemoura/Nemurella* in the channels.

Individual biomass of *Nemoura/Nemurella* sp. in the channels increased from 0.1 mg in September and October to 6 mg in the less disturbed and 4.6 mg in the more disturbed channel in April (Figure 42). Despite of lower abundance in the less disturbed channel, growth was very similar in both channels and data points are very close to the calculated exponential growth curve. In contrast to Chironomids, there was no decline in individual biomass of *Nemoura/Nemurella*.

6.6.3 *Baetis muticus*

The mayfly *Baetis muticus* shows abundances similar to those of *Nemoura/Nemurella* in the channels, as pictured in Figure 43. However, there is no clear difference between the channels. Average abundance was increasing in both channels, but remained very low compared to abundance in the Seebach (Figure 44). The figures further indicate that specimens were distributed very patchily, in the channels as well as in the Seebach. Individual biomass of *Baetis muticus* could not be calculated, since accuracy of the scale was not sufficient to scale these small animals and they were scaled together with other *Baetis* species and *Siphonurus lacustris*.

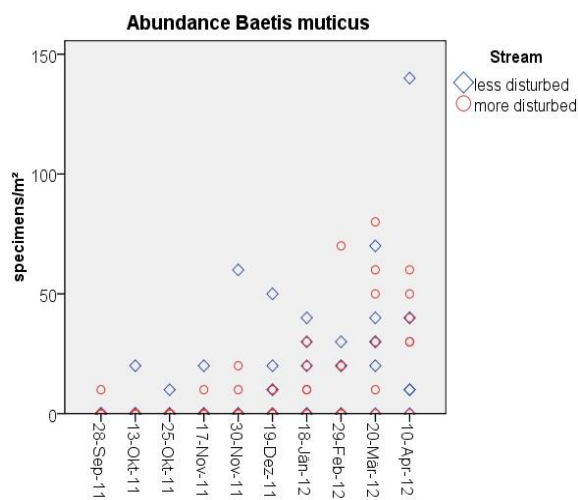


Figure 43: Scatter-plot - abundance of *Baetis muticus* without Seebach.

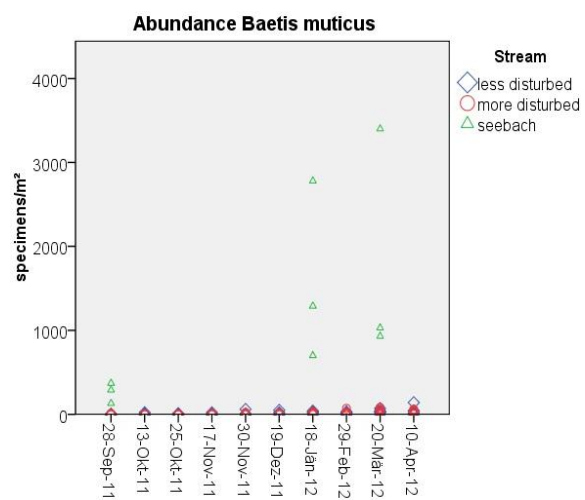


Figure 44: Scatter-plot - abundance of *Baetis muticus* with Seebach.

6.6.4 Oligochaeta

As already mentioned above, Oligochaetes were more abundant in the less disturbed channel throughout the study period, which is reinforced by Figure 45. Abundance was fluctuating and distributed very patchily between the box-samples. There was neither a distinct growth nor decline in Oligochaete abundance.

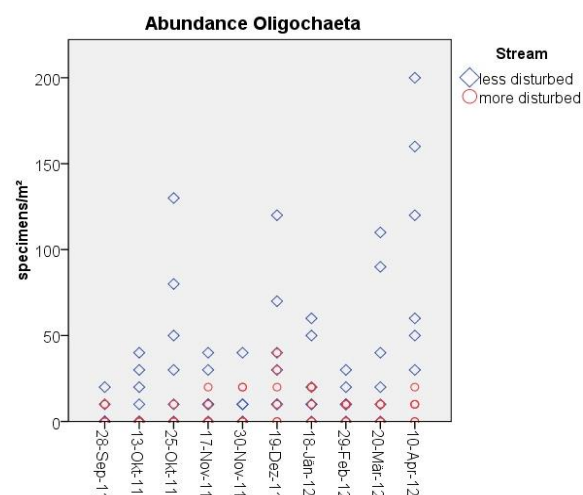


Figure 45: Scatter-plot – abundance of Oligochaeta.

Individual biomass of Oligochaetes, as displayed in Figure 46, was strongly fluctuating, although the large *Eiseniella tetraedra*, of which three specimens were found in both channels together, was not taken into account. Hence, no growth could be observed, which might be due to the lack of further identification of Oligochaetes.

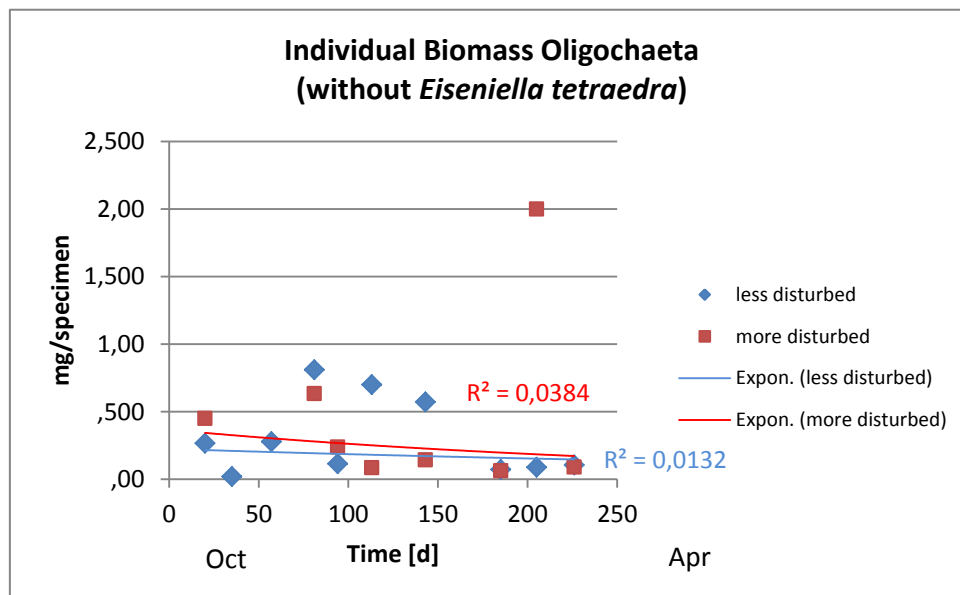


Figure 46: Individual biomass of Oligochaeta excluding *Eiseniella tetraedra*.

6.6.5 Taxa found regularly but with low abundances

All other taxa apart from the four described above were found in the samples only with abundances of maximum five specimens per sample. However, some of them still occurred constantly, at least from a given sampling date onwards.

One example is the caddisfly *Ecclisopteryx guttulata*, which was first found in the samples from the 17th of November (Figure 47). Interestingly, this species was not found in any of the Seebach-samples examined in the context of this study, but constantly some specimens appeared in both channels. Furthermore, Figure 48, although calculated on the basis of few specimens, suggests *E. guttulata* gaining weight over the winter.

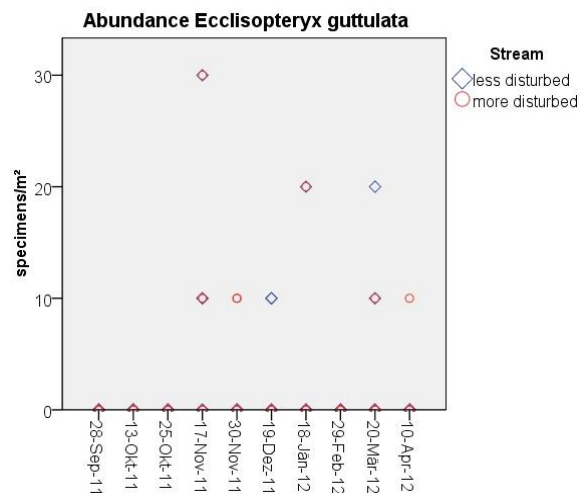


Figure 47: Scatter-plot – abundance of *Ecclisopteryx guttulata*.

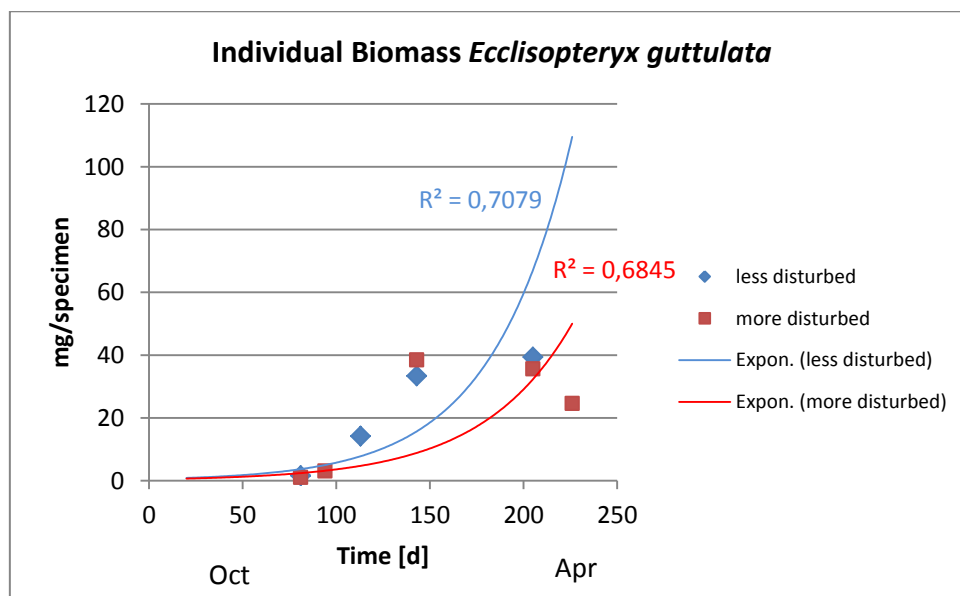


Figure 48: Average individual biomass of *Ecclisopteryx guttulata* in the channels.

Another species that was not found in the Seebach-samples although in the channels is the mayfly *Siphonurus lacustris*. It first appeared in both channels on the 29th of February and was constantly abundant in the more disturbed channel in March and April, but did not reappear in the less disturbed channel (Figure 49).

Further taxa appearing constantly in the channels were the stonefly *Amphinemura* sp. and the mayflies *Baetis alpinus* and *Baetis rhodani*, the latter with one exception only in the more disturbed one (Figure 50, Figure 51 and Figure 52). *Amphinemura* and *B. rhodani* were first found in the samples from the 17th November. In contrast, three specimens of *B. alpinus* were found already on the 28th September. Every three of them appeared in the Seebach-samples with abundances of 600 to 800 specimens per m².

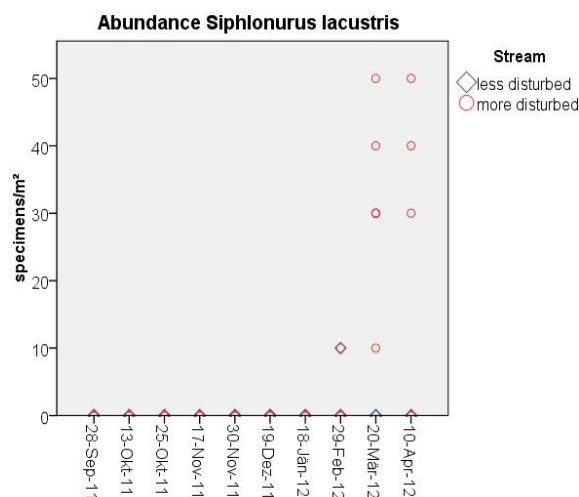


Figure 49: Scatter-plot – abundance of *Siphonurus lacustris*.

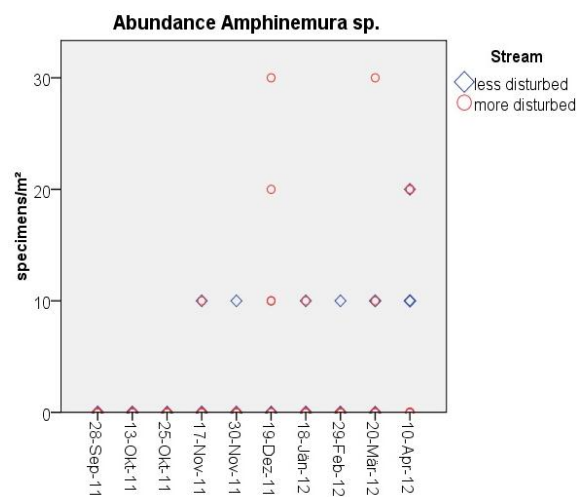


Figure 50: Scatter-plot – abundance of *Amphinemura* sp.

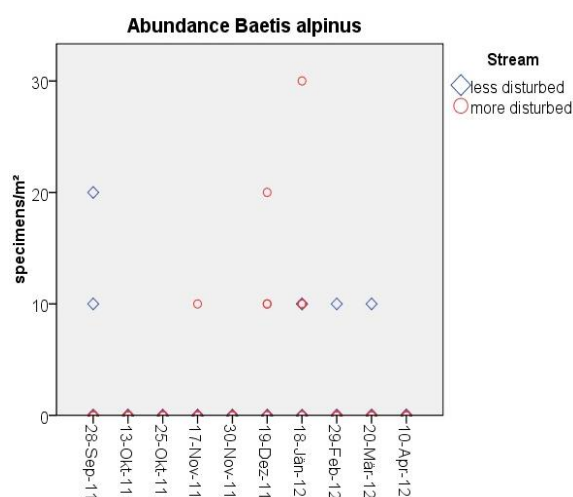


Figure 51: Scatter-plot – abundance of *Baetis alpinus*.

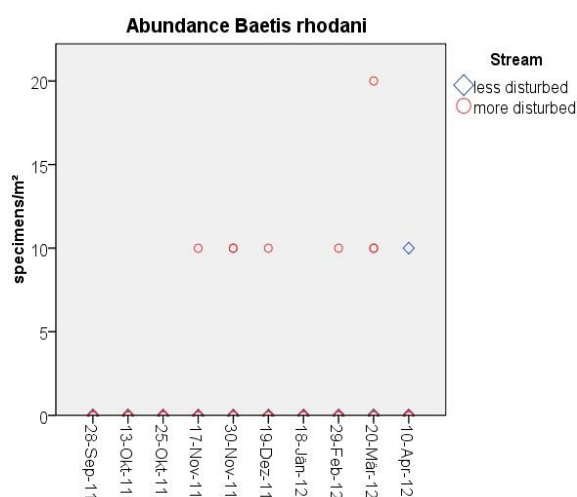


Figure 52: Scatter-plot – abundance of *Baetis rhodani*.

6.6.6 Other taxa

Most of the taxa recorded in this study appeared only sporadically in the samples taken from the experimental channels. The taxa described in Figure 53 and Figure 54, *Ecdyonurus* sp. and *Micrasema setiferum*, may give an example for presence and absence of various species of Gastropoda, Diptera and the beetle family Elmidae, the stonefly genera *Isoperla*, *Leuctra* and *Protonemura*, the caddisflies *Rhyacophila sensu strictu* and *Hydropsyche* sp., *Hydra* sp., *Gammarus fossarum* and *Asellus aquaticus*. It has to be noted that ten specimens per m² refer to one specimen in a box-sample. On the contrary, *M. setiferum*, *Ecdyonurus* sp., Elmidae, *Hydropsyche* sp. and three Gastropod genera were highly abundant in the Seebach (see Appendix).

An exceptional case is the sudden appearance of the mayfly *Rhithrogena* sp. in November and the rapid decrease of its abundance until total absence from February onwards, pictured in Figure 55.

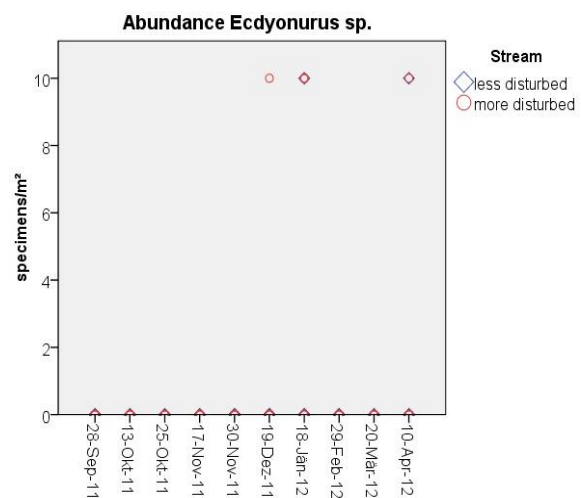


Figure 53: Scatter-plot – abundance of *Ecdyonurus* sp.

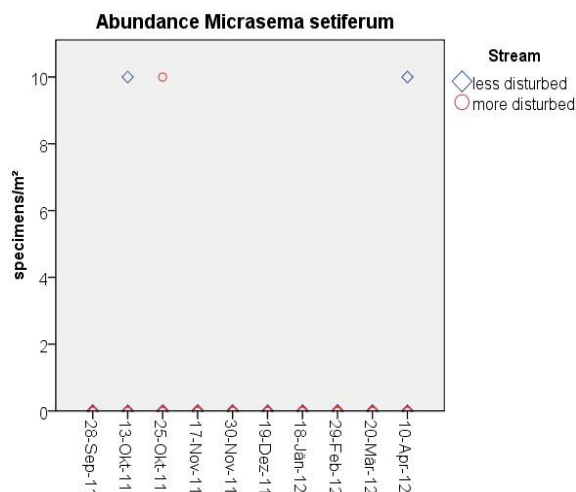


Figure 54: Scatter-plot – abundance of *Micrasema setiferum*.

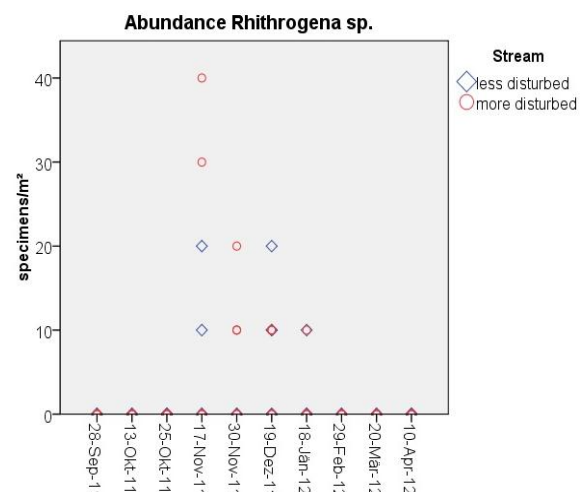


Figure 55: Scatter-plot – abundance of *Rhithrogena* sp.

6.6.7 *Hydrurus* algae

From the end of November the filamentous alga *Hydrurus* sp. dominated the flora in the experimental channels. Especially in December and January, it covered a big percentage of the wetted channel area with thick brown layers (Figure 57). By the 10th of April 2012, however, *Hydrurus* covered area and the thickness of coverage had declined (see Figure 58 and Figure 59).



Figure 56: *Hydrurus* coverage 28-Sep-11.



Figure 57: *Hydrurus* coverage 19-Dec-11.



Figure 58: *Hydrurus* coverage 20-Mar-12.

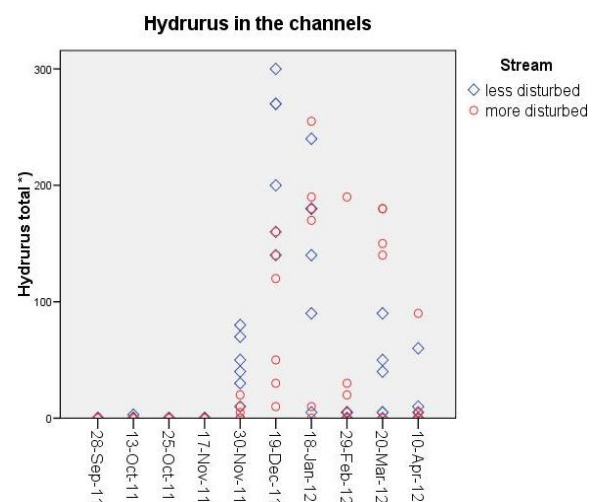


Figure 59: Scatter-plot: *Hydrurus* alga in the channels.
*) *Hydrurus* total is calculated through multiplying *Hydrurus* coverage [%] by a given factor [1;3] for the thickness of the *Hydrurus* cover (see Appendix).

Total *Hydrurus* amount was estimated through multiplying the percentage of *Hydrurus* coverage by a factor between 1 and 3, referring to the thickness of coverage. Correlation analysis between total *Hydrurus* amount and total biomass or abundance of various taxa did not indicate any correlation. Low correlation coefficients between total *Hydrurus* amount and total biomass (Table 3), Chironomid abundance (Table 4) and abundance of *Baetis muticus* (Table 5) were significant at the level 0.01. For the correlation of Oligochaete abundance with *Hydrurus* amount (Table 6), coefficients were not significant. Furthermore, no significant correlation coefficients were found for *Hydrurus* coverage with total biomass (Table 7) and total abundance (Table 8).

The graphs for growth of *Hydrurus* coverage and total abundance or total biomass pictured in one diagram back up the finding, that these variables do not correlate (Figure 60 - Figure 63). *Hydrurus* coverage had a main peak in both channels in December and January and a secondary peak in March. In contrast, total biomass and total abundance of invertebrates increased constantly until March and April respectively.

Table 3: Correlation between total biomass and *Hydrurus* amount.

			Total Biomass	Hydrurus total
Kendall-Tau-b	Total Biomass	Correlation coefficient	1,000	,232(**)
		Sig. (2-sided)	.	,001
		N	129	120
	Hydrurus total	Correlation coefficient	,232(**)	1,000
		Sig. (2-sided)	,001	.
		N	120	120
Spearman-Rho	Total Biomass	Correlation coefficient	1,000	,320(**)
		Sig. (2-sided)	.	,000
		N	129	120
	Hydrurus total	Correlation coefficient	,320(**)	1,000
		Sig. (2-sided)	,000	.
		N	120	120

** The correlation is significant at the level 0.01 (2-sided).

Table 4: Correlation between Chironomid abundance and *Hydrurus* amount.

			Hydrurus total	Chiro. L+P
Kendall-Tau-b	Hydrurus total	Correlation coefficient	1,000	,232(**)
		Sig. (2-sided)	.	,001
		N	120	120
	Chiro. L+P	Correlation coefficient	,232(**)	1,000
		Sig. (2-sided)	,001	.
		N	120	129
Spearman-Rho	Hydrurus total	Correlation coefficient	1,000	,320(**)
		Sig. (2-sided)	.	,000
		N	120	120
	Chiro. L+P	Correlation coefficient	,320(**)	1,000
		Sig. (2-sided)	,000	.
		N	120	129

** The correlation is significant at the level 0.01 (2-sided).

Table 5: Correlation between abundance of *Baetis muticus* and *Hydrurus* amount.

			Hydrurus total	Baetis muticus
Kendall-Tau-b	Hydrurus total	Correlation coefficient	1,000	,204(**)
		Sig. (2-sided)	.	,008
		N	120	120
	Baetis muticus	Correlation coefficient	,204(**)	1,000
		Sig. (2-sided)	,008	.
		N	120	129
Spearman-Rho	Hydrurus total	Correlation coefficient	1,000	,250(**)
		Sig. (2-sided)	.	,006
		N	120	120
	Baetis muticus	Correlation coefficient	,250(**)	1,000
		Sig. (2-sided)	,006	.
		N	120	129

** The correlation is significant at the level 0.01 (2-sided).

Table 6: Correlation between Oligochaete abundance and *Hydrurus* amount.

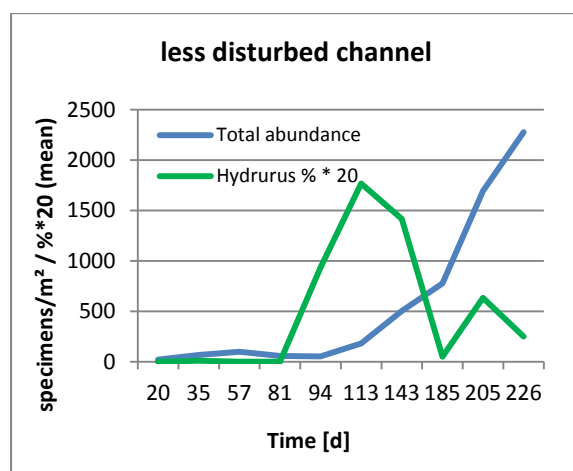
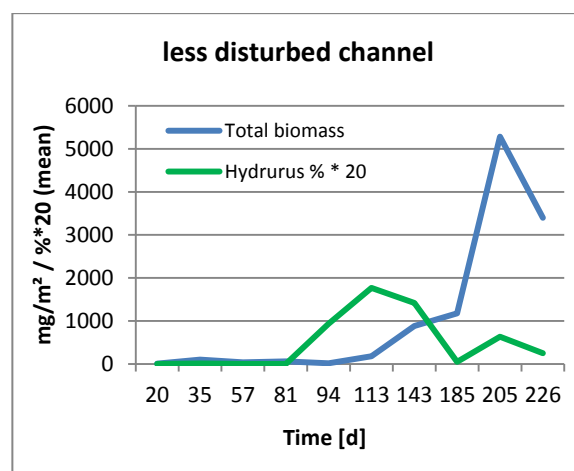
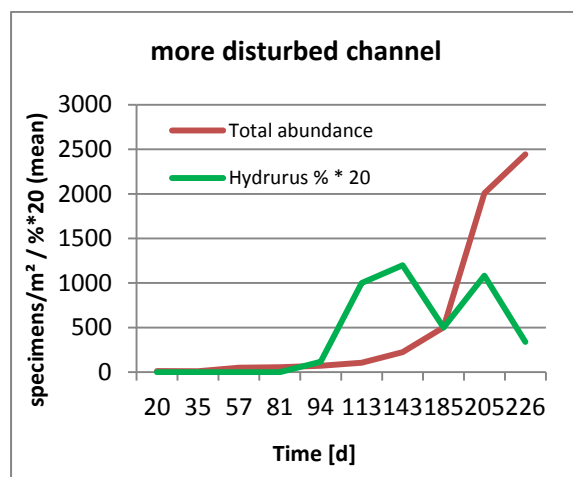
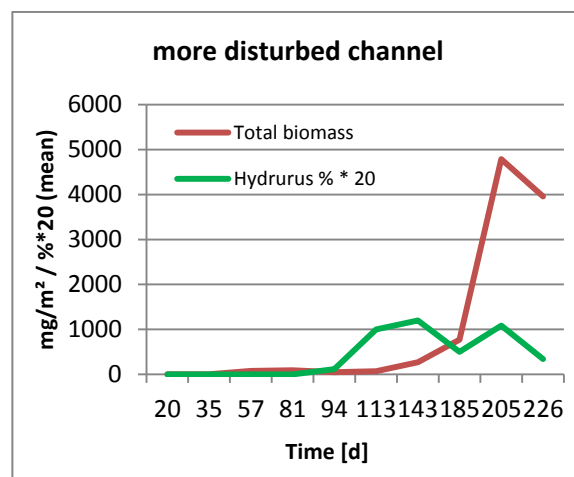
			Hydrurus total	Oligochaeta
Kendall-Tau-b	Hydrurus total	Correlation coefficient	1,000	,118
		Sig. (2-sided)	.	,113
		N	120	120
	Oligochaeta	Correlation coefficient	,118	1,000
		Sig. (2-sided)	,113	.
		N	120	129
Spearman-Rho	Hydrurus total	Correlation coefficient	1,000	,142
		Sig. (2-sided)	.	,121
		N	120	120
	Oligochaeta	Correlation coefficient	,142	1,000
		Sig. (2-sided)	,121	.
		N	120	129

Table 7: Correlation between total biomass and *Hydrurus* coverage.

			Total Biomass	Hydrurus [%]
Kendall-Tau-b	Total Biomass	Correlation coefficient	1,000	-,029
		Sig. (2-sided)	.	,764
		N	55	55
	Hydrurus [%]	Correlation coefficient	-,029	1,000
		Sig. (2-sided)	,764	.
		N	55	55
Spearman-Rho	Total Biomass	Correlation coefficient	1,000	-,053
		Sig. (2-sided)	.	,700
		N	55	55
	Hydrurus [%]	Correlation coefficient	-,053	1,000
		Sig. (2-sided)	,700	.
		N	55	55

Table 8: Correlation between total abundance and *Hydrurus* coverage.

			Hydrurus [%]	Total Abundance
Kendall-Tau-b	Hydrurus [%]	Correlation coefficient	1,000	-,053
		Sig. (2-sided)	.	,582
		N	55	55
	Total Abundance	Correlation coefficient	-,053	1,000
		Sig. (2-sided)	,582	.
		N	55	55
Spearman-Rho	Hydrurus [%]	Correlation coefficient	1,000	-,080
		Sig. (2-sided)	.	,563
		N	55	55
	Total Abundance	Correlation coefficient	-,080	1,000
		Sig. (2-sided)	,563	.
		N	55	55

Figure 60: *Hydrurus* coverage and total abundance – less disturbed channel.Figure 61: *Hydrurus* coverage and total biomass – less disturbed channel.Figure 62: *Hydrurus* coverage and total abundance – more disturbed channel.Figure 63: *Hydrurus* coverage and total biomass – more disturbed channel.

7 Discussion

7.1 Ecological profile of pioneer species

According to TOWNSEND & HILDREW (1994) characteristic traits of quickly recolonizing species are:

- High mobility (active or passive - aquatic / aerial / terrestrial)
- Polyvoltinism – more than 1 generation per year
- Small body size
- Unspecific habitat requirements (habitat generalists)
- Resistance to change in environmental conditions (drought, pH, water temperature...)

Table 10 displays the taxa which were most abundant in the samples with selected traits. Only winged taxa are mentioned, since others were rare in the channels and Oligochaetes were not identified more precisely. The traits body size and voltinism are given in one column each and mobility is represented by the flight periods shown in Table 9, since other aspects of mobility are unknown for several taxa and are not that important considering the set-up of the experimental channels. Listed flight periods are based on various observations and do not necessarily reflect actual flight periods of the respective species in Lunz am See in the season 2011/2012. The red box in Table 9 marks the sampling period. The remaining columns altitude preference, thermal preference, saprobity, longitudinal distribution, habitat specialist and current preference in Table 10 describe the habitat requirements.

Moreover, Table 10 lists the four criteria, which were considered important to define pioneers in the channels: whether they completed their life cycle in the channels, high abundance during or in the end of the sampling period, constant occurrence throughout the sampling period (from the moment of first capture) and growth of biomass per specimen. According to these criteria, *Diamesa hamaticornis*, *D. insignipes*, *D. cinerella/tonsa*, *Nemoura/Nemurella* sp., *Micropsectra atrofasciata*, and *Baetis muticus* are termed pioneers. *M. atrofasciata* and *B. muticus* might not have been very abundant compared to the samples from the Seebach, but they were abundant compared to other taxa in the channels. Further taxa appearing constantly in the samples but with low

abundance were *Baetis rhodani*, *B. alpinus*, *Amphinemura* sp. and *Ecclisopteryx guttulata*. It is now of interest whether these pioneers show specific traits that enhance colonization and which other, less abundant taxa do not have.

Table 9: Flight periods of the most important taxa in the study.

Winged taxa found in channels	Flight period											
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
<i>Amphinemura</i> sp. ¹												
<i>Baetis alpinus</i> ²												
<i>Baetis muticus</i> ²												
<i>Baetis rhodani</i> ²												
<i>Diamesa cinerella</i> ³												
<i>Diamesa hamaticornis</i> ⁴												
<i>Diamesa insignipes</i> ⁵												
<i>Diamesa tonsa</i> ⁵												
<i>Ecclisopteryx guttulata</i> ⁶												
<i>Ecdyonurus</i> sp. ²												
<i>Elmis</i> sp. ¹												
<i>Hydropsyche</i> sp. ⁶												
<i>Lepidostoma hirtum</i> ⁶												
<i>Leuctra</i> sp. ¹												
<i>Micrasema setiferum</i> ⁶												
<i>Micropsectra atrofasciata</i> ⁵												
<i>Nemoura / Nemurella</i> sp. ¹												
<i>Isoperla</i> sp. ¹												
<i>Rhithrogena</i> sp. ²												
<i>Riolus</i> sp. ¹												
<i>Siphonurus lacustris</i> ²												
Winged taxa very abundant in Seebach but not found in channels												
<i>Habroleptoides confusa</i> ²												
<i>Micrasema minimum</i> ⁶												
<i>Orectochilus villosus</i> ¹												
<i>Polycentropus</i> sp. ⁶												

Sources: 1) NILSSON (ed.), 1996; 2) BAUERNFEIND & HUMPESCH, 2001; 3) SERRA-TOSIO, 1972;

4) RINGE, 1974, CASPERS, 1983 and LEHMANN, 1971; 5) GUSICH, 2003; 6) WARINGER & GRAF, 1997

Table 10: Species traits of the most important taxa in the study.

Winged taxa found in channels	body size ¹	number of generations	altitude preference	thermal preference	saprobity	longitudinal distribution	habitat specialist	current preference	complete life cycle	high abundance	constant occurrence	growth of biomass
<i>Amphinemura</i> sp.	m	1 ¹			x-bm ¹	kr-hr (4) ¹		rh			X	
<i>Baetis alpinus</i>	m	2	n (3)	c	x-bm	kr-hr (4)		rh			X	
<i>Baetis muticus</i>	s	2	n (3)	e	x-bm	kr-mp (6)	macroph.	rl		~	X	
<i>Baetis rhodani</i>	m	2	n (3)	e	o-am	kr-mp (6)		rh			X	
<i>Diamesa cinerella</i>	s	2+ ²	n (3)	e	x-am	kr-hr (4)			X	X	?	X
<i>Diamesa hamaticornis</i>	s	2	n (3)	e	x-am	kr-mr (3)			X	X	?	X
<i>Diamesa insignipes</i>	s	2	n (3)	e	o-am	er-ep (4)			X	X	?	X
<i>Diamesa tonsa</i>	s	2	n (3)	e	x-am	kr-hr (4)			X	X	?	X
<i>Ecclisopteryx guttulata</i>	l	1	>200 (2)		x-bm	er-hr (3)	lithal	rh			X	X
<i>Ecdyonurus</i> sp.	m	1 ³			x-am ¹	kr-mp (6) ¹		rh				
<i>Elmis</i> sp.	s	1 ¹			x-bm ¹	kr-ep (5) ¹		rh				
<i>Hydropsyche</i> sp.	l	1+ ¹	<800 (2)		x-am ¹	kr-mp (6) ¹		rh				
<i>Lepidostoma hirtum</i>	m-l	1	n (3)	e	o-am	mr-hr (2)	CPOM	lr				
<i>Leuctra</i> sp.	m	1 ¹			x-bm ¹	kr-ep (5) ¹		rh				
<i>Micrasema setiferum</i>	s	1 ¹	< 800 (2)		o-am	mr-ep (3)						
<i>Micropsectra atrofasciata</i>	s	2	n (3)	e	x-am	kr-ep, lit (6)			X	~	?	?
<i>Nemoura / Nemurella</i> sp.	m	1 ¹			x-bm ¹	kr-mp (6) ¹				X	X	X
<i>Isoperla</i> sp.	m-l	1 ¹			x-bm ¹	kr-mp (6) ¹		rh				
<i>Rhithrogena</i> sp.	m	1 ³			x-bm ¹	kr-ep (5) ¹		rh				
<i>Riolus</i> sp.	s	1 ¹			o-bm ¹	kr-ep (5) ¹						
<i>Siphonurus lacustris</i>	m	1 ³	n (3)	e	x-bm	er-mp, lit (6)		li				
Winged taxa very abundant in Seebach but not found in channels												
<i>Habroleptoides confusa</i>	m	1 ³	n (3)	e	o-am	kr-ep (5)		rl				
<i>Micrasema minimum</i>	s	1 ¹	>200 (2)		o-bm	er-hr (3)	mosses	rhb				
<i>Orectochilus villosus</i>	m	1 ¹			o-am	mr-mp (4)		rh				
<i>Polycentropus</i> sp.	m-l	1+ ¹	<800 (2)		x-am ¹	kr-hp (7) ¹		rl				
Sources: If not marked with a footer, sources are: Ephemeroptera: BUFFAGNI et al., 2007 & 2009; SCHMEDTJE & COLLING, 1996; Plecoptera: GRAF et al., 2007 & 2009; Trichoptera: GRAF et al., 2006 & 2008; GRAF & SCHMIDT-KLOIBER, 2011; SCHMEDTJE & COLLING, 1996; Chironomidae: JANECEK et al., 2002; BRABEC et al., 2007; Coleoptera: JÄCH et al., 1995; SCHMEDTJE & COLLING, 1996; Footers: 1) TACHET et al. (2000); 2) SERRA-TOSIO, 1972; 3) BAUERNFEIND & HUMPESCH, 2001 Legend: body size: s=small, m=medium, l=large; generations: 1=1per year, 1+=1 or more, 2=2 per year, 2+=2 or more; altitude preference: n=no restriction (all 3 classes), >200=over 200 m.a.s.l., <800=below 800 m.a.s.l. (2 of 3 altitude classes); thermal preference: c=cold, e=euryoecious; saprobity: x=xenosaprobic, o=oligosaprobic, bm=beta-mesosaprobic, am=alpha-mesosaprobic; distribution: kr=krenal, er=epirhithral, mr=metarhithral, hr=hyporhithral, ep=epipotamal, mp=metapotamal, hp=hypopotamal, lit=littoral, number in brackets=number of zones; current preference: rhb=rheobiont, rh=rheophilic, rl=rheo- to limnophilic, lr= limno- to rheophilic, li= limnophilic												

Table 11: Calculation of a percentage of pioneer traits.

Explanation of the table: Every trait stated by TOWNSEND & HILDREW (1994) to be characteristic for pioneers is weighted with 25 % to add up to 100 % (resistance is not taken into account due to a lack of data). Flight period, which stands for mobility, body size and number of generations are weighted with 25 % each, habitat requirements consist of six different aspects, which are weighted with 25/6 % (about 4,17 %) each, to add up to 25 %. A taxon gets weight for a certain trait if this trait agrees with the characteristic pioneer traits. If there is no data for a trait, no weight is added, except for habitat specialism, where no specialism is assumed for taxa without data. The sum of these weights is given in the last column and indicates the pioneer potential of the respective taxon.

An example: *Baetis muticus* is given weight for a flight period which coincides with the sampling period, for a small body size, for bivoltinism and for unspecific requirements concerning altitude, temperature, saprobity, longitudinal zonation, and current. Exclusively its habitat specialism disagrees with typical pioneer traits and thus, the sum of weights is about 96 %.

Winged taxa found in channels	flight period	body size ¹	No. of gen.	altitude pref.	therm. pref.	saprobity	longitud. dist.	habitat specialist	current pref.	potential pioneer [%]
<i>Amphinemura</i> sp.	25							4,17		29
<i>Baetis alpinus</i>	25		25	4,17				4,17		58
<i>Baetis muticus</i>	25	25	25	4,17	4,17		4,17		4,17	92
<i>Baetis rhodani</i>	25		25	4,17	4,17		4,17	4,17		67
<i>Diamesa cinerella</i>	25	25	25	4,17	4,17	4,17		4,17		92
<i>Diamesa hamaticornis</i>	25	25	25	4,17	4,17	4,17		4,17		92
<i>Diamesa insignipes</i>	25	25	25	4,17	4,17			4,17		88
<i>Diamesa tonsa</i>	25	25	25	4,17	4,17	4,17		4,17		92
<i>Ecclisopteryx guttulata</i>										0
<i>Ecdyonurus</i> sp.	25					4,17	4,17	4,17		38
<i>Elmis</i> sp.	25	25					4,17	4,17		58
<i>Hydropsyche</i> sp.	25					4,17	4,17	4,17		38
<i>Lepidostoma hirtum</i>	25			4,17	4,17				4,17	38
<i>Leuctra</i> sp.	25						4,17	4,17		33
<i>Micrasema setiferum</i>		25						4,17		29
<i>Micropsectra atrofasciata</i>	25	25	25	4,17	4,17	4,17	4,17	4,17		96
<i>Nemoura / Nemurella</i> sp.	25						4,17	4,17		33
<i>Isoperla</i> sp.	25						4,17	4,17		33
<i>Rhithrogena</i> sp.	25						4,17	4,17		33
<i>Riolus</i> sp.	25	25					4,17	4,17		58
<i>Siphonurus lacustris</i>				4,17	4,17		4,17	4,17		17
Winged taxa very abundant in Seebach but not found in channels										
<i>Habroleptoides confusa</i>				4,17	4,17		4,17	4,17	4,17	21
<i>Micrasema minimum</i>	25	25								50
<i>Orectochilus villosus</i>								4,17		4
<i>Polycentropus</i> sp.	25					4,17	4,17	4,17	4,17	42

If a certain trait of a taxon matches the typical traits of pioneers listed above, the respective field in Table 10 is marked with orange color. A trait was defined to match in the following cases: Small body size, a number of generations higher than one per year, occurrence in all three altitude classes from below 200 m to over 800 m, no thermal preference, saprobity tolerance ranges over more than two saprobic classes, longitudinal distribution ranges over more than three zones, no habitat specialization, current preference not restricted just to limnophilous or rheophilous. The probability for each taxon to be a pioneer in the channels was then calculated as shown in Table 11.

Table 10 shows that all taxa which were pioneers in the experimental channels, except for *Nemoura/Nemurella* sp., are characterized by small body size and have more than one generation per year. Moreover, they mostly appear to be euryoecious in terms of altitude- and thermal preference, saprobic condition and longitudinal distribution, show no distinct habitat specialism and their potential flight period coincides with the sampling period. In Table 11, traits of all pioneers except for *Nemoura/Nemurella* sp. are weighted with 88 to 96 %, which is not the case for any other taxa in the table. This indicates a high relevance of the traits stated by TOWNSEND & HILDREW (1994) for primary colonization of the experimental channels. Percentages might have been different, if data had been available for all traits and all taxa. However, the effect of this lack of data can be assumed to be little, since data was constantly available for all three traits weighted with 25 % and other traits have less weight.

For some taxa there was a lack of data for certain traits that form the habitat requirements. This is due to a lack of knowledge and to a lack of identification to the species level. However, this incompleteness has no influence on the statement which taxa have more pioneer traits and which have less: Longitudinal zonation and saprobic preference show no difference between pioneers and non-pioneers. Thermal- and current preference are lacking data but existing data does not indicate a difference between pioneers and non-pioneers. Only the existing data for habitat specialization and altitude preference differ between the taxa but they account for less than 9 % of the weight and hence cannot change percentages significantly.

It is pointed out that several taxa in these tables, which were rare in the channels, have unspecific habitat requirements and therefore small body size and polyvoltinism appear to be the most characteristic features of pioneers in this study. These traits are part of the ecological strategy described by the term “r-strategy” (PIANKA, 1970). Ecological r-strategists are characterized by high and early reproduction, quick development, short

life time, low body weight and low competitiveness, which is typical for pioneers. In contrast, K-strategists are characterized by low and later reproduction, a tendency to slower development, long life time, relatively high body weight and high competitiveness. Distinction between r- and K-strategists is relative, since every species has a position in an r-K-continuum (SCHAEFER, 2003). Nevertheless, the proportion of r-strategists to K-strategists is used as a metric for river status assessment, for instance in Germany (SCHÖLL et al., 2005).

Besides small body size and polyvoltinism, there were other factors which potentially supported the dominance of *Diamesa* spp. Firstly, temperature regime in the channels was quite stable between 2.5 and 6.9°C from November until April, which is comparable to a krenal brook (JUNGWIRTH et al., 2003). This region is a typical habitat for *Diamesa* spp. of the groups *cinerella* and *zernyi* (JANECEK et al., 2002). Secondly, *Hydrurus* algae covered a large part of the gravel in the channels from December until April. *Hydrurus* was probably promoted by the temperature regime in the channels, since this alga prefers temperatures between 0.2 and 13.3°C (BURSA, 1934; KAWECKA, 1981) and by lacking riparian vegetation that would shade the channels. *Hydrurus* density was shown by MOOG & JANECEK (1991) to correlate with Macrozoobenthic biomass and abundance, which reinforces the importance of *Hydrurus* as a habitat and potential food resource for *Baetis* spp. and Diamesini. In this study, however, *Hydrurus* coverage and total amount of *Hydrurus* did not correlate with invertebrate biomass or abundance. Nevertheless, the presence of *Hydrurus*, widely independent of its amount, may have supported *Diamesa* and *Baetis* species, on one hand through food supply, and on the other hand as a habitat for smaller stages in the winter, when *Hydrurus* layers were thicker. Other species, which feed on leaf litter for instance, probably had no chance to grow due to a lack of food.

There are further possible theories for the absence or rareness of the taxa which did not or only singularly colonize the channels. Some of them do not emerge between September and May, but others potentially do. Their absence might be due to unsuitable habitat features, i.e. disturbances in the channels, or their actual flight period was not identical to the one derived from the literature.

7.2 Sources of colonists

In a study with four different traps (size 30 x 60 cm) WILLIAMS & HYNES (1976) tried to find out, what percentage of total colonization was contributed by each of the different

sources. In the drift trap they caught an average number of 14 074 individuals in 28 days. In the upstream migration trap the average number was at 6 195 individuals, in the vertical migration trap at 6 505 individuals and in the aerial colonization trap at 9 577 individuals in 28 days. Thus, colonization by drift was the most considerable mechanism, followed by aerial colonization. Nevertheless, upstream and vertical migration contributed a notable part to total colonization.

The importance of **drift** for colonization is also emphasized by SAGAR (1983) and BILTON et al. (2001). In the HyTEC experimental channels, however, drift does not play an important role for colonization, since the channels are supplied directly with lake water. A few specimens of *Heterotrissocladius marcidus* and *Hydra* sp. as well as *Siphonurus lacustris*, taxa known to populate stagnant waters (MOOG (ed.), 2002; JANECEK et al., 2002 and BAUERNFEIND et al., 2002) most probably arrived via the pipelines. This may also be the case for some part of Oligochaeta, a group that has not been identified to a higher taxonomic level.

Aerial colonization is quantitatively probably the most important source of colonists in the experimental channels. Chironomids and *Baetis muticus* most probably colonized by oviposition, since they show all the traits for aerial colonization as described in Table 9 and Table 10. Moreover, *Diamesa* spp. and Orthocladiinae were observed flying in swarms over the channels in February 2012 and one adult specimen of *Diamesa* sp. appeared in the box-samples of the 10th April (verified by B. Janecek). For *Baetis rhodani* and *B. alpinus* it is not clear whether they colonized via upstream migration or oviposition. Their physiology (BAUERNFEIND & HUMPESCH, 2001) and low abundances support the former theory, the finding of two adult specimens of *B. rhodani/alpinus* (verified by T. Huber) in the samples from the end of November supports the latter.

Upstream migration into the channels from the Seebach is handicapped by the wooden ramps at the ends of the channels and therefore not comparable to the upstream traps of WILLIAMS & HYNES, however, it is not impossible. Nemourids, *Baetis rhodani* and *B. alpinus*, *Ecdyonurus* sp. and *Rhithrogena* sp., among others, might have arrived in the channels via this path, taking into account their main form of mobility (see NILSSON (ed.), 1996 and BAUERNFEIND & HUMPESCH, 2001). Interestingly, *Nemoura/Nemurella* sp. and *Baetis rhodani* were more abundant in the samples from the more disturbed channel. This may simply be due to the fact that this channel is closer to the Seebach. However, it is also supposable that the pipe culvert which conducts the discharge from the left channel beneath the pathway is difficult to pass for these benthic organisms.

Very low abundances of *B. rhodani* in the channels compared to the Seebach relativize the apparent difference between the channels, but abundances of *Nemoura/Nemurella* sp. in the channels were in the same range as those in the Seebach.

Recolonization of the streambed **from within the substrate** as described by WILLIAMS & HYNES (1976) is not possible for the channels, since they are no natural gravel beds. Nevertheless, some benthic invertebrates most probably have survived in the gravel from the river Ybbs that was introduced into the experimental channels and were therefore found in the samples. The gravel added in the end of October was stored only for two days before introduction into the channels. In autumn with low air temperatures and fog it is possible that it did not dry completely and larval or egg stages survived. This is very likely to be the case for *Ecclisopteryx guttulata*, which did not occur in the samples taken from the Seebach and first appeared in the samples from the channels on the 17th November, about two weeks after the introduction of new gravel. It could also be an explanation for the sudden appearance of *Rhithrogena* sp. in the samples of the 17th November and its steady decline in abundance. That theory is reinforced by the peak in individual biomass (calculated without Gastropods and *Eiseniella tetraedra*) on the 17th November (see Figure 23), which indicates the occurrence of larger invertebrates for a short time span. Moreover, eggs of *Isoperla* (identified by W. Graf) and several dried out invertebrates appeared in the samples from that date. Oligochaeta probably also colonized via this path, if not by drift from the lake, since the ones found in the channels (except for *Eiseniella tetraedra*) were small and due to their physiology not likely to achieve far upstream movements.

TRONSTAD et al. (2005) indicate high relevance of survival of drought in a dormant or active state for the recolonization of a southeastern U.S. floodplain. Over 45 000 invertebrate individuals per m² recovered after rehydration of floodplain soil, only 5 % of which were terrestrial taxa. After six weeks of rehydration, density decreased at high floodplain elevation but increased to 110 000 individuals per m² at low elevation sites.

In contrast, CHESTER & ROBSON (2011) suggest that resilience traits, expressed through recolonization after drought by individuals that lived in perennial reaches or pools, are constitutive for the species composition and abundance of the recolonizing community. Resistance traits, the survival of drought on site, seemed to have less quantitative importance for recolonization, but anyway contribute to species diversity.

WILLIAMS & HYNES (1976) concluded that the composition of species as well as the number of specimens in a habitat, where not all four ways of colonization are possible, differs from an equal habitat without restrictions in colonization mechanisms. For example in the upstream migration trap there were about 50 % more non-Chironomids than Chironomidae. In the vertical movement trap it was the other way around. In the drift trap, abundance of Chironomidae exceeded the abundance of all other taxa together by the factor four and in the aerial colonization trap by the factor 90. These findings support the conclusion that Chironomids dominated aerial colonization and since the other three paths of colonization are limited this family dominated the benthic fauna in the experimental channels in Lunz am See.

7.3 Time scale of colonization

Statements in the literature about the time scale of colonization vary, depending on the study design or the type of disturbance and on the characteristics of the investigated river. Communities on small patches of disturbed sediment or artificial sediments recover faster than larger areas of riverbed disarranged by a spate (MATTHAEI et al., 1996; ELSER, 1999).

OFENBÖCK & MOOG (2000) tested the so-called Danube-Net-Basket-Sampler and found the number of taxa, biomass and abundance in the exposed substrate to be similar to the surrounding natural benthos after three weeks. FOWLER (2002) used two types of colonization baskets, one that could exclusively be colonized via the surface and another that could only be invaded via the hyporheic zone. Species diversity in the baskets compared well to that in the surrounding sediments within the first seven days of colonization. However, density and community composition were comparable to the original sediments only after 28 days in the surface colonization baskets and in the non-restricted control baskets and after 42 days in the hyporheic colonization baskets. Consequently, complete colonization could be achieved on the long term even though sources were restricted.

In contrast to such experiments with basket-samplers or patch disturbance, recovery times after natural spates seem to be much longer. Invertebrate density and the total number of species recovered within 132 days after a major flood in a braided river (SCRIMGEOUR et al., 1988). However, community composition fluctuated over the whole period. This seems quite logical, since recolonizing invertebrates were very probably coming from different sources, depending on the species, and this also varied

by time. For instance, high numbers of early-instar larvae of the mayfly *Deleatidium* indicated intense recolonization by egg-hatching in deeper sediment layers or by oviposition. Nevertheless, drift or active movement from undisturbed braids also seemed to play an important role for recovery.

A common event disturbing benthic communities in streams and rivers is the input of pesticides. Documented recovery times after such incidents range from less than one month to more than 25 years. This depends on various factors like amount, toxicity and persistence of the introduced substance, duration and spatial scale of the disturbance as well as position in the watershed and the life cycles of populations affected by the pollution (WALLACE, 1990).

Recovery time spans after in-stream habitat enhancement ranged from 1 to 50 years with a median of 2.5 years in a review by FELD et al. (2011). Long time spans are also required for recolonization after removal of weirs, since fine sediments that had accumulated in the impoundment are then deposited downstream. Thus, biological recovery is expected to occur after several years or decades, when these fine sediments have been removed further downstream (FELD et al., 2011).

SAGAR (1983) investigated invertebrate recolonization of previously dry channels of a braided river with sporadic desiccation of some channels in New Zealand. In winter, complete colonization took 33 days compared to 15 days in summer. According to WALLACE (1990), invertebrate recolonization of previously dry sediment appears to proceed in less than two months if drought is happening periodically. In contrast, benthic community in normally permanent streams is less adapted to drought and hence, recolonization can take more than one year.

The present study investigated colonization of two whole channels instead of small patches or sampling baskets within a natural stream. Therefore, and because three of the four sources of colonization are limited, a longer time scale for colonization must be expected. It is likely to take at least one year before all potential colonists arrive and much longer until species composition is stable, not taking into account the various disturbances in the channels.

For the sampling period from the 9th of September 2011 until the 10th of April 2012, colonization can be divided into three phases. The first phase lasts from the beginning until the end of November and is characterized by first attempts to colonize the channels. This is expressed by low abundances, appearance of single specimens of

various taxa and strongly fluctuating individual biomass (see Figure 16, Figure 22 and Figure 25). It is followed by a phase of growth in abundance and individual biomass from December until March, which indicates that some species had colonized the channels successfully. However, this phase is still accompanied by colonization attempts of new species. Finally, this phase is interrupted by a decline in individual biomass in April, most probably resulting from Chironomid emergence. This hypothesis is supported by the observation of adult specimens of *Diamesa* spp. and by a decline in abundance of Chironomid pupae in spite of increasing Chironomid larvae abundance (Figure 34 and Figure 35).

7.4 Seasonal aspects

The composition of species colonizing a habitat varies with season and the seasonal development of colonization is determined by the climate. Whereas in subtropical areas some insects are emerging and dispersing during the whole year, this can be restricted to some months for many species in colder climate zones and therefore limiting colonization in the cold season (TRONSTAD et al., 2007). Moreover, there is an indirect influence of season on colonization. SAGAR (1983) found recolonization rates depending strongly on flow fluctuations, being highest during spates. This is fleshed out by MATTHAEI et al. (1996, 1997), who regarded drift to be the most important source of recolonization in summer, the flood season in Switzerland, but not in winter.

Hence, primary colonization might have occurred in a fundamentally different pattern if the experimental channels had been started up in spring instead of late summer. Since a good part of species occurring in the samples from the Seebach start their flight periods in April or May, dominance of *Diamesa* spp. might be a result of the seasonal aspects described above.

7.5 Colonization and disturbance

Since disturbance is an inherent factor in stream ecosystems (TOWNSEND, 1989), it cannot just be seen as a limiting factor for invertebrate community. In fact, disturbance also has to be regarded as a chance for less competitive species to recolonize and thus persist in the system. Hence, moderate intensity and frequency of disturbance contribute to diversity, whereas undisturbed and extremely disturbed habitats are inhabited by less diverse communities. This is known as the “intermediate disturbance hypothesis” (TOWNSEND et al., 1997).

In this study, dominance conditions in the experimental channels suggest high disturbance, although this is not likely to be the only reason for low diversity. It was not possible to test the impact of particular disturbances on colonization, as both channels were disturbed by various means. Apart from Oligochaeta and *Nemoura/Nemurella* sp., there was no clear difference between communities in the channels which would indicate an effect of the difference between the two hydrologic regimes. Thus, influence of the hydrologic regime should be tested with a hydrologically undisturbed reference channel.

Nevertheless, the constantly higher abundances of Oligochaetes in the less disturbed channel indicate an influence of pulse-release frequency on worms. The most plausible theory is, that Oligochaetes (except for *Eiseniella tetraedra*, which did also occur in the Seebach) entered the channels via the pipelines or the introduced gravel and are not adapted to strong current and were therefore more often dislodged by the current in the more disturbed channel.

Nemoura/Nemurella also showed a clear difference in abundance between the channels, however, this taxon was more abundant in the more disturbed channel. This is not likely to be an effect of the hydrologic regime, since base flow was similar in both channels and there is no sign of stronger competition in the less disturbed channel, which would reduce abundance of Nemourids.

MATTHAEI et al. (1996) compared recolonization by benthic invertebrates between two sites of a Swiss prealpine river. One of the sites was more frequently disturbed by natural spates than the other. At the less disturbed site, total invertebrate densities were more than twice as high as those at the other site. In contrast, invertebrate density did not differ between the two experimental channels in the present study.

MATTHAEI et al. (1996) further found that benthic invertebrate density at the more frequently disturbed site recovered faster after experimental disturbance by kicking and stirring than at the less disturbed site. Hence, they concluded that benthic community at the more disturbed site was better adapted to recolonize after spate.

The dominance of *Diamesa* spp. can not only be seen as a result of source limitation, but also be interpreted as an effect of discharge fluctuations. The abundance of Chironomids was shown to increase in reaches affected by pulse-release in four Austrian rivers (MOOG, 1993). Moreover, *Diamesa* spp. are known to populate glacial streams very

close to the glacier snout, where discharge is characterized by naturally strong fluctuations (LODS-CROZET et al., 2001).

The question, whether source limitation or disturbance is the main reason for this noticeable community structure, might be answered through further data collection in the channels throughout the next few years. Nevertheless, it can be assumed that continuous pressures probably have a negative influence on colonization. Such pressures can be discharge fluctuations and mechanical sediment disturbance, but also the input of fine sediments, pesticides and fertilizers due to land use practice, which constantly disturb populations (JÄHNIG et al., 2010; FELD et al., 2011).

8 Summary and Conclusions

This study documented primary colonization of two newly created experimental channels, which essentially differed from natural streams in the following aspects:

- They were subject to strong discharge fluctuations, one of them more frequently and more intensely than the other.
- The bottom of the channels is sealed with a bentonite blinding and the channels are filled with gravel extracted from a nearby river.
- They have no upstream connection to a body of flowing water, but are supplied by pipelines from a lake.
- Downstream connection to a semi-natural stream is hindered by wooden dam- and ramp constructions.
- There is no riparian vegetation along the channels.
- They were frequently disturbed by construction works and electro-fishing.

Monthly sampling with a box-sampler in both channels from the start-up on the 9th September 2011 until the 10th April 2012 showed a high similarity between benthic communities in the channels. However, there were big differences to the community in the Unterer Lunzer Seebach, a semi-natural stream which parallels the channels. Three phases of succession could be observed: First colonization attempts from September until November 2011, a phase of growth from December until March/April 2012 and, referring to Chironomids, a phase of emergence that started in February but was explicit in the end of March. Few taxa occurred constantly in the samples taken in the channels and Chironomids from the genus *Diamesa* were the most important pioneers.

Five hypotheses, phrased in chapter 2, were investigated in the course of this study. In the following, they are verified or falsified.

The hypothesis, that pioneers show specific traits, which distinguish them from other species that colonize later, can be verified. The traits small body size and polyvoltinism appeared to be the ones distinguishing pioneers from other taxa that did not or rarely colonize. Other traits did not show differences between pioneers and non-pioneers or they are unknown for part of the taxa. Taxa that were abundant in the channels, showed growth of biomass, occurred constantly and completed their life cycle in the channels (*Diamesa cinerella/tonsa*, *D. hamaticornis*, *D. insignipes*), appeared to have a high accordance with typical traits of pioneers. Moreover, *Diamesa* spp. were probably

supported by the temperature regime in the channels, which was comparable to a krenal brook, and by extensive hydrurus coverage in December and January.

It can also be verified, that aerial colonization is quantitatively the main source for colonization of the experimental channels, since other sources are restricted. Flight periods found in the literature of all pioneers coincided to some extent with the sampling period. Moreover, polyvoltinism and small body size, i.e. low body weight, are characteristic for r-strategists, which are described further by high reproduction, quick development, short life time and low competitiveness. Hence, it is quite plausible that the first pioneers in the experimental channels are r-strategists and colonized via position of large quantities of eggs which developed very quickly into larvae and in some cases to pupae and adults. There were only few indications for upstream migration, drift and colonization via the introduced substrate. Consequently, colonization of the channels is limited, which is another possible reason for dominance of a few species of the Chironomid genus *Diamesa*. Aerial colonization alone can be quantitatively very effective, even during the cold season, but species diversity needs other migratory pathways too.

The third hypothesis, that it will take a longer time to achieve an equilibrium benthic community composition in the channels than in other habitats, where colonization is less restricted, cannot be verified. There was by far no indication for a relatively stable community composition in the channels after the study period of seven months. However, colonization times of habitats in streams and rivers found in the literature ranged from 15 days (SAGAR, 1983) to 50 years (FELD et al., 2011), depending on spatial scale, season, study design or type of disturbance and other factors.

It is verified, that primary colonization of the experimental channels was restricted by the start-up in late summer. Several taxa, which showed high abundance in the reference samples from the Seebach, generally emerge between April or May and September and thus, they could not colonize the channels aurally during the study period.

The fifth hypothesis was that benthic colonization was influenced by discharge fluctuations and thus, colonization patterns differed between the two channels, since they were subject to pulse-releases of different frequency and intensity. This assumption can only be verified partly. Oligochaeta were more abundant in the less disturbed channel on all sampling dates, which indicates an influence of the hydrologic

regime. Nevertheless, total abundance and biomass showed a similar development in both channels. Moreover, no plausible relation between the hydrologic regime and differences between the channels in abundance of *Nemoura*/*Nemurella* could be found. Hence, the hypothesis is falsified concerning an overall effect, but verified concerning Oligochaeta. To investigate the effect of discharge fluctuations on colonization, a hydrologically undisturbed reference channel would be an important methodological pre-requisite.

In summary, benthic community in the experimental channels fundamentally differed from the one in a close near-natural stream throughout the sampling period. On the contrary, there were only minor differences between benthic communities in the two channels. Few taxa occurred constantly in the samples taken in the channels and Chironomids from the genus *Diamesa* were the most important pioneers. Pioneer taxa in the channels showed characteristic traits, of which small body size and polyvoltinism appeared to be the ones distinguishing them from other species that did not or rarely colonize. Moreover, flight periods found in the literature of all pioneers coincided to some extent with the sampling period. Thus, oviposition most probably was the main source for colonization of the experimental channels. There were only few indications for upstream migration, drift and colonization via the introduced substrate. Consequently, colonization of the channels is limited, which is a possible reason for dominance of a few species of the Chironomid genus *Diamesa*. An alternative explanation is the high disturbance frequency in the channels, which prevents competition between species and therefore supports r-strategists. Hence, both source limitation and disturbance most probably determined community composition in the experimental channels.

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All photos, if not declared otherwise, were taken by the author.

Appendix

Legend: l1 – l6 = box-samples of the left (less disturbed) channel
 r1 – r6 = box-samples of the right (more disturbed) channel
 s1 – s6 = box-samples of the Seebach

Abundance 28-09-11

Sample	l1	l2	l3	l4	l5	l6	Σ l	r1	r2	r3	r4	r5	r6	Σ r
Allogamus auricollis*	1						1							0
Asellus aquaticus			1				1							0
Baetis alpinus	2	1					3							0
Baetis muticus							0	1						1
Chironomidae	1	1	1				3				1			1
Hydropsyche sp.					1		1				1			1
Liponeura sp.							0	1						1
Nemoura/Nemurella sp.							0				1			1
Oligochaeta	1		2				3		1				1	2
Riolus sp.							0	1						1
Zooplankton aprox.	3		2	4			9	27	28	74	120	80	30	359
Notes	Hydropsyche in l5 corrupt, *)Allogamus auricollis dried out													

Biomass [mg] 28-09-11

Sample	l1	l2	l3	l4	l5	l6	Σ l	r1	r2	r3	r4	r5	r6	Σ r
Asellus aquaticus			2,8				2,8							0
Baetis spp.	0,6	0,5					1,1	0,05						0,05
Chironomidae+Liponeura	0,3	0,05	0,05				0,4				0,05			0,05
Hydropsyche sp.					0,05		0,05				0,05			0,05
Nemoura/Nemurella sp.							0				0,05			0,05
Oligochaeta	0,3		0,5				0,8		0,6				0,3	0,9
Riolus sp.							0	0,05						0,05
Sum	0,9	0,55	2,85	0	0,05	0	4,35	0,1	0	0	0,15	0	0	0,25

Habitat features 28-09-11

Sample	l1	l2	l3	l4	l5	l6	r1	r2	r3	r4	r5	r6
Depth [cm]	18	14	12	5	14	6	16	15	12	7	7	9
Current velocity [m/s]	0,15	0,1	0,35	0,5	0,5	0,4	0,15	0,3	0,35	0,5	0,8	0,5
Substrate	akal / micro	micro / meso	meso / macro	meso	micro / macro	meso	akal / micro	akal / micro	micro / meso	meso	meso	micro / meso
Micro algae [%]	20	5	3	3	3	3	1	1	1	3	1	1
Filamentous algae [%]	0	0	0	0	0	0	0	0	0	0	0	0
Moss [%]	0	0	0	0	0	0	0	0	0	0	0	0
Sum of aufwuchs [%]	20	5	3	3	3	3	1	1	1	3	1	1
Hydrurus [%]	0	0	0	0	0	0	0	0	0	0	0	0
Hydrurus thickness	0	0	0	0	0	0	0	0	0	0	0	0
Hydrurus total	0	0	0	0	0	0	0	0	0	0	0	0
electric conductivity [µS/cm]	240						240					
Water temperature [°C]	14,2						13,8					
Oxygen [mg/l]	9,93					10,22	9,93					10,22
Oxygen [%]	103					105	103					105

Abundance Seebach 28-09-11

Sample	s3	s4	s5
Baetis muticus	29	37	13
Baetis rhodani	8	2	0
Baetis alpinus	18	37	1
Ecdyonurus sp.	10	24	2
Habroleptoides confusa	20	2	21
Ephemera danica	0	0	1
Rhithrogena sp.	4	16	0
Ephemerella major	1	3	0
Epeorus sp.	0	13	4
Amphinemura sp.	32	60	24
Nemoura sp.	1	8	0
Isoperla sp.	4	0	0
Leuctra sp.	0	0	4
Hydropsyche sp.	48	42	12
Hydroptila sp.	0	0	4
Polycentropus	12	7	11
Micrasema setiferum	722	172	13
Lepidostoma hirtum	10	49	8
Micrasema minimum	2	0	8
Rhyacophila dorsales pupae	0	1	1
Rhyacophila sensu strictu	1	0	3
Rhyacophila tristis	0	2	0
Glossosomatidae	45	17	24
Orectochilus villosus	5	22	1
Elmis sp. Larvae	74	28	10
Elmis adult	0	1	0
Platambus maculatus larvae	1	0	0
Scirtidae larvae	29	57	12
Riolus sp. Larvae	182	70	134
Riolus sp. adult	30	8	12
Simuliidae	8	2	4
Chironomidae	136	71	63
Diptera Pupae	13	0	0
Athericidae - Ibis marginata	1	0	2
Brachycera	1	0	0
Empididae	9	0	9
Limoniidae - Antocha sp.	1	0	0
Oligochaeta	30	12	39
Ancylus fluviatilis	5	13	6
Gyraulus sp.	6	1	6
Hydrobiidae	20	4	13
Sum	1518	781	465

Biomass [mg] Seebach 28-09-11

Sample	s3	s4	s5
Coleoptera	180,9	33,8	35,8
Chironomidae	12,9	10,8	10,6
Ephemeroptera	48,6	42,9	31
Gastropoda	76,4	41,4	42,1
Oligochaeta	337,8	199,8	42,2
Plecoptera	5,2	27,2	5,2
Trichoptera	564,8	386,1	182,9
Rest	35	134,9	180,4
Sum	1261,6	876,9	530,2

Habitat features Seebach 28-09-11

Sample	s1	s2	s3	s4	s5	s6
Depth [cm]	18	18	14	18	20	14
Current velocity [m/s]	0,8	0,8	0,35	0,5	0,35	0,3
Substrate	micro / meso	meso	meso / macro	micro	micro / meso	meso / macro
Micro algae [%]	10	90	85	90	90	50
Filamentous algae [%]	30	0	10	5	5	0
Moss [%]	60	10	5	5	5	50
Sum of aufwuchs [%]	100	100	100	100	100	100
electric conductivity [μ S/cm]	241					
Water temperature [°C]	16,3					
Oxygen [mg/l]	8,8					
Oxygen [%]	95					

Chironomidae abundance 28-09-11

[illegible]

MHS Seebach 28-09-11

[illegible]

Proportions of habitat-types for the MHS of the Seebach, 28-09-11.
Scheme by MOOG (2004).

Abundance 13-10-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Baetis muticus	2						2							0
Chironomidae	7	7	4	2	3		23	1			5			6
Gastropoda - Radix sp.				1			1							0
Micrasema setiferum						1	1							0
Nemoura/Nemurella sp.			2			1	3							0
Oligochaeta	3	4	1		2		10							0
Zooplankton	100	10	39	17	40	35	241	10	8	16	110	120	10	274

Biomass [mg] 13-10-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Baetis muticus	0,2						0,2							0
Chironomidae	0,3	1,1	2,2	0,7	0,3		4,6	0,05			1,3			1,35
Gastropoda - Radix sp.				54,6			54,6							0
Micrasema setiferum						0,05	0,05							0
Nemoura/Nemurella sp.			0,05			0,1	0,15							0
Oligochaeta	0,05	0,05	0,05		0,05		0,2							0
Sum	0,55	1,15	2,3	55,3	0,35	0,15	59,8	0,05	0	0	1,3	0	0	1,35

Habitat features 13-10-11

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	16	12	15	10	14	8	15	17	14	11	10	10
Current velocity [m/s]												
Substrate	micro/ meso	micro/ meso	micro/ meso	meso	micro/ meso	macro	micro/ meso	micro	micro/ meso	meso	micro/ meso	meso/ macro
Micro algae [%]	5	5	5	3	10	5	3	3	5	8	3	3
Filamentous algae [%]	0	8	3	0	0	0	0	0	0	0	0	0
Moss [%]	0	0	0	0	0	0	0	0	0	0	0	0
Sum of aufwuchs [%]	5	13	8	3	10	5	3	3	5	8	3	3
Hydrurus [%]	0	3	0	0	0	0	0	0	0	0	0	0
Hydrurus thickness	0	1	0	0	0	0	0	0	0	0	0	0
Hydrurus total	0	3	0	0	0	0	0	0	0	0	0	0
electric conductivity [µS/cm]	244						244					
Water temperature [°C]	10,3						10,3					
Oxygen [mg/l]												
Oxygen [%]	95						95					

Abundance 25-10-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Baetis muticus		1					1							0
Chironomidae	8	5	6		3	3	25	4	3	1	3	8	5	24
Eiseniella tetraedra							0						1	1
Hydra sp.							0	1			1			2
Limnius sp.							0		1					1
Micrasema setiferum							0				1			1
Nemoura/Nemurella sp.			3				3							0
Oligochaeta	5	13	8	1	3		30							0
Valvata piscinalis							0	1						1
Zooplankton	110	45	105	26	70	65	421	15	13	45	65	210	56	404
Notes	Chironomidae adult in r5													

Biomass [mg] 25-10-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Baetis spp.		0,1					0,1							0
Chironomidae	4	2,4	4		1,5	0,4	12,3	2,3	0,3	0,3	0,1	0,4	0,6	4
Eiseniella tetraedra							0						20,8	20,8
Hydra sp.							0	0,1			0,1			0,2
Limnius sp.							0		1,5					1,5
Micrasema setiferum							0				0,1			0,1
Nemoura/Nemurella sp.			1				1							0
Oligochaeta	3	2,9	1,4	0,05	1		8,35							0
Valvata piscinalis							0	19,5						19,5
Sum	7	5,4	6,4	0,05	2,5	0,4	21,75	21,9	1,8	0,3	0,3	0,4	21,4	46,1

Habitat features 25-10-11

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	18	12	13	4	12	5	12	12	11	8	7	8
Current velocity [m/s]	0,15	0,1	0,35	0,5	0,5	0,4	0,15	0,3	0,35	0,5	0,8	0,5
Substrate	micro / meso	micro / meso	micro / macro	micro / meso	akal / meso	meso / macro	akal / micro	akal / micro	micro / meso	micro / meso	micro / meso	akal / micro
Micro algae [%]	0	30	15	15	10	15	20	10	10	5	3	3
Filamentous algae [%]	95	10	10	0	0	0	0	0	0	0	0	0
Moss [%]	0	0	0	0	0	0	0	0	0	0	0	0
Sum of aufwuchs [%]	95	40	25	15	10	15	20	10	10	5	3	3
Hydrurus [%]	0	0	0	0	0	0	0	0	0	0	0	0
Hydrurus thickness	0	0	0	0	0	0	0	0	0	0	0	0
Hydrurus total	0	0	0	0	0	0	0	0	0	0	0	0
electric conductivity [μS/cm]	240						240					
Water temperature [°C]	8,9						8,9					
Oxygen [mg/l]	9,9					10	9,6					10
Oxygen [%]	93					94	91					94

Abundance 17-11-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.					1		1	1						1
Ancyclus fluviatilis							0			1		1		2
Baetis alpinus							0	1						1
Baetis muticus						2	2				1			1
Baetis rhodani							0			1				1
Chironomidae	1	1			5	1	8	1		1	1	3	5	11
Chloroperlidae*					1		1							0
Ecclisopteryx guttulata			1	1	3		5	1		3	1			5
Elmidae - Elmis sp.					1		1							0
Elmidae - Esolus sp.					2		2							0
Lepidostoma hirtum					1		1							0
Leuctra sp.*					1		1			2				2
Oligochaeta	1	1	1		3	4	10					2	1	3
Rhithrogena sp.				1	2		3			4	3			7
Sericostomatidae*							0					1		1
Taeniopteryx sp.*					1		1							0
Tipulidae							0					1		1
Valvata cristata					1		1							0
Zooplankton				2	4	4	10				2	3		5
Notes	lots of empty caddis; Chironomidae adult in I4; Isoperla eggs in I5; Sample taken 2,5 weeks after deposition of new gravel; *) marked taxa were dried out													

Biomass [mg] 17-11-11

Sample	u1	u2	u3	u4	u5	u6	Σ u	b1	b2	b3	b4	b5	b6	Σ b
Amphinemura sp.					0,05		0,05	2,6						2,6
Ancylus fluviatilis							0			6,7		3,8		10,5
Baetis spp.						0,1	0,1	0,2		1,3	1,2			2,7
Chironomidae	0,9	0,7			0,5	0,05	2,15	0,05		0,1	0,1	0,1	0,8	1,15
Ecclisopteryx guttulata			3,1	0,5	4,5		8,1	0,6		3,4	1,3			5,3
Elmidae - Elmis					0,3		0,3							0
Elmidae - Esolus					0,5		0,5							0
Lepidostoma hirtum					11,9		11,9							0
Oligochaeta	5,9	0,1		0,1	0,6	1,4	8,1					0,9	1	1,9
Rhithrogena sp.				0,9	2,1		3			2	2,7			4,7
Tipulidae							0					26,7		26,7
Valvata cristata					1,1		1,1							0
Sum	6,8	0,8	3,1	1,5	21,55	1,55	35,3	3,45	0	13,5	5,3	31,5	1,8	55,55

Habitat features 17-11-11

Sample	l1	l2	l3	l4	l5	l6	r1	r2	r3	r4	r5	r6
Depth [cm]	3	6	6	6	9	10	6	7	8	5	7	10
Current velocity [m/s]	0,2	0,25	0,3	0,4	0,3	0,3	0,15	0,2	0,5	0,4	0,3	0,2
Substrate	micro/meso	micro/macro	micro/meso	akal/micro	akal/micro	akal/micro	akal/micro	micro/meso	akal/micro	akal/micro	micro/meso	micro/meso
Micro algae [%]	5	5	10	5	3	10	5	10	3	3	5	20
Filamentous algae [%]	3											
Moss [%]												
Sum of vegetation cover [%]	8	5	10	5	3	10	5	10	3	3	5	20
Hydrurus [%]	0	0	0	0	0	0	0	0	0	0	0	0
Hydrurus thickness	0	0	0	0	0	0	0	0	0	0	0	0
Hydrurus total	0	0	0	0	0	0	0	0	0	0	0	0
electric conductivity [μS/cm]	no valid data											
Water temperature [°C]	6,9						6,8					

Abundance 30-11-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.					1		1							0
Asellus aquaticus							0				1			1
Baetis muticus						6	6			1		2		3
Baetis rhodani							0			1		1		2
Bithynia tentaculata		1					1							0
Chironomidae	1	1				11	13	1	2	5	1	7	4	20
Diptera: Empididae						1	1							0
Ecclisopteryx guttulata							0		1	1				2
Elmidae - Elmis sp.						2	2			1				1
Elmis sp. Adult							0			1				1
Hydra sp.			1				1							0
Leuctra sp.							0	2						2
Nemoura/Nemurella sp.							0						3	3
Oligochaeta	1	1	1			4	7			2			2	4
Rhithrogena sp.							0		1	2	1			4
Zooplankton		1	4			10	15	3	2	2		5	3	15
Notes	Baetis rhodani/alpinus adult in I1 and I4													

Biomass [mg] 30-11-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.					0,05		0,05							0
Asellus aquaticus							0				0,05			0,05
Baetis sp.						0,4	0,4			0,6		0,6		1,2
Bithynia tentaculata		4,4					4,4							0
Chironomidae	0,05	0,05				1,8	1,9	0,05	4,2	4,8	0,1	5,4	0,2	14,75
Diptera: Empididae						1,3	1,3							0
Ecclisopteryx guttulata							0		1,3	4,9				6,2
Elmidae - Elmis sp.						0,5	0,5			0,05				0,05
Elmis Adult							0			0,05				0,05
Hydra sp.			0,05				0,05							0
Leuctra sp.							0	0,05						0,05
Nemoura/Nemurella sp.							0					2,7		2,7
Oligochaeta	0,05	0,05	0,3			0,4	0,8			0,9			0,05	0,95
Rhithrogena sp.							0		0,8	1	0,5			2,3
Sum	0,1	4,5	0,35	0	0,05	4,4	9,4	0,1	6,3	12,3	0,65	6	2,95	28,3

Habitat features 30-11-11

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	6	7	6	4	6	9	6	7	6	4	8	10
Current velocity [m/s]	0,1	0,3	0,2	0,05	0,2	0,25	0,1	0,3	0,2	0,05	0,3	0,2
Substrate	micro/ meso	akal/ micro	micro/ meso	akal/ micro	micro/ meso	akal/ micro	akal/ micro	akal/ micro	akal/ micro	micro/ meso	micro/ meso	akal/ micro
Micro algae [%]	50	30	50	10	60	40	90	20	95	70	70	95
Filamentous algae [%]	40	70	10	80	30	50	5	10		20		
Moss [%]												
Sum of vegetation cover [%]	90	100	60	90	90	90	95	30	95	90	70	95
Hydrurus [%]	40	70	10	80	30	50	5	10	0	20	0	0
Hydrurus thickness	1	1	1	1	1	1	1	1	1	1	1	1
Hydrurus total	40	70	10	80	30	50	5	10	0	20	0	0
electric conductivity [μS/cm]	237						238					
Water temperature [°C]	5,5						5,5					

Abundance 19-12-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.							0		3	1	1		2	7
Baetis alpinus	1						1	1	2	1				4
Baetis muticus			5		2		7			1			1	2
Baetis rhodani						1	1			1				1
Chironomidae	10	12	18	1	17	4	62	4	6		4	2	14	30
Chironomidae Pupae	3						3		1	1			1	3
Diptera: Limoniidae			1				1							0
Ecclisopteryx guttulata	1		1				2							0
Ecdyonurus sp.							0			1				1
Hydra sp.							0			1				1
Nemoura/Nemurella sp.							0					1	1	2
Oligochaeta	1	1	12	3	4	7	28			2	1	3	4	10
Rhithrogena sp.	1		2		1		4	1	1	1				3
Zooplankton	152	31	96	27	49	72	427	21	73	37	55	85	125	396

Biomass [mg] 19-12-11

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.							0		1,6	0,1	0,9		0,3	2,9
Baetis spp.	0,05		0,6		0,1	0,05	0,8	0,05	0,5	1,4			0,05	2
Chironomidae	13,8	4,7	13,2	0,05	11,9	3	46,65	0,6	8,1		0,8	7,6	1,3	18,4
Chironomidae Pupae	5,8						5,8		2,8	5			3,7	11,5
Diptera: Limoniidae			1,2				1,2							0
Ecclisopteryx guttulata	24,5		3,9				28,4							0
Ecdyonurus sp.							0			1,5				1,5
Hydra sp.							0			0,3				0,3
Nemoura/Nemurella sp.							0					1,1	1,8	2,9
Oligochaeta	0,05	0,05	18,4	0,2	0,1	0,8	19,6			0,3	0,05	0,3	0,2	0,85
Rhithrogena sp.	1,2		4,3		0,6		6,1	0,1	0,5	1				1,6
Sum	45,4	4,75	41,6	0,25	12,7	3,85	108,55	0,75	13,5	9,6	1,75	9	7,35	41,95

Habitat features 19-12-11

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	6	4	9	5	10	6	4	8	8	6	7	7
Current velocity [m/s]	0,5	0,1	0,5	0,3	0,3	0,1	0,15	0,5	0,4	0,4	0,5	0,6
Substrate	micro/ meso	micro/ meso	akal/ micro	meso/ makro	micro/ meso	micro/ meso	micro/ meso	akal/ micro	micro/ makro	akal/ micro	micro/ meso	micro/ meso
Micro algae [%]	10		10	20		25	10	20	40	70	60	40
Filamentous algae [%]	90	100	90	80	100	70	80	70	60	10	30	50
Moss [%]												
Sum of vegetation cover [%]	100	100	100	100	100	95	90	90	100	80	90	90
Hydrurus [%]	90	100	90	80	100	70	80	70	60	10	30	50
Hydrurus thickness	3	3	3	2	2	2	2	2	2	1	1	1
Hydrurus total	270	300	270	160	200	140	160	140	120	10	30	50
electric conductivity [μS/cm]	241						240					
Water temperature [°C]	4,2						4,3					

Abundance 18-01-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.						1	1		1					1
Ancylus fluviatilis			1				1							0
Baetis alpinus		1		1			2			3	1	1	1	6
Baetis muticus			3		4	2	9		3		2	1	1	7
Chironomidae	6	63	30	6	116	33	254		9	34	8	4	21	76
Chironomidae Pupae		5		1	4	5	15			1				1
Diptera: Simuliidae		1					1							0
Ecclisopteryx guttulata					2		2			2				2
Ecdyonurus sp.						1	1			1	1	1		3
Eiseniella tetraedra					1		1							0
Nemoura/Nemurella sp.	1						1						26	26
Oligochaeta	1	5		2	5	1	14	2	2			2	1	7
Rhithrogena sp.					1		1		1					1
Zooplankton	16	52	60	36	102	30	296	12	34	99	116	218	104	583

Biomass [mg] 18-01-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.						0,8	0,8		1,4					1,4
Ancylus fluviatilis			10				10							0
Baetis alpinus							0			0,05		0,05	0,05	0,15
Baetis sp.		0,05	0,1	0,1	4	0,3	4,55		0,8	0,05	0,1	0,05	0,3	1,3
Chironomidae	10,7	75,2	16,3	15,3	120,4	55,9	293,8		7,8	13,9	13,8	7,8	3,7	47
Chironomidae Pupae		18,8		2,8	21,9	11,5	55			4,3				4,3
Diptera: Simuliidae		0,3					0,3							0
Ecclisopteryx guttulata					66,8		66,8			76,9				76,9
Ecdyonurus sp.							0			2,3	1,3	0,1		3,7
Eiseniella tetraedra					88,7		88,7							0
Nemoura/Nemurella sp.	0,4						0,4						21,5	21,5
Oligochaeta	0,05	7,4		0,3	0,2	0,05	8	0,6	0,05			0,05	0,3	1
Rithrogena sp.					1,2	0,5	1,7		4,1					4,1
Sum	11,15	101,75	26,4	18,5	303,2	69,05	530,05	0,6	14,15	97,5	15,2	8,05	25,85	161,35

Habitat features 18-01-12

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	3	6	7	7	10	11	7	5	9	8	6	11
Current velocity [m/s]	0,3	0,4	0,5	0,4	0,2	0,3	0,5	0,4	0,4	0,3	0,4	0,4
Substrate	micro/ meso	micro/ meso	micro/ makro	akal/ micro	micro/ meso	meso	meso/ makro	micro/ meso	micro/ makro	micro/ meso	micro/ makro	micro/ meso
Micro algae [%]	10	10	20	5	10	95	5	5	10	10	60	90
Filamentous algae [%]	80	90	70	90	90	5	90	95	85	85	5	0
Moss [%]												
Sum of vegetation cover [%]	90	100	90	95	100	100	95	100	95	95	65	90
Hydrurus [%]	80	90	70	90	90	5	90	95	85	85	5	0
Hydrurus thickness	3	2	2	2	1	1	2	2	2	3	2	1
Hydrurus total	240	180	140	180	90	5	180	190	170	255	10	0
electric conductivity [µS/cm]	242						242					
Water temperature [°C]	2,5						2,5					

Abundance Seebach 18-01-12

Sample	s1	s5	s6
Amphinemura sp.	179	48	36
Ancylus fluviatilis	2	3	2
Athericidae - Ibisia marginata	7	1	1
Baetis alpinus	288	39	38
Baetis muticus	278	129	70
Baetis rhodani	247	49	44
Chironomidae	565	159	109
Dictyogenus sp.	2		
Dinocras sp.	1		
Ecdyonurus sp.	1	13	7
Eiseniella tetraedra	3		
Elmis sp. adult	2	8	2
Elmis sp. Larvae	17	19	21
Empididae	84	42	30
Epeorus assimilis	19		
Ephemerella major		1	9
Glossosomatidae	9	45	4
Goeridae		1	
Gyraulus sp.		7	4
Habroleptoides confusa	1	5	5
Hydrachnidia	4		12
Hydrobiidae	3	1	7
Hydropsyche sp.	109	17	27
Isoperla sp.	10		
Lepidostoma hirtum	3	1	19
Leuctra sp.	4	5	5
Limoniidae - Antocha sp.		4	
Micrasema minimum	22	16	11
Micrasema setiferum	141	211	182
Nemoura/Nemurella sp.		2	
Oligochaeta	7	10	23
Orectochilus villosus	19	17	25
Perla sp.	1		
Planaria	7		
Polycentropus sp.		1	6
Propappus volki	3		
Rhithrogena sp.	35	4	2
Rhyacophila sensu strictu	5	1	1
Rhyacophila tristis	7		
Riolus sp. adult	17	23	11
Riolus sp. Larvae	126	114	68
Scirtidae larvae	63	1	29
Sericostoma sp.			1
Simuliidae	140	11	9
Sum	2431	1008	820

Biomass [mg] Seebach 18-01-12

Sample	s1	s5	s6
Coleoptera	70,6	96,2	87,7
Chironomidae	76,3	20,7	19,6
Ephemeroptera	425,8	151,8	148,8
Gastropoda	11,1	35,2	26,7
Oligochaeta	212,5	26,2	78,2
Plecoptera	539,2	45,7	39,6
Trichoptera	2435,5	342,9	543,2
Rest	670,9	141	125,2
Sum	4441,9	859,7	1069

Habitat features Seebach 18-01-12

Sample	s1	s2	s3	s4	s5	s6
Depth [cm]	21	10	15	22	21	20
Current velocity [m/s]	1	0,5	0,5	1	0,6	0,4
Substrate	micro/makro	makro	meso/makro	makro	meso	micro/meso
Micro algae [%]	95	65	80	20	95	95
Filamentous algae [%]	0	5	0	0	0	0
Moss [%]	5	30	20	80	5	5
Sum of vegetation cover [%]	100	100	100	100	100	100
electric conductivity [μ S/cm]	245					
Water temperature [°C]	2,4					

Abundance 29-02-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.	1						1							0
Baetis muticus	1	2			3		6		7	2	2			11
Baetis rhodani							0	1						1
Chironomidae	141	56	2	8	178	35	420	5	86	87	27	10	43	258
Chironomidae Pupae	13	2		3	7	1	26		4	7		3	4	18
Diptera: Simuliidae							0		1					1
Eiseniella tetraedra					1		1							0
Elmidae - Elmispis	1						1							0
Elmispis Adult	1				1		2							0
Nemoura/Nemurella sp.							0			1	1	5	1	8
Oligochaeta	2	1	1		2	1	7		1	1	1	1		4
Riolus sp. Adult			1				1							0
Siphonurus lacustris	1						1	1						1
Zooplankton	205	50	6	15	306	38	620	6	64	49	10	28	19	176
Notes	Swarming midges observed around the channels, 2 Adult Diamesa sp. + 1 Ad. Orthocladiinae caught													

Biomass [mg] 29-02-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.	0,9						0,9							0
Baetis and Siphonurus	1	0,7			0,2		1,9	1,3	0,5	0,1	0,05			1,95
Chironomidae	266,4	62,6	3,3	7,1	203,4	34,1	576,9	5,9	127,7	110,6	25	19	94,2	382,4
Chironomidae Pupae	68,2	2,1		17,2	6,7	1,4	95,6		15,9	20		11,4	12	59,3
Diptera: Simuliidae							0		0,1					0,1
Eiseniella tetraedra					29,5		29,5							0
Elmidae - Elmispis	0,7						0,7							0
Elmispis Adult	0,5				0,05		0,55							0
Nemoura/Nemurella sp.							0			3,8	6,5	7,5	2	19,8
Oligochaeta	0,1	0,2	0,05		0,05	0,1	0,5		0,05	0,05	0,1	0,05		0,25
Sum	337,8	65,6	3,35	24,3	239,9	35,6	706,55	7,2	144,25	134,55	31,65	37,95	108,2	463,8

Habitat features 29-02-12

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	7	4	7	3	8	8	5	8	8	6	5	8
Current velocity [m/s]	0,1	0,05	0,3	0,2	0,3	0,2	0,35	0,3	0,35	0,2	0,3	0,2
Substrate	micro/meso	akal/micro	micro/meso	micro/meso	micro/meso	micro/meso	akal/micro	akal/meso	micro/meso	akal/micro	meso/macro	micro/meso
Micro algae [%]	85	90	60	85	50	50	5	90	70	70	50	75
Filamentous algae [%]	5	5	0	5	0	0	95	5	0	30	0	20
Moss [%]												
Sum of vegetation cover [%]	90	95	60	90	50	50	100	95	70	100	50	95
Hydrurus [%]	5	5	0	5	0	0	95	5	0	30	0	20
Hydrurus thickness	1	1	0	1	0	0	2	1	1	1	0	1
Hydrurus total	5	5	0	5	0	0	190	5	0	30	0	20
electric conductivity [µS/cm]	243						243					
Water temperature [°C]	3						3					

Abundance 20-03-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.	1	1					2		1	3				4
Baetis alpinus	1						1							0
Baetis muticus	2	3	3	7		4	19	6	3	8	5	1		23
Baetis rhodani							0		2	1		1		4
Chironomidae	84	164	73	139	162	178	800	50	157	165	158	176	149	855
Chironomidae Pupae	37	51	23	40	71	55	277	20	73	43	58	55	25	274
Diptera: Diceranota sp.							0			1				1
Diptera: Limoniidae							0						1	1
Diptera: Simulium sp.				1			1			1				1
Ecclisopteryx guttulata			2	1			3					1		1
Eiseniella tetraedra					1		1							0
Gammarus fossarum		1					1							0
Leuctra sp.							0		1					1
Nemoura/Nemurella sp.		2				1	3			4	5	12	11	32
Oligochaeta	4		1	2	8	11	26		1	1		1		3
Siphonurus lacustris							0	5	3	3	4	3	1	19
Zooplankton	33	195	107	195	165	320	1015	132	335	230	255	205	47	1204
Notes	Chironomidae Adult in I1 and I5													

Biomass [mg] 20-03-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.	2,7	2,4					5,1		0,9	5,6				6,5
Baetis + Siphonurus	0,8	1,8	0,2	0,9		0,8	4,5	1,9	5,7	8,1	3,2	2,2	0,6	21,7
Chironomidae	298,3	485,2	193,5	316,5	275,3	417,7	1986,5	60,5	438,1	300,8	315,8	315	252,9	1683,1
Chironomidae Pupae	195,8	208,9	74,4	107	260,6	137,9	984,6	48,3	294	112,3	235,8	213,3	77,4	981,1
Diptera: Diceranota sp.							0			18,7				18,7
Diptera: Limoniidae							0						1,5	1,5
Diptera: Simuliidae				0,5			0,5			0,5				0,5
Eiseniella tetraedra					40,2		40,2							0
Gammarus fossarum		16,4					16,4							0
Leuctra sp.							0		2,5					2,5
Nemoura/Nemurella sp.		7,7				4,9	12,6			26,3	14,2	38,8	35,2	114,5
Oligochaeta	0,2		0,05	0,05	0,5	1,5	2,3		5,8	0,1		0,1		6
Sum	497,8	722,4	268,15	424,95	576,6	562,8	3052,7	110,7	747	472,4	569	569,4	367,6	2836,1

Habitat features 20-03-12

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	5	7	8	7	6	8	5	8	8	6	7	9
Current velocity [m/s]	0,15	0,2	0,5	0,4	0,25	0,3	0,2	0,3	0,35	0,3	0,35	0,4
Substrate	meso/ macro	akal/ micro	meso/ macro	meso	micro/ meso	meso	micro/ meso	micro/ meso	meso/ macro	micro/ meso	micro/ meso	micro/ meso
Micro algae [%]	45	40	50	65	5	70	5	25	95	5	20	95
Filamentous algae [%]	50	40	5	5	90	0	90	70	0	90	75	0
Moss [%]												
Sum of vegetation cover [%]	95	80	55	70	95	70	95	95	95	95	95	95
Hydrurus [%]	50	40	5	5	90	0	90	70	0	90	75	0
Hydrurus thickness	1	1	1	1	1	0	2	2	0	2	2	0
Hydrurus total	50	40	5	5	90	0	180	140	0	180	150	0
electric conductivity [μS/cm]	245						242					
Water temperature [°C]	4,6						4,9					

Abundance Seebach 20-03-12

Sample	s1	s4	s5
Amphinemura sp.	16	33	133
Baetis alpinus	96	32	80
Baetis muticus	103	93	340
Baetis rhodani	55	83	102
Bithynia tentaculata	8		
Caenis sp.		1	
Chironomidae	75	202	247
Ecdyonurus sp.	3		3
Eiseniella tetraedra	2		13
Elmis sp. adult		5	
Elmis sp. Larvae	1	50	3
Empididae			8
Epeorus assimilis			1
Ephemera danica		3	
Ephemerella major		1	
Gammarus fossarum	1		3
Goeridae		2	
Habroleptoides confusa	10	17	27
Hydraenidae			1
Hydrobiidae	1	7	6
Hydropsyche sp.	3	6	2
Lepidostoma hirtum	12	31	13
Leptoceridae		7	
Leuctra sp.	1	7	12
Megaloptera		3	
Micrasema minimum		6	1
Micrasema setiferum	14	341	154
Nemoura/Nemurella sp.	6	2	21
Oligochaeta		12	75
Orectochilus villosus		2	1
Planorbidae - Gyraulus sp.	8	25	3
Polycentropus		11	1
Psychodidae	2		1
Radix sp.		3	
Rhithrogena sp.			1
Riolus adult		11	1
Riolus sp. Larvae		170	8
Simuliidae	105	13	336
Tabanidae	1		
Tipulidae		1	
Valvata cristata		1	
Sum	523	1181	1597

Biomass [mg] Seebach 20-03-12

Sample	s1	s4	s5
Coleoptera	0,6	72,8	38
Chironomidae	11,8	34,3	16,7
Ephemeroptera	153	440,7	429,8
Gastropoda	135	327,3	15,5
Oligochaeta	93,5	21,5	1225,7
Plecoptera	55,5	62,8	377,1
Trichoptera	174,6	414,4	135,2
Rest	230,9	354,6	291,1
Sum	854,9	1728,4	2529,1

Habitat features Seebach 20-03-12

Sample	s1	s2	s3	s4	s5	s6
Depth [cm]	12	18	15	25	16	17
Current velocity [m/s]	1	1,2	0,8	1	1,4	0,9
Substrate	meso	meso/macro	micro/meso	meso	micro/meso	micro/meso
Micro algae [%]	100	85	90	100	70	100
Filamentous algae [%]	0	0	0	0	0	0
Moss [%]	0	15	10	0	0	0
Sum of vegetation cover [%]	100	100	100	100	70	100
electric conductivity [μ S/cm]	241					
Water temperature [°C]	5,7					

Abundance 10-04-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.	1	2	1	1	1	1	7		2	2				4
Baetis muticus	1	4	14	4	1		24		6	3	5	3	4	21
Baetis rhodani			1				1							0
Chironomidae	260	193	197	225	91	164	1130	64	316	270	213	230	219	1312
Chironomidae Pupae	26	9	39	21	12	13	120	7	25	19	9	14	7	81
Diptera: Limoniidae							0						1	1
Diptera: Simuliidae	1		3		1		5			1		2		3
Ecclisopteryx guttulata							0		1					1
Ecdyonurus sp.					1		1		1					1
Elmidae - Esolus sp.		1					1						1	1
Esolus sp. Adult							0	1						1
Isoperla sp.							0			1				1
Leuctra sp.		3	1	3	1		8					1		1
Micrasema setiferum		1					1							0
Nemoura/Nemurella sp.			3	1		1	5		3	3	3	5	6	20
Oligochaeta	12	3	16	6	5	20	62		1		1	2	1	5
Protonemura sp.			1				1							0
Rhyacophila sensu strictu							0	1						1
Siphonurus lacustris							0	4	5		3			12
Zooplankton	14	7	16	17	11	68	133	55	20	3	2	1	5	86
Notes	Diamesa sp. Adult in I1													

Biomass [mg] 10-04-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Amphinemura sp.	0,4	3,3	1,6	1,8	2	1,5	10,6		4,9	4,1				9
Baetis + Siphonurus		0,4	2,7	0,6	0,1		4,9	7,7	4,3	0,5	0,5	0,6	0,7	16,6
Chironomidae	479,7	340,2	324,25	286,7	126,2	161,5	1718,55	80,9	501,3	479,1	286,7	277,4	344,1	1969,5
Chironomidae Pupae	44,8	17,1	91,6	63,1	17,8	21,8	256,2	13,8	54,1	38,1	13,4	22,4	9,2	151
Diptera: Limoniidae							0						1,2	1,2
Diptera: Simuliidae	0,1		0,05		4,1		4,25			0,2		1,6		1,8
Ecclisopteryx guttulata							0		24,6					24,6
Ecdyonurus sp.					2,6		2,6		0,2					0,2
Elmidae - Esolus sp.		0,1					0,1						0,05	0,05
Esolus sp. Adult							0	0,1						0,1
Leuctra sp.		3,4	0,3	0,6	0,05		4,35					0,1		0,1
Micrasema setiferum		0,05					0,05							0
Nemoura sp. + Isoperla		0,3	13,8	4,5		11,4	30,3		20,4	19,9	12,3	21,3	17,8	91,7
Oligochaeta	1,1	0,1	1,8	0,5	0,5	2,5	6,5		0,1		0,05	0,1	0,2	0,45
Protonemura sp.			0,8				0,8							0
Rhyacophila sensu strictu							0	97,1						97,1
Siphonurus sp.	0,05						0,05				14,3			14,3
Sum	526,15	364,95	436,9	357,8	153,35	198,7	2037,85	199,6	609,9	537,8	327,25	323,5	373,25	2371,3

Habitat features 10-04-12

Sample	I1	I2	I3	I4	I5	I6	r1	r2	r3	r4	r5	r6
Depth [cm]	7	5	5	6	8	8	4	6	5	4	7	8
Current velocity [m/s]	0,2	0,15	0,25	0,4	0,3	0,2	0,1	0,25	0,1	0,2	0,3	0,15
Substrate	micro/ meso	akal/ micro	meso	micro/ meso	micro/ macro	meso	meso	micro/ meso	micro/ meso	micro/ meso	micro/ meso	micro/ meso
Micro algae [%]	20	60	40	50	95	95	5	85	80	20	70	70
Filamentous algae [%]	60	5	0	10	0	0	90	5	5	0	0	0
Moss [%]												
Sum of vegetation cover [%]	80	65	40	60	95	95	95	90	85	20	70	70
Hydrurus [%]	60	5	0	10	0	0	90	5	5	0	0	0
Hydrurus thickness	1	1	0	1	0	0	1	1	1	0	0	0
Hydrurus total	60	5	0	10	0	0	90	5	5	0	0	0
electric conductivity [μ S/cm]	237						237					
Water temperature [°C]	5,8						5,8					

Abundance Chironomidae Larvae+Pupae 10-04-12

Sample	I1	I2	I3	I4	I5	I6	Σ I	r1	r2	r3	r4	r5	r6	Σ r
Chironomidae indet.		1			1		2				1	2		3
Corynoneura / Thienemanniella sp.			1				1							0
Corynoneura sp.			2	3		3	8	1	3					4
Cricotopus sp.							0		1					1
Cricotopus/Orthocladius/Paratrichocladius indet.	1		1	1	1	2	6		1					1
Diamesa cinerella gr.	13	7	10		1	5	36		6	7	2			15
Diamesa cinerella gr. /zernyi gr.							0			15		15		30
Diamesa cinerella gr. w\insignipes (early stages)	49	37	26	57	8	28	205	19	51	45	60	27	27	229
Diamesa cinerella/tonsa	62	43	56	29	25	29	244	4	41	63	28	25	36	197
Diamesa hamaticornis	71	49	114	63	17	63	377	21	147	83	92	145	116	604
Diamesa insignipes	57	48	15	52	25	28	225	13	64	45	24	7	32	185
Diamesa zernyi Gr.		1	7	1		3	12						2	2
Eukiefferiella sp.	1						1					1		1
Heleniella ornaticollis		1					1					1		1
Heterotrissocladius marcidus					1		1	1						1
Heterotrissocladius sp.					1		1							0
Micropsectra / Tanytarsus sp.	1						1	3			3		1	7
Micropsectra atrofasciata	5	12	6	13	5	3	44	5	6	11	4	4	1	31
Micropsectra notescens					1		1				1	1		2
Micropsectra pallidula	1						1							0
Micropsectra sp.						2	2							0
O. (Euorthocladius) ashei			2	1			3					1		1
O. (Euorthocladius) rivicola			1				1							0
O. (Euorthocladius) thienemanni				1			1							0
Orthoclaadiinae indet.	2	1			1		4							0
Orthocladius frigidus	1		1			1	3		3		1			4
Orthocladius ob. umbratus			1				1							0
Orthocladius rivulorum			1				1			3				3
Orthocladius rubicundus	1	1					2		3					3
Orthocladius sp.				2		1	3		3	3	1			7
Parachironomus arcuatus Gr.							0				1		1	2
Parametriocnemus stylatus							0						1	1
Paratanytarsus austriacus							0			1				1
Paratanytarsus sp.							0			1		1	1	3
Paratrichocladius rufiventris	1	2		1	2	2	8		1	2		1	1	5
Paratrichocladius skirwithensis	1		1	2			4					2		2
Paratrichocladius veronicae							0		1					1
Parorthocladius nudipennis	4		4	3	3		14	2	3		1	1	2	9
Pentaneurini indet.				1			1							0
Phaenopsectra flavipes							0						1	1
Phaenopsectra sp.							0	1						1
Polypedilum albicorne		1		1			2						1	1
Polypedilum laetum	2			1			3							0
Rheocricotopus chalybeatus							0		1					1
Rheocricotopus effusus							0			1				1
Rheocricotopus fuscipes					1		1							0
Stempellina bausei			1	1			2						1	1
Synorthocladius semivirens	2			1		1	4	1						1
Tanytarsini indet.		3		2		3	8							0
Tanytarsus brundini / curticornis						1	1	1				1	3	5
Tanytarsus sp.				2			2					9		9
Thienemanniella sp.				1			1						1	1
Tvetenia calvescens	1		1	1	2		5			2	2			4
Sum	276	207	251	240	95	175	1244	72	335	282	221	244	228	1382

Sample pictures 28-09-11



l1



l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 13-10-11

l1



l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 25-10-11



l1



l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 17-11-11

		
l1	l2	l3
	No photo l5	
l4		l6
		
r1	r2	r3
		
r4	r5	r6

Sample pictures 30-11-11



l1



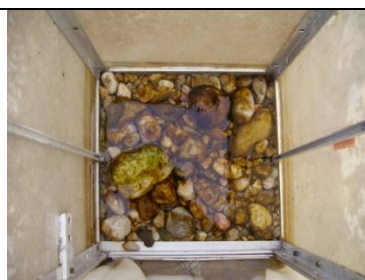
l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 19-12-11

 l1	No photo l2	 l3
 l4	 l5	 l6
 r1	 r2	 r3
 r4	 r5	 r6

Sample pictures 18-01-12



l1



l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 29-02-12



l1



l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 20-03-12



l1



l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6

Sample pictures 10-04-12



l1



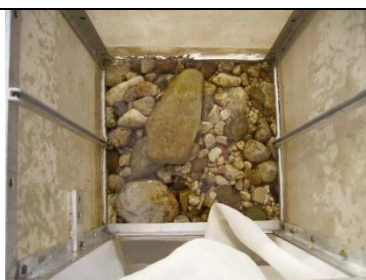
l2



l3



l4



l5



l6



r1



r2



r3



r4



r5



r6